

Dare to be the Best - Chapter 1

Dare to be the
best

Looking into the future

The dream of winning a major championship motivates many of us. To be a winner you need to be able to see into the future, and the only thing we can know about the future is that is not going to be like today. Things that are true today do not need to be the same tomorrow, totally the opposite in fact. The only thing we know with any certainty is that there will be change. As an elite athlete you have to be ready for that change.

Daring to be different

A winner can be different. Perhaps you could even say that a winner must be an original thinker. Winners do not copy their rivals. They find their own way to success. A winner dares to think differently. There are many examples of new ways of thinking and creativity in elite sport. Björn Borg's double-handed grip in tennis, the Fosberry flop technique in high jumping, or Graeme Obree's extreme "superman" aerodynamic position in cycling are examples of how a winner thinks differently and looks beyond accepted wisdom. A winner also knows that common goals, shared with leaders or coaches, release the energy needed to focus on hard work. If both the leader or coach and the athlete have shared goals it is easier to understand each other and this can help build a good relationship between leader and athlete. If goals are not shared it is much harder to understand each other, which can lead to a poor relationship between the athlete and leader. We should remember this when trying to help an athlete release the energy needed to strive to do their best. This is especially true when we face setbacks and when things are not working out as planned.

Release the energy needed to be a winner

Athlete

Shared goals Communication Relationships

Leader

Relationships are the basis of success

If a good relationship has been built up between an athlete and leader it is much easier to communicate with each other and also to solve any problems that arise. It is

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therefore essential that relationships between athlete and leader be based on honesty, trust, security, fun, and camaraderie. These factors are extremely important for elite athletes if they are to achieve their goals. Many coaches, leaders and athletes, believe that this kind of teamwork helps the athlete to perform optimally. So we need to place a lot of emphasis on the social competence of leaders if we want to help and support elite athletes towards successful performances. By social competence we mean the ability to create enthusiasm in a group, to inspire, to care, to be honest and fair and to be a role model – to practice what we preach. A winner:

- Has the ability to think in new ways
- Knows when to compromise and when something is worth fighting for
- Understands the need to experience something for real in order to master it
- Thinks in the present
- Sees possibilities
- Has self-confidence
- Knows that success breeds success.
- Knows that you can succeed if you want to

- Knows that good team spirit creates a winning attitude
- Knows that shared goals release the energy needed for hard work
- Sees the struggle as a challenge
- Has greater inner motivation than external motivation
- Knows that you need to give everything all the way to the finish line
- Works harder than a loser
- Always takes full responsibility for his or her performance
- Has fun
- Is modest
- Loves to compete

Top performance

On the top of the podium, a medal round your neck, head held high, looking at the Union Jack, tears in your eyes, a lump in your throat. The feeling of standing on the podium, the feeling of singing the national anthem is something all elite athletes long for. The dream of success in the sports arena both culminates and ends as the flag is hoisted to the top of the pole. This intensive moment of extreme happiness is so very short. The national anthem lasts perhaps half a minute. If you are lucky they might play the second verse and your moment of glory will last a little longer. This sort of success does not come by chance. Those who succeed have many things in common. Years of hard work, training and more training have eventually given results. Training weekends, camps abroad, endless sessions at home and careful preparation have resulted in the success that they longed for. Even if physical form, which is essential to many sports, is at a peak before the competition it does not always bring an optimal performance. Many international

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successes by elite athletes can be explained in other ways. Enthusiasm, self-confidence, attitude, mental strength, a winning instinct, the capacity to prepare thoroughly, the capacity to give everything in the race and to take responsibility for your own performance are some of the factors which have by far the greatest effect on your result. Producing a great performance involves much more than the last few weeks before the championship. Top performance begins at the point where you decide that you will be the best at one special, chosen time. Getting on that day and achieving that successful performance requires a strategy that takes you there. Strategic thinking

- The competition, for example BOC or WOC, will dictate the content of my plan.
- Every training session will have relevance and meaning for that competition. If I want to succeed at a championship, how will my training look – technical, tactical, physical, mental – after my decision to achieve a certain result at a competition in the future? How can I work or study? How can I strike a good social balance with family and friends? What support can I get from my club or region? I must start by taking stock of my current situation in its entirety and then draw up a strategy to achieve my goals. I need a plan. If you want to peak at a chosen point in the future, you could have a plan based on the following factors: My surroundings Which factors are going to influence me? Development of physical, mental, technical and social competence happens simultaneously. The competition What can I expect? An analysis of the demands I am likely to face in the race. Me Where do I stand? An analysis of my strengths and weaknesses. Training How can I keep my goal alive during my training? Competition-like model training. Strategy How can I achieve my goal? Build

up my own model, which fits into my life and suits my personality, but can also be changed and developed with time. My preparation for a top performance begins when I have decided on my goal. The process of preparation can be split into four periods:

- A long time before the competition: 1-4 years before the race.
- Before the competition: 4-10 weeks before the race.
- Shortly before the competition: the last weeks before the race.

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- Just before the competition: the last day before the race.
- During the competition: actually competing in the race.
- Between competitions: often the competition will consist of several races, for

example a championship with qualification and final races.

- After the competition: immediately after the race.

Camp - Race - Camp - Race - Camp Strategy towards my goal Major
Race Training Training Training Long time before Before Shortly before Just before During
Between After

A long time before the competition

This is when I lay the groundwork for my strategy. I decide that this really is what I want to do. I change my goals from something verbal and external into a desire deep inside my heart. I try to create an all-encompassing vision that sets all the jigsaw pieces in place. Work, studies, family, friends, club, coach, leaders will all contribute to my struggle to achieve my goal. I get acceptance from my surroundings and I can focus on the coming championship. From now on my strategy is internal and my drive comes from inside me. The desire to win is not as important as the desire to prepare yourself to want to win. Preparation for the championship begins right now with, for example, gathering information about things like the terrain, technical and physical requirements, course setter, the mapper and other important facts. I make a list of everything that is new or can affect my performance capability. I look for things and ideas that I can introduce into my everyday training as model training. A lot of my preparation involves informing myself about what is coming, and accepting this as something positive for my preparations to achieve the planned result. When I have realised and accepted what I have to do reach my goals I “just” need to develop or change my training to reflect this. It is at this stage that I build on my strategy and try to create thoughts and images that influence my capability to perform, through belief in the route I have chosen to achieve my goals. I will attend training camps and competitions near the championships area so that I can get to know the competition and accommodation environment. This gives me experiences, images and feelings that both strength my goal and make it more alive. It is also important that I compete in major fixtures where I can test my capacity fully. These competitions are model competitions: They are a testing ground where I can prepare for the race or carry out, evaluate and improve different models for the essential ingredients in my training.

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Before the competition

It is now that I move away from basic training and start to prepare myself before the peaking process in the last few weeks. This part of my training contains more or less only specific sport-related training. The technical and physical parts of my performance need to become internal and automatic here. Then I can work more with mental preparation, thinking about situations I might face in the competitions. Those might be extremely warm weather, very heavy rain, meeting the race favourite in qualification, a bad course, or an unfavourable start position. I go through every distraction that I could face and mentally turn each one into something positive. I also need to think about social aspects at this stage, for example flexible working times, leave from work or studies, support from my family or club. Shortly before the competition

The competition is getting closed and my preparations become more focused. My lifestyle is organised very carefully. Training and my attitude to performance have a very high priority now. Quality sessions, diet, rest and sleep are important factors in staying illness free. I concentrate more and more on how things are going to be when I am competing. The vision is becoming clearer and clearer and I begin to long for that perform-when-it-counts atmosphere. During this time I try to only think positive thoughts: “Eor-ism” and “can’t” has no place in my thoughts. It is important to only look for possibilities now. It is also important that I am able to carry out what I want to do. It is therefore important that leaders understand that each person must be treated a little more individually now than earlier. My physical and mental fine-tuning process has begun. Just before the competition

Just one day left. I know the conditions for tomorrow’s competition now. It is time to set my pre-race routine into action and begin to psych up for the competition. Some of the questions I now need to address are: What is my last training session going to be? When am I going to eat? Am I going to jog in the morning? When should I start to warm up? Which coaches are going to be at the start? When am I starting? Is there anything new that has come up at the last minute? In my concentration I can see myself execute a perfect ace, I have flow, I think positively, I am strong, I can, I want to, I dare to and I think this is great fun. My opponents are there, but this is my competition and I am totally dependant on my own efforts and my own will power. It is me who has total responsibility for my performance. During the competition

I stand on the start line and it is now that I will perform optimally and give my absolute best. At BOC, WOC or a World Cup many things surrounding the competition are different but actually competing, carrying out my race, is very like a normal competition. My technique, my speed and my thoughts are totally concentrated on the task I will carry out. My technique is automatic, sitting deep in my reflexes, while my concentration and motivation are totally focused on the task. I only think about the present and the future. I dare to be a winner!

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Between competitions

Total exhaustion – you gave everything. Enjoy the sweet taste of victory. Relaxation and recovery. Time to charge the batteries again. The medal can sit on the side. I can read the newspapers when I come home. What is my next task? Is there something I can learn and take with me from the previous race. Any changes to the conditions? Back into my own routine. After the competition

One of the most important parts of a top performance is to evaluate the result. Why did it go the way it did? Success or failure? Analyse how successful I was – write down and improve the things that went well, note and change the things that went less well. How can I develop in the future before the next exciting competition? Keywords for athletes

The following keywords and phrases are a summary of interviews with athletes in different sports:

- Enthusiasm and commitment
- Think about the whole picture
- Planning in advance
- Take responsibility
- 24-hour athlete
- Performance focused
- Personal routines
- Performance analysis
- Attitude Commitment

Elite sport today is exceptionally demanding. If you want to succeed nationally, you need to understand that you are really competing at an international level. You need to prioritise what is more and less important. You have to “choose sides.” Do I really want to be British Champion or get a place in the national team, and sometime in the future perhaps even be a world champion? What consequences will that have for the rest of my life? Or, if I become world champion, what happens then? What does that mean for me – socially, for my work and my finances? Those elite athletes who can easily and quickly answer the question: Why elite sport? find it much easier to set goals and plan their way to achieving their goals. The content of their training is very clear for them. Being able to answer that question creates determination in an athlete. The vision of standing on the top of the podium releases the energy that is needed to train every day, year after year for 15-20 years. The figure below illustrates how much time you need to spend answering each question for yourself:

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Why?

Because...

Requirements?

Capabilities?

What? When? Where? How? With

whom? The distance between the two dark areas symbolises how long each question needs to be considered. The more time you spend answering the big question Why? the easier it is to answer the easy questions later. What sort of training shall I do? When should I train?

Where should I train? How should I train? Who can help and support me? It is in my answer to the big question Why? that I create the passion to compete and develop the will to win.

Passion, the spark inside, is the lynch pin in the process of achieving success. Terry Orlick, a sports psychologist from Canada, states clearly that passion and commitment, together with strong self-confidence are the building blocks of success. By commitment we mean a strong desire to achieve a goal - the athlete must be determined and have a positive attitude to the work to be carried out. The athlete works with her heart and has a strong inner motivation.

One way to strengthen and increase your commitment is to work with vision and goal setting. Commitment is also related to the level and importance of a particular competition. The more challenging a goal is, the more commitment is required. Commitment is often about the will to prioritise more and train harder, longer and more often. I need to analyse orienteering, its demands and my own capabilities. What is orienteering like today, and how might it be in the future? What should I be prepared for? What is required of me as a future orienteering champion? What capabilities do I have now? When these questions are answered it is

relatively easy to build a personal training program that is effective, varied and balanced. A committed sports person is curious. He or she looks for new ways to approach a task, listens to coaches and other competitors, discusses and questions different solutions. She remembers the gems that others come up with and then creates her own approach to the task, in a way that suits her own personality. A committed sports person thinks that an important competition is a challenge, which she enters into with enthusiasm. A committed sports person loves her sport and, loves to compete and loves to test her capacity to perform. A committed sports person has strong will power, believes in herself, is positive and always takes the initiative in her actions. A committed sports person is a winner.

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Thinking about the whole picture

What is it that makes a winner different from a loser? Apart from the earlier list of qualities that make up a winner, "a winner..." it is also likely that a winner is willing to make elite sport a central part of their life. Many athletes who have succeeded have had the ability to look both at small details, but also the whole picture. By looking at their sporting life from both perspectives, it is probably much easier for them to get structure both in their training and in life outside sport.

Performance

Genes Environment

Finance Other Training Social Work General Specific In order to perform optimally and achieve success you need to consider all the factors that can affect you and your progress towards your goals. You cannot change your inherited talent, but you can influence, change or develop many of the factors in your environment. Areas which do not come under training but are more to do with health, family, work, studies, leisure or finance have a lot of influence on the whole for elite sportspeople. Most athletes who have stood on the podium at WOC or world cup have thought through these things very carefully. Before an elite athlete spends time planning the details of her training programme, That your partner accepts her commitment to elite sport, that your employer understands and supports your commitment, and that you are clear in your own mind that this is what you have chosen to do are some of the areas an athlete should think about before you start spending time planning the details of your training programme. Feeling good, being free from injuries and being in harmony with yourself and your surroundings creates solid platform for winning performances. When the big picture is solved, it is time to further with a detailed plan for general and specific training. What type of general training can I carry out? What other sports apart from my own can raise or complement my physical, mental and technical capabilities? Which specific training methods will I use to be a better competitor in my own event? By analysing your own sport in detail, listening to other people, thinking about your sport's future requirement and possibilities it becomes much easier to answer questions such as when, where, how and with whose help you should train. Getting down to this detailed level of planning helps you to get the most out of your training.

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Long-term planning

A successful elite athlete makes an early mental decision – 1-4 years before the event. Peaking for a championship begins when you have decided to focus on that event. An early decision is important so that you can make a strategic plan that leads to the event. The chosen event therefore steers the content of your training. Things that are relevant to the event, physically, mentally, technically and tactically, dictate what sort of training you should do. At training camps, both at home and abroad, competition-like training should be a key

component. That which you expect to experience during the competition should also have been experienced in your training. For that reason, competition-like training should also be a regular and important part of your everyday training. When we talk about competition-like training it is important to remember that both physical and mental aspects are included in the idea. In summary we can say that every training session has significance for the approaching competition. The nearer the competition comes, the more competition-like training should be included in the elite athlete's preparations. Initiative and responsibility
Far too many less successful elite athletes believe that resources, in the form of sponsors, coaches, training facilities, equipment or money are the limiting and decisive factors. Success is no coincidence for those who believe that their own initiative is worth much more than resources. Successful athletes take responsibility for their own performance. When things do not go so well in a competition, they look to themselves for the reasons and things they did wrong, not at their resources. Complaining about external factors is an escape from reality and creates excuses that do not help the athlete forwards.

Own initiative Many believe that the figure on

Own initiative the left shows the right relationship between resources and initiative. The Resources one on the right is of course much Resources more likely to lead to success. A lot of this reasoning stems from the fact that when the athlete is actually standing on the start line, it is only her who can influence her performance. It is me, just me, who can run a good race. Coaches and leaders can really only support and help me before the start and after I have arrived in the finish field. During the race it is me, and my thoughts which steer my body towards an optimal performance. It is my responsibility, no anyone else's, to succeed.

24-hour athlete

An elite athlete, in this case a winner, is systematic in almost everything. When she has decided something, she does it. A winner has the ability to put turn thoughts into

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actions, or plans into reality. A winner can also use her time more effectively than others. She is effective and has very little "energy leakage", which means that she focuses on the things are best done right now. In her life, she strives to give 100 percent the whole time. By this we mean that when you are training or competing, you give it 100 percent. All your thoughts are focused on sport, it is here and now that matters. What happened at work, or what is happening at home with the family are of no interest just now. When you are work, you are a 100 percent worker, and when you are at home, you are a 100 percent family member. If your thoughts leave what you are doing now, your focus and concentration on the task will also be less. A winner is focused on the moment, therefore he or she gives 100 percent to everything he or she does. Whether it is sport, work, family, leisure or partying there is only one thing that is important for a winner: Giving 100 percent! Performance focused

Everyone wants to win the competition. Victory is a signal that you have performed extremely well and that you were best when it counted. On the other hand, you could also produce an optimal performance near your maximum capacity but still not win the competition. You are still a winner in that you have given the best you can. "Spiking" all the controls as quickly as possible in a championship is a "victory" that we should all strive towards. As a runner, the important thing to remember is that I can influence myself a great deal, my opponents very little. In my vision and goal, which perhaps functions more as a source of inspiration for my performance, I stand on the top of the podium. But before and during the competition I am totally focused on doing my best with every stride, every route choice, every control.

Thoughts only exist in the present, not in the past or the future – only right now. The only thing I can influence is the present, and I am going to do that to the best of my ability. A

winner is performance focused. The diagram below shows how a winning elite athlete thinks before an important competition.

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This is I'm in good I take BIG fun! shape responsibility RACE

This is a challenge

I feel I believe in I am in harmony myself motivated I will do I am my best focused

PERFORM WIN!

Personal routines

Everyone is unique. We all have different requirements for support. Everyone wants to do things in their own way. Many believe that it is important that a leader knows how each athlete reacts at the big race. Athletes have their own rituals before championships which can often seem totally absurd to the uninitiated, but can be crucial for the runner to get herself into the right frame of mind. A winner has the right to be different or unique in a winning environment.

Performance analysis

- What went well?
- What can I do better?
- When, where, how? With whom? A winner analyses these things that went well during the competition. The analysis of what went well is strengthened and stored in her reflexes for competitions in the future. Things that went less well are dealt with through positive ideas to make them better. A winner analyses her successes because she knows that success breeds success. Analysing mistakes in detail does not make a winner. Think how much time we spend talking about mistakes we made out on the course.

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Attitude

How you approach the task, or your attitude, is decisive for success. You can never change badly planned courses, cold and rainy weather, early start times, steep terrain, thick green areas, a bad map and so on, but you can change your attitude to the circumstances. If you compete in other countries, you cannot count on the conditions being the same as home, but you can learn to accept and like the situation and even see it as an advantage for yourself. The importance of attitude has been expressed in the following way: On a scale of 1-5, very weak to very strong, we rate natural talent, current capacity and attitude. These factors combine to create the final result, performance.

$(\text{Natural talent} + \text{Current ability}) \times \text{Attitude} = \text{PERFORMANCE}$ In the first example we have a person with normal natural talent, who is very skilled in their area, but a weak attitude towards the task. In the second example, the person has normal natural talent, normal ability, but a very strong attitude to tackling the task. Example 1: $(3 + 4) \times 2 = 14$ Example 2: $(3 + 3) \times 5 = 30$ With these examples we can illustrate that the elite sportsperson's attitude is one of the most important factors in achieving success. Because thought steers a person's actions, perhaps the extract below can help us to think in a new way:

Attitude

The longer I live, the more I realize the impact of attitude on life. Attitude, to me, is more important than facts. It is more important than the past, than education, than money, than circumstances, than failures, than successes, than what other people think or say or do. It is more important than appearance, giftedness, or skill. It will make or break a company ... a

church ... a home. The remarkable thing is we have a choice every day regarding the attitude we will embrace for that day. We cannot change the inevitable. The only thing we can do is play on the one string we have, and that is our attitude. I am convinced that life is 10 percent what happens to me, and 90 percent how I react to it. And so it is with you - we are in charge of our Attitudes. Attitude, Charles Swindoll

Chaper 2 – All roads lead to Rome

All roads lead to

Rome

There is never only one way or one truth that leads to success. The route there is often very different. Individuals are unique; nobody is the same as anyone else. Every person has different inherited talents, has grown up in their own environment and has a different starting point on the route to success. The job of the coach is to free the energy that is stored in every athlete. If the coach can help the athlete with inspiration, involvement, enthusiasm, motivation and self-confidence there is a good chance that this energy will be released positively and powerfully. Encouraging the athlete, being sensitive toward the athlete, getting the athlete to think for herself, creating enthusiasm in the team, letting the individual be unique, seeing possibilities and being good judge of human nature are some of the qualities that successful orienteers have identified as important in a coach. The following pages describe two international orienteers' paths from juniors to extremely successful seniors. Both of them have stood on top of the podium at WOC or world cup. Both of them have achieved a goal that many people strive towards – being the best. The ways to success that they have taken are very different. Both have found their own solution, chosen their own route to success. The physical, technical and mental training they carry out is in many respects totally different. However there are several things that they have in common: Both are excellent map-readers, often include competition-like training in their schedules and are physically strong runners in forest terrain. Other characteristics they share in common are more to do with personality and psychology. That both are focused, disciplined, determined, plan carefully and have good self-confidence is no coincidence. These factors probably lie at the heart of any attempt to succeed in any task, be it at work or in an orienteering race. Beyond that, the actual route that you find to success can be very individual. It is just a case of finding a way of doing things that is right for you...

Chaper 2 – All roads lead to Rome

Johan Ivarsson

Overall World Cup Winner 1996 What vision of your future as a senior orienteer did you have as a junior?

I am a very cautious person, a careful general, so to speak. I never had the vision like now I am going to be world champion. I have aimed towards it gradually. Every year since I started I have got a little better, right up until 1996. This year, 1997, things have not gone quite as well as last season. During my last year as a junior the national team coach said that we juniors would have a hard time establishing ourselves in the senior team in our first years as seniors. Many of the best runners in the team at that time were born in 61, 62 and 63 - some of the best orienteering years Sweden has had in a long time. I decided of course that I would go out and show them... But I also realised that it was important to bide my time and plan for the future. In my first year as a senior I decided to focus on the competitions where I thought I could perform best - the district champs and the Swedish night champs. I taught myself to succeed just when I wanted to. I controlled my own goals. For example, there is a points system in the different disciplines of the Swedish Champs. One of my goals was to get 10 points during my time as a senior, a mark of respectability. As it happened, I achieved

this in my first year thanks to a victory at the night champs. I had to move the goalposts of course, so my long-term goal became to get 50 points instead. I managed that in 1994. I have to have sensible goals that I can accept. Trying to be a bit better every year for example. Perhaps that is a passive stance to take, but that's the sort of person I am. The motivation for me has always been that it is such fun to orienteer, fun to train and fun to succeed with what you are striving towards. It gives me a kick. The process of achieving a goal is just as important as the goal itself. Feeling "this is a really good run here" as you punch the last control and then getting confirmation on the run-in that it was a great race - that is an amazing feeling. It gives a huge sense of satisfaction, a feeling of joy which goes right to your heart, a great experience. I could arrive at a competition far out in the forest, with no prizes, no commentator but all the best runners there. Run the race, put up my time on a little paper slip on a homemade results list. That would be enough for me. The runner with the strongest inner motivation would win. How have you combined studies and work with training and competing?

Thanks to a good base with a smaller Swedish club like OK Illern, then the elite club IK Hakarpspojkarna and going to a sixth form college in Eksjö that specialised in orienteering, I got a fantastic start to my orienteering career. In Eksjö I learnt how to train, developed my technique and socialised with like minded friends. After my military service year, I worked for one year before I started studying to be a civil engineer in Gothenburg. I finished my studies with one year in Trondheim. I finished my degree in the normal time, studying full time, and managed to develop as an orienteer at the same time. Despite being big universities I found that both institutions were receptive to combining studies and competitions. There was

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also a good gang who trained together. I had, I found, loads of time to train. If you prioritise and structure your life, if you are systematic and disciplined, it is no problem to train 15-20 hours per week. I would find it hard to be like the top cross country skiers and train and compete full time. There would be far too much time to kill. But sometimes I wonder why I was in such a rush with my studies, why I stressed and rushed through my studies at full speed. Perhaps I could have taken a year out, or taken a bit longer to finish my degree by spreading my courses out more. Today I work for a firm of consultants in Oslo who deal with road projects. I have an understanding boss, also an orienteer, who helps and supports me in combining work with training and competing. I work full time, but when you consider extra leave I guess I work about 85 percent over the whole year. Training programme – physical? During the build-up period training happens every day, week in week out. I try to get a good balance of different types of training into the week: Two faster sessions, a real long session, mostly in the forest and then also some strength. A week during my time in Gothenburg might look something like this: Monday Steady running Tuesday Intervals Wednesday Faster run + circuits Thursday Fartlek Friday Steady running Saturday Long run Sunday Long run When I arrived in Trondheim I got a real surprise. My training week, above, contained about 10-12 hours. Most people here, including the girls, trained much more than I did. I decided that I would try to increase the total training volume but still keep the quality sessions in my programme. To achieve this, I started to do a lot of cross-country skiing. I got a real kick from training more during my time in Trondheim. It was during this time that things really started going well, and my results got better. As a junior I trained 8-9 hours per week, sometimes 10-12 hours. As a senior I train between 12-15 hours per week with peaks of 20 hours. In those weeks cross-country skiing, or other alternative training like cycling, makes up a large percentage of the volume. I feel however, that I did better quality training when I was a junior

than I do know. Different environments have different philosophies. Sometimes it is difficult to make your own training fit in with club training. I try to train 2 times a week with the club, 1-2 times with someone else and the rest of the time I train on my own. Running in water and cycling when I have been injured has been an important barrier for me to break. In the beginning it was exceptionally tedious training. But, after a while I got into it and even got a kick out of the training. I saw that the training was actually beneficial and I became stronger.

Chapter 2 – All roads lead to Rome

During the competition season, with an important competition at the weekend, a week can look something like this: Monday Easy running on paths Tuesday 90 min steady pace in terrain Wednesday 60 min orienteering technique Thursday Interval training, “2 minutes” or “70-20” Friday Rest Saturday 30 min easy running Sunday Competition I don’t do any alternative training during the competition season. I try to make the orienteering training I do relevant to the coming race or my aim goal for the season so I like to do a lot of model training. I look at course planning and old maps which are relevant to the competition area. I go through everything mentally: technique and tactics, orienteering technique controlling the speed and also that I will dare to stop if uncertain, “grab the hand brake” if I need to. During a competition I can sometimes be a bit passive on the last part of the course, but on the other hand I very rarely blow it near the end. Moreover, I start more strongly than most. My tactic is based on taking one control at a time. I rarely look at the whole course and note that it is tough near the end. Past world champion, Kent Olsson once said: “Push it right from the start, and if things go well you’ll be able to keep going all the way to the finish.” I try to check route choices for the next control if I can, but not at all costs. If I need to I will take ten seconds extra at the control. Training philosophy - orienteering technique?

During my time at orienteering sixth form college I enjoyed nothing more than technique training. Lots of people thought that I was a bit strange as I did so very much technical training. I often set off on my own to train in a detailed area, just to get a feel for the terrain and how it was represented on the map. I learned how to interpret contour shapes and how the map reflects reality. I trained up my map-reading skills into a very sharp tool. During my time as a junior I did a lot of exercises focusing on a particular technique, for example control picking or compass. Today I run whole courses more, often from important competitions like past Swedish champs or old selection races. If I run a whole course like this, then I’ll do it at competition speed. Other orienteering sessions can be done at a slower tempo. Training philosophy – mental?

I am a calm and focused person both at work and in the forest. I think I am good at tolerating stress when the day of the competition arrives. I am not at all nervous, but the evening before when it’s time to sleep (especially before relays) I often find it difficult to relax. When I run the race though, I dare to trust in myself - I think I have strong self-confidence. I am disciplined, whenever I am uncertain, I stop.

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Last year I was the best. That little spark that you need to be the best has died a bit during this year. I think I need to make new goals to strive for otherwise I will lose that decisive motivation. I make a list of all my competitions. I note down different factors which influence my competition result, using a simple code +,0,-. Things like hassle at school, stressful journeys, or a feeling of harmony before the race with everything clicking into place - those are perhaps the most important factors which can influence my result. Motivation and physical shape are other important aspects of course, if I am going to arrive at the race ready for a good performance. Best training session?

Running fast and getting tired out is a kick, for example interval sessions. Or when a fast technique session a few days before an important race goes well, everything flows, then I get a real feeling of self-confidence for the coming race. That is a really great feeling. Five qualities which make you a top orienteer?

Determined - if I start something, then I want to see it through to the end. I have retired in two competitions, the Swedish long champs because of the heat, and a night race when my head-lamp went out. Systematic and structured - I am good at prioritising and planning my work and training time. I am very systematic about planning work and my competition schedule, so that I get every-thing fitting into place before a race. I think however, that it is in my training that I am most systematic and thorough. Accurate - I am best at night orienteering. Perhaps I am a little too careful and accurate some-times, to the extent that I become passive and cautious. Enthusiasm - orienteering is fun! Enthusiasm and joy in what you do are some of the most important building blocks for success, be it at work in sport. Self-confidence - I dare to trust in myself. I dare to do what I believe in, especially during orienteering competitions. Advice to coaches

Roger Glanefors meant a great deal to me during my time at orienteering gymnasium. I learnt a lot about technique and how to train from him. Running with a big club like IK Hakarspojarna with Göran Öhlund at the helm was also very important for me. Taking the responsibility in decisive situations and being the start in the club gave me self-confidence and taught how to rise to the occasion and be the best when it counted. A good coach starts with the athlete and discusses things to find out what they want. The relationship is extremely important. A good relationship makes the athlete feel very secure. There also has to be a balance between listening and giving sensible advice. Of course, when it comes to selection you have to be professional, separate the facts from personalities. Advice to juniors aiming higher?

Dare to stand still in the forest. Try the tactic that Kent Olsson and I have used: Stop as soon as you are unsure. You need to learn to orienteer at the speed that you compete at. Let the map reading dictate your speed, not the other way round. During 1994-96 I guess I ran ten competitions where I missed less than half a minute in total.

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When you stand on the start line, regardless of how your training has gone, or what injuries you may have had, do your best. Forget about things in past. In 1986 I had a string of bad results that made me stop and think again about my technique. I had won the O-ringen and had a good season in general. I was a little doubtful and with hindsight I didn't do what I wanted to do myself. I ran the district champs straight after the O-ringen and should have stayed at home. The selection races for the Nordics next weekend also went badly, and even the weekend after that. I decided that it wasn't cool any more. It was more fun to be a steady fifth all the time and run a good orienteering race, than it was to win a competition then run badly for four or five races in a row. My new tactic became: I am going to stop at every control and before I run on to the next control I am going to think a "mantra". My mantra or trigger phrase was "Micke Wehlin". He was a runner in the Swedish senior team and at the time he was the best orienteering technician I knew. He basically never made mistakes! So, at every control I thought "Micke Wehlin" and my thoughts focused on the job again: Orienteering. This was a turning point for me. Now I was thinking about technique all the time during races. Motivation gives concentration. My motivation is to have a perfect run. That is what pays dividends in the long run. Of course, you have to be pleased when you have good runs. My idea of a perfect run includes a few stops. I let the technique control my running speed the whole time. When I start thinking about trying to run faster, or that I need

to fight a bit, I start to lose focus on the most important thing (orienteering technique) and think about less important things (physical feelings). Your speed is always high enough when the orienteering is going well. Even if the speed isn't there at first, it will come, be it in three weeks, three months or three years. Always put technique before speed. Learn from others. Read the orienteering literature that is out there. For example, I have read books by Kent Olsson, Marita Skogum and Christer Rolf and found loads of gems in them. Would you change anything that you regret?

One thing I think I have done which I think is good is moving around: Småland, Göteborg, Trondheim and Oslo. I have a wider experience thanks to the different training environments, both terrain and training partners. I wish that I had been more committed to orienteering when I was studying and that I was a bit more focused with my training earlier as a junior. I wish that I had given even more to orienteering then, and aimed a bit higher.

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Heather Monro

Overall bronze medallist World Cup 2000

What vision of your future as a senior orienteer as a junior?

I don't really think I had a vision of how things would be in the future when I was a junior.

There is no way that I ever imagined that I would do so well: I didn't have a mental picture of myself standing on the World Cup podium this year or anything like that when I was 17. I think the goalposts move all the time. I was also lucky to get a chance to taste of senior international competition early – I was picked for a World Cup when I was 19. I did terribly of course, but I suppose it gave me some insight into what was required and inspired me to come back and do better. The difference between now and then is enormous - in my training, my commitment to orienteering, even the country I live in! But all of those things have come gradually. Each lifestyle change, whether it was taking a year out before school, or deciding to work in Norway for a while after university was quite a small decision at the time. My training has also developed steadily so that even though I do about twice as much as I used to when I was a junior, and train in a totally different way, there has been no single massive change. Success has followed each change that I have made, but not straight away. Improved results have usually come some time after I have changed my training or lifestyle. If there is a vision that has followed me, I guess it is that orienteering is really great fun. I enjoy it so much that I can't ever see myself stopping, even though I won't be an elite orienteer forever. I always want to be a little bit better than last time, improve a little on the last year. It was the same when I was a junior – I never thought about giving up because there was always another race to go to or another camp to meet friends on. I couldn't give up now either – it's just too much fun! How have you combined work and studies with training and

competition?

I think this has been one of the most important aspects for me. Even though the changes have come slowly, I can definitely link all the steps I have made in my orienteering to changes in my environment, studies or work. For example, I decided to take a year out after school and spend a lot of time orienteering. I'm really glad I did that now – I travelled around and got a lot of experience of different countries and racing abroad. I worked a bit and stayed at home between travelling. It was the first time that I had really put orienteering first, and I had a great time doing it. When I went to Cambridge the year after I found it tough. I had to be very disciplined to fit everything in. In some ways a university like that is very supportive of sport. They approve of excellence, so if someone is good at sport they really appreciate it. But on the other hand there is a very rigid framework and it can be hard for a

sportsperson to fit into that. If there have been Saturday lectures in your college for the last 300 years they are not going to move them

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just because you want to go training. Tradition makes the system inflexible, but it also taught me to use my time effectively. The good thing about Cambridge was the people who were around. I trained a lot with the athletics and cross-country clubs during my time there. I even rowed a bit in my first year, but that was just for fun. I developed a good training routine and enjoyed being part of a team as well as just an individual. Although I look back at some of the training I did and question its relevance now, I certainly developed a good base of fitness and improved my running. I won-der if I could run some of the times I ran on the track now! After finishing university I worked at an outdoor centre in the Lakes. Really it was just a chance to think about what I wanted to do in the longer term. I had a few romantic ideas about what fun it would be to live and train in the Lakes, but I actually found it quite lonely. I did most of my training on my own. During that year I was also in contact with a Norwegian club - Yvette Baker and Dickie Jones were also running for them at that time. I was really inspired when I got to run in the big relays such as Tio-Mila and Jukola, and when I later got the chance to work as an au pair in Norway I leapt at it. However, things still felt quite temporary at that time. I had not decided what I would do in the longer term and was applying for jobs and courses in Britain. The town I was in at first in Norway was beautiful and great for training, but I was still quite isolated in many ways. My plans became more concrete when I moved to Halden. It seemed a perfect combination - bet-ter training opportunities, but also a big club and plenty of contact with other orienteering friends, like I had at Cambridge. When I later started a three-year teaching degree there in 1996 I guess I had really decided that my longer-term plans lay in Norway. The difference between my two degrees was enormous. In Norway, there was a much better understanding of elite sport and I was able to move exams or ask for extensions on projects if I needed extra time to train or compete. In my final year, I even delayed the last part of my course by six months so that I could focus on World Champs in Scotland. I have also got a lot of flexibility in my current job. I work part time, and I am able to plan my work to some ex-tent to take account of important races or training camps. I really feel that I have got a good balance between orienteering and work now. Training philosophy – physical?

When I was at Cambridge I trained mainly with runners. I was very fit, but I don't think I was really training to be a good runner in the forest. There was also a big focus on miles. The measure of how hard you trained was how many miles you had in your training diary. A lot of the running I did was probably junk mileage. But I learnt how to push myself and build up a routine and the discipline that I would need later to carry out a hard training programme. I week in Cambridge usually looked a bit like this: Monday jogging + circuits Tuesday intervals on a football pitch (5 x 4 mins) Wednesday long run on paths Thursday intervals (6 x 800m) Friday jogging Saturday XC race

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Sunday orienteering race I would often jog in the morning for half an hour or so, with other runners, or even go out for a steady jog in the evening after a race on Saturday. I don't really think that those jogs con-tributed anything to my training, apart from making me feel better about my weekly total! Now I try to train much more in the forest, try to include a lot more technique, and have made my quality sessions much longer. Instead of jogging to pass the time, I go to the gym and work on my strength and flexibility. When I'm doing long sessions, I either go for proper long runs in the terrain or I cross train, biking or cross-country skiing for example. By cross train-ing I can give my body a rest and still get a good aerobic session.

Even though my training volume is much higher than it used to be, I think there is much less junk mileage in my training now. A hard week in the winter might look something like this: Monday (am) Interval training with a map (50 x "45-1 ") (pm) Spinning class + Gym work Tuesday Interval training on the roads (12 x "180-60") Wednesday Spinning class + Gym work Thursday (am) Rowing (pm) Intervals on a hilly sawdust trail (12 x "180-6 ") Friday Night orienteering Saturday (am) Rowing (pm) Gym work Sunday Long orienteering course I have really gone in for what we call Foxdal intervals. By slightly reducing the intensity of my interval training, I am able to do much longer quality sessions and this has had a big effect on my fitness. I spend more a lot of time in the forest running at competition speed. Things like the spinning class are good for aerobic fitness, and I find it a sociable way to train that gives me a break from running, so that I've got more energy left for the really important tough sessions. The rowing is just a bit of fun – our orienteering club are going to enter a race next spring. But is also a nice way to build up a bit of team spirit and train at the same time. Later in the winter I might ski instead. The core of the week is the running sessions. Pretty much all the running sessions I do are hard so I make sure that I warm up and warm down thoroughly. Training Philosophy – Technique?

I do a lot more technique training now. Technique training when I was a junior was just something I did at the weekends but now it has a much higher priority in my weekly programme. The biggest difference however, isn't the amount that I do, but the way I do it. I think there is much more quality in my technique training now. I really consider what I want to get out of each session and then concentrate on doing it right. I am very goal orientated. When I went on training weekends before, I used to run round courses without thinking about what I was really trying to practise. I used to make lots of mistakes and never really analyse why things sometimes went right and sometimes went wrong. A lot of the time I was training in bad habits.

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Now I have a plan for different times of the year and know exactly what I want to focus on when. I make sure that I do things correctly, so that I practise good habits and learn the behaviour that I want to reproduce in a race. I also think about my training afterwards. I don't mean that I analyse every course in detail, but I reflect on what went well and why. Last year I kept a little graph of how many controls I was happy with on my main technique training session each week. I watched my percentage success rate climb through the year until, by the time the most important races came round, I was really happy with my technique. Better quality in my technique training has made a massive difference - I miss far less now and that is the key to doing well. Training Philosophy – Mental?

I am a real geek actually! I could sit and study maps for hours. I become absorbed in preparations for important races in the future, thinking about the demands of the orienteering and planning my physical and technical training to meet those demands. I am a very thorough person who doesn't like to leave things to chance. If we discuss terrain or a race at a club or team meeting I often get very involved in the discussion or even sit and take notes. I think I am quite structured in my daily life as well. I am a lists person, even if they are only in my head. I never get out of bed and just float around; I always know what I am going to do with the day. That means I am quite effective and I guess that is very important as an elite sportsperson – you need to be effective if you are going to make time for everything. But sometimes I am aware that my need for preparation and structure is a little bit of a weakness. A lot of things in orienteering are unexpected for example and require flexibility. On the other hand the more you plan for the unexpected the better prepared you are – a lot of the surprises in orienteering are really quite predictable. I like to have harmony in my life

and my orienteering. My environment contributes a lot to how I feel about my training and performance. I like to feel that people around me such as my club mates, boyfriend or family support what I am doing for example. I also like to feel that the people who I am working with have the same goals and want to work in the same way as I do to achieve them. Sometimes in a team situation that can be a bit difficult - not everyone wants to do things in the same way. I think my need for structure is a bit of a weakness in those situations. I also work with what most people would regard as mental training: Relaxation, concentration, visualisation and so on. I think this helps me focus on what I need to do in stressful situations. As I have become more experienced I have become better at leaving external factors behind when I stand on the start line. Although harmony in my life is still important to me, I am good at focusing on job in hand when I actually get into the forest.

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Best Training Session?

I don't look forward to interval training or particularly enjoy it as I do it. But there is a definite rush afterwards, a feeling of having worked hard, that gives me a real sense of satisfaction. I think my favourite sessions are with maps actually. There is nothing better than running fast in open, detailed forest and feeling in total command as you spike every control. Five qualities that make you a top orienteer?

Thorough – I plan thoroughly and don't leave anything to chance. My training and preparations make sure I am ready to perform when I stand on the start line. Discipline – I never have any problems training hard. I can always get off the sofa and out the door, even on the darkest winter night. Enthusiasm – I enjoy what I do. I love to orienteer and that gives me a lot of positive energy to train and race. I rarely get tired of orienteering, even when I do it a lot. Running – I am physically strong. I have always been a fast runner, and now I am also a strong runner in the forest, even in tough terrain. Determined – I work steadily towards my goals. I am good at setting targets and working out a strategy to achieve them. If I decided on something, I can and will achieve it. Advice to coaches?

Focus on the positive things. We are very good at analysing mistakes or finding problems – not fit enough, not enough training, not concentrated. Especially with girls I think it is important to motivate and encourage. I have learnt much more when the things I have done well in the forest have been pointed out and reinforced, than I have when people have analysed my mistakes. Advice to juniors aiming higher?

You need to have patience! There are a couple of trend breakers in British orienteering, but for most of us international success has been a long and slow process. Concentrate on where you are now and how you can be better next race, next month, next year, not how you can be World Champion in five years. Most importantly, do it because you enjoy it. Let the goalposts move naturally and enjoy the smaller successes that you may have along the way. So many of my friends from junior years have given up. There are a lot of different reasons, but if you are only motivated by success then of course not everyone can win and you may be disappointed. Never forget just how much fun orienteering is! Any regrets?

No actually. Now that I think of it, I don't think I really regret anything. Of course I have made plenty of mistakes in and out of the forest, but I have learnt a lot from all of them. Everything I have done contributes to better performances in the future. I definitely don't regret any of the big changes I have made in my life either. Leaving my Cambridge degree and all its possibilities behind and moving to a little town in Norway to orienteer might seem like a crazy decision. But, I have really lived life and followed what I wanted to do deepest down in my heart.

Chapter 3 – What is Orienteering

What is
orienteering?

The "forest sport", "rallying on foot" or orienteering as it is actually called is a technically demanding sport. Both the navigation and running aspects of the sport are technical. To keep

up with split second decisions in the forest like a skilled rally driver requires a brain that can work in the red zone of the rev counter. Orienteering is a complex balance between incisive thinking and hard physical activity. Brains and brawn must be working in harmony before you can hope to achieve a good result.

To find out what is needed to be successful in orienteering we need to analyse the demands of

the sport. That analysis needs to include both the general demands of orienteering, but also the

specific demands of particular competitions. For example, to achieve a high general performance level we must work with the whole picture so that physical, technical, mental and

social factors combine to give the basic requirements for success. Then important competitions like WOC might also present specific challenges. These may influence the content of athlete's training programme as he or she prepares for that race. Whether the race is

in Nordic or continental terrain, if the terrain is fast or tough, steep or flat - these factors will naturally influence training and preparation during that particular season.

There are several questions that we need to ask: What happens between the start and finish?

What do we do? How long is the course? How fast do we run? What is terrain we run through

like? Do we run in the forest or on paths? What sort of running style do we have in the terrain? What muscles are working and how are they working? What heart rate do we have during the competition? How much lactic acid do our muscles produce? What can disturb concentration during the race? What can stress a runner during a competition? What technical

challenges are there in different types of terrain?

Analysis of the demands of orienteering

Orienteering today consists of several different disciplines: short distance, classic distance, long distance, night orienteering and relays. The running time for each discipline depends on which class you compete in. BOF has guidelines for course planning based on age and sex.

Physical

- Orienteering is a physical sport. During an orienteering race an athlete works at about 90

percent of their maximum oxygen uptake capacity (VO₂ max). Carbohydrate, mainly in the form of muscle glycogen, is the main energy source for this activity.

High Speed

- The athlete runs at high speed. Speed varies between 4-10 minutes per kilometre, but can

dip under 3 minutes per kilometre at certain points on a course. Men run faster than women. In very runnable terrain the difference is only 10-20 seconds per kilometre, but in steep or tough terrain with heavy undergrowth women lose more time in relation to men. Here the difference can be 50-75 seconds per kilometre.

Borg 17-18

Chapter 3 – What is Orienteering

- One way of rating effort and speed is using the Borg scale. In orienteering races athletes

rate their effort at 17-18 out of 20, which corresponds to a very high work rate. This means that there is significant lactic acid production in the leg muscles.

Tough terrain

- The terrain is often steep, marshes can often be wet and areas of felling or brashed forest

can reduce runnability. Open pine forest can change into deep heather or thick bracken. Underfoot conditions are usually soft. This is especially true in Nordic orienteering, but even in Britain small paths, forest or rough open fell side account for the larger part of most courses. No one step is like the other and if we compare the quick foot strike of a track runner with the deliberate, powerful stride needed in the forest our feet are in contact with the ground for a relatively long time.

Individual

- Orienteering is essentially an individual sport. The individual must make all the decisions

needed to get from start to finish. The runner takes responsibility for everything that happens in the race.

Quick decisions

- The athlete must continually make new decisions: "I am here. I want to get there."
Map

contact, compass contact, rough orienteering, fine orienteering, round or over route choices, extending and simplifying the control feature.

Stress

- Athletes can catch other runners, be overtaken and pass public controls or film cameras in

the forest. The desire to win or fear of not succeeding can distract from the immediate task of getting to the next control. Athletes face external and inner distractions the whole time and this stress can lead to incorrect decisions. Concentration can be especially difficult in relays or mass and chasing starts when the orienteer is placed in a man-to-man situation. The above is an attempt to describe the sport of orienteering. Try describing the sport yourself.

Think about how the sport is, the different disciplines you compete in and the demands of the sport. You can even think about what orienteering might be like in the future. How are maps, course planning, distances and competition formats going to develop in the next 5, 10 and 20

years? What significance does this have for how you should train, if you want to perform well? Will classic distance orienteering exist in the same form as today, or will short distance and park orienteering be the main events in orienteering? If orienteering becomes an Olympic

sport, what type of “TV friendly” orienteering will be in the Olympic programme?

Regardless of how the future looks an athlete can always look at the current situation. What is

orienteering like today? What is required to be successful, and what skills and capacity do I already have? These questions are simple, but it can be difficult to find meaningful answers. However the answers are important if the athlete is to shape his or her training and work towards his or her goals.

The demands of

orienteering

Psychological demands

Thoughts turn into actions. All the thoughts that we think daily have reactions in our bodies. These reactions can be the physical feelings associated with an emotion or nerve impulses to a specific muscle. Hormones and nerves react to signal from the brain and carry messages through the body. It is therefore natural that both brain and body are involved in our studies, solving problems at work, relationships, or in the case of sport – how we succeed in our event. To train optimally we have to know what is required of an orienteer in the competition situation. We know that when things go right and we run a whole course cleanly, all the controls just seem to appear in front of us. When we “spike” all the controls, there is a flowing feeling, as if we are running on “autopilot”. For all this to happen, something has to come first – a thought. It is that thought or series of thoughts that are of interest if we want to build up a psychological profile of a good run. What are the psychological factors that lie behind a successful performance in a competition? A decisive factor in our ability to perform is personality. Roughly speaking, personality is partly the result of inheritance and partly the result of the environmental influences we have experienced.

Together, these factors make up a personality. As far as orienteering performance is concerned the essential mental qualities are self-confidence, motivation and the ability to concentrate. In addition the ability to tolerate stress, control nerves, focus on a task and maintain a positive attitude are important. Profile of psychological requirements

Together with your coach, a leader or someone else who you have a good relationship with, you could attempt to fill in the profile below. It rates both psychological requirements and your personal strengths and weaknesses. It could be used as the basis of a discussion between you and your coach about strengths and weaknesses, the way you think, and the way you carry out different tasks. This process can help both to give a better picture of how you react in a competition situation and also how you can develop and change your training and behaviour. Quality Importance (1-5) My own capacity (1-5) How can I train this?

Self-confidence Motivation Concentration Attitude Being positive Stress tolerance Controlling nerves Focus

Being systematic Vision Setting goals Taking responsibility Commitment Enthusiasm

Harmony Original thinking Being a team player Seeing the whole

Questionnaires

Tests such as the TOPS (Test of Performance Strategies) or the Psychological Strategies in Orienteering questionnaire can be a useful way to get more concrete information on strengths and weaknesses. These questionnaires consist of a list of questions which the athlete answers spontaneously. By analysing the answers it is possible to gain information on different psychological skills and strategies. It is important that the athlete answers honestly and without considering the questions for too long – the best answer is usually the one which comes to mind first. The aim of these questionnaires is to encourage awareness of the factors which are important for performance in competition. The results can be the basis for individual discussions with a coach afterwards and give the athlete confirmation of strengths and a better understanding of weaknesses. When working with psychological skills the inform-accept-change model is the most appropriate. The athlete first fills in the questionnaire and then discusses the results with a coach to try to understand what sort of strategies they have currently. This process demands that the athlete analyses and accepts the current situation. When these first two stages have been carried out it is relatively simple to think of creative, constructive solutions for change that can improve training and performance. When working with individual development the inform-accept-change model is the most appropriate way to work. Psychological questionnaires such as the two above can be a good starting point for the athlete to begin to understand how they perform.

Physiological profile

Orienteering is a highly demanding sport. Reaching the top of the sport and competing at international level requires very high levels of fitness and physical form. During recent years there has been increasing interest in how both fitness and strength can be trained at the same time. This is partly necessary because of the tougher and increasingly physically demanding courses runners face at major championships. In addition, course planners strive to plan courses with as much variety as possible. Often, the runner must tackle steep climbing, where lactic acid tolerance will be pushed to the limit, and flatter, faster terrain within one race.

Oxygen uptake and lactic acid

We know from previous studies on elite orienteers that a top orienteer today must have both a high aerobic and a high anaerobic capacity. A study carried out on the Swedish national team directly after the 1991 World Champs in Czechoslovakia showed that elite runners had very high values for both maximal oxygen uptake (VO_2 max) and maximal lactate level. Women in the team had an average VO_2 max value of 68 ml/kg x min with a range of 62-71 ml/kg x min, while the men had an average value of 78 ml/kg x min with a range of 75-81 ml/kg x min.

0

10

20

30

40

50

60

70

80

90

ml/kg

x

m in

1 3 5 7 Average

Maximum Oxygen Uptake, WOC 1991 Team

Men

Women

The most surprising result of the study was that orienteers had very high lactate levels at maximum exertion, which suggested that they were used to eliminating and tolerating lactic acid during exertion. Women had an average value of 12 mmol/litre with a range of 8.4-14.0 mmol/litre, while the men had an average value of 12 mmol/litre with a range of 10.0-17.0 mmol/litre.

0

10

20

30

40

50

60

70

80

90

m l/k

g x

m in

1 3 5 7 Average

Maximum Oxygen Uptake, WOC 1991 Team

Men

Women

It has also been shown in another Swedish study, that blood lactate levels during a simulated competition lie on or over the lactate threshold (nominally 4 mmol/litre) during practically the entire course. Actual measured values for the runners varied between 3.5-15.5 mmol/litre at different points of the course. The average value was 8 mmol/litre. These results show that orienteers are forced to work anaerobically for long periods: Classic courses last 65-70 minutes for women and 90-100 minutes for men. Additionally, the runner is working at approximately 90 percent of his or her maximal oxygen uptake capacity, and the heart is pumping at 89-99 percent of its maximum frequency. When we consider these factors, an orienteering runner could be seen as a combination of a cross-country skier and a middle distance runner. An interesting result from the same Swedish study was that the athletes' own rating of their aerobic effort and leg tiredness reflected measured lactate levels very accurately, while there was no significant correlation between heart frequency and lactate.

Measured Blood Lactate - Men

0

2

4

6

8

10

12

14

16

Start Up Down, flat Up Flat Up

m m

o l/l

it re

Measured Blood Lactate - Women

0

2

4

6

8

10

12

14

Start Up Down, flat Up Flat Up

m m

o l/l

it re

Orienteering, in common with cross-country skiing and running distances longer than 5000 metres, is an endurance sport. The main factors that separate orienteering from other fitness and endurance sports are the variety of terrain that the runner must negotiate and the steep climbs and descents that may be encountered.

Uphill Downhill, flat

Uphill Flat Uphill Downhill 0

2

4

6

8

10

12

14

16

18

B o

rg R

P E

, m m

o l/l

it re

Relationship between HLa and Borg RPE

Borg RPE

HLa

Threshold

These factors lead to a running speed which varies between 4-10 min/km and a continually changing stride, where no step is the same as the one before. To cope with this a very high aerobic capacity is required, both centrally in the heart and lungs and locally in the "orienteering muscles". This capacity is usually measured by maximal oxygen uptake ability,

VO₂ max, and we know that elite runners have had high values for a long time. Lactic acid is also a performance limiting factor: It has been shown that blood lactate can average 5-7 mmol/litre for a whole race in Norwegian, Swedish and continental studies. This shows that a well-trained orienteer is dependant on lactate production during a race. An elite orienteer needs to have good anaerobic capacity to be able to run in the undulating and rough terrain encountered in orienteering races.

Aerobic and anaerobic metabolism for different competition distances

10000

400 100 200 800

1500 3000 5000 0

10

20

30

40

50

60

70

80

90

100

0 10000

Distance

P e r

c e n

t a g

e Energy from aerobic processes (%)

Energy from anaerobic processes (%)

Anaerobic capacity has to be developed in the specific muscle groups which work when we run in steep and rough terrain, such as the hip-flexors and quads. We can only really get this capacity by recreating the actual competition situation in our training - fast running in the forest. However, cycle training can also help to build p muscle mass, and develop aerobic and anaerobic capacity in these muscle groups. There is also good reason to include running on paths and faster, harder surfaces in our training, as courses often include such runnable sections. A good summary is that "we become good at the things we train" and therefore we should let the character of the terrain at our most important competitions determine what sort of training we do. As blood lactate levels are well correlated with our rating of effort during a run, using for example the Borg scale, we can use this as a good measure of training intensity. Heart rate is a less useful tool for judging training intensity in orienteering training. By studying effort levels during competition, we can use this to guide competition-like training in our programme so that we are prepared for the real competition situation. General and specific strength

Strength requirements for orienteers can be divided, roughly speaking, into two areas – general and specific strength. General strength is required everywhere in the body for posture and the control and coordination we need to carry the body during a competition. General strength helps both to prevent injury and provide a base for further training and intensive competition and training. Abdominal, side and back muscles form a supportive corset round the torso and make it possible for the legs to work intensively for long periods. This support has to be strong and stable if we are to run further, faster and more often. Core

stability must be continually maintained and developed to cope with increased training. Even the shoulders, chest and upper back contribute to this supportive corset, though not as much. Being strong

and well balanced, so that you can lift your own weight and stand on your hands can only be an advantage for an orienteer. Specific strength is strength in the muscles that we use in orienteering for the sport specific movements that we make. The leg muscles work hardest in orienteering of course. The large and small muscles in the legs and the hip flexors work together to move us quickly through steep forest terrain. The muscles work dynamically for long stretches of time which means that we need dynamic strength, both eccentric and concentric, to drive our body weight forwards. The muscles must have excellent strength endurance, but also be light. In this way we can get the most from our strength. Specific strength is best trained through fast running with resistance, such as uphill running (concentric strength), downhill running (eccentric strength), running on marshy ground or in snow. Where are we today?

The following questions can be a good starting point for discussion between athlete and coach:

- What is orienteering really? Short, classic, park, today and in the future?
- What demands does an orienteer face?
- What capacity do I have now?
- How can I train to be better or even the best?
- What sort of ambition, attitude and motivation do I have to working towards the above?
- Do we have shared goals as athlete and coach? What are they?

Analysis of requirements

Physical capacity

Poor Excellent Aerobic, central 1 2 3 4 5 Aerobic, local 1 2 3 4 5 Anaerobic, central 1 2 3 4 5 Anaerobic, local 1 2 3 4 5 Running economy 1 2 3 4 5 Strength, general 1 2 3 4 5 Strength, specific 1 2 3 4 5 Flexibility, general 1 2 3 4 5 Flexibility, specific 1 2 3 4 5 Running ability
Poor Excellent Running on paths 1 2 3 4 5 Running in flat terrain 1 2 3 4 5 Running in gentle undulating terrain 1 2 3 4 5 Running in undulating terrain 1 2 3 4 5 Running in steep terrain 1 2 3 4 5 Running in Nordic terrain 1 2 3 4 5 Running uphill in Nordic terrain 1 2 3 4 5 Running downhill in Nordic terrain 1 2 3 4 5 Running in continental terrain 1 2 3 4 5 Running uphill in continental terrain 1 2 3 4 5 Running downhill in continental terrain 1 2 3 4 5 Running in marshes 1 2 3 4 5 Running on stony ground 1 2 3 4 5 Running in heavy undergrowth 1 2 3 4 5 Running over felled areas 1 2 3 4 5 Running through green areas 1 2 3 4 5 Getting past obstacles in the terrain easily and quickly 1 2 3 4 5 Capability to change tempo when the terrain changes 1 2 3 4 5 My own rating of my ability as a "forest runner" 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5

Analysis of my strengths and weaknesses

Physical capacity

Poor Excellent Aerobic, central 1 2 3 4 5 Aerobic, local 1 2 3 4 5 Anaerobic, central 1 2 3 4 5 Anaerobic, local 1 2 3 4 5 Running economy 1 2 3 4 5 Strength, general 1 2 3 4 5 Strength, specific 1 2 3 4 5 Flexibility, general 1 2 3 4 5 Flexibility, specific 1 2 3 4 5 Running ability
Poor Excellent Running on paths 1 2 3 4 5 Running in flat terrain 1 2 3 4 5 Running in gentle undulating terrain 1 2 3 4 5 Running in undulating terrain 1 2 3 4 5 Running in steep terrain 1

2 3 4 5 Running in Nordic terrain 1 2 3 4 5 Running uphill in Nordic terrain 1 2 3 4 5 Running downhill in Nordic terrain 1 2 3 4 5 Running in continental terrain 1 2 3 4 5 Running uphill in continental terrain 1 2 3 4 5 Running downhill in continental terrain 1 2 3 4 5 Running in marshes 1 2 3 4 5 Running on stony ground 1 2 3 4 5 Running in heavy undergrowth 1 2 3 4 5 Running over felled areas 1 2 3 4 5 Running through green areas 1 2 3 4 5 Getting past obstacles in the terrain easily and quickly 1 2 3 4 5 Capability to change tempo when the terrain changes 1 2 3 4 5 My own rating of my ability as a "forest runner" 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5

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The Brain

The brain is a fantastic creation. Every person has a brain, which is in principal without limits. There are endless fascinating facts about the brain's capacity, everything from how it is comparable to a supercomputer or how today we only use a seventh of its capabilities. Much of the complicated system of impulses and pathways is concentrated on the surface of the brain, which is incredibly large due to its complex twists and folds. If the surface of the brain were flattened out it would cover an area the size of a football pitch. The body functions as a whole; physical and mental reactions are mixed together. Mind and body work together closely and influence each other in many different ways. Often it is difficult to know or determine if a reaction or behaviour depends on signals from the brain or the body. The brain has however the controlling function in our actions. It is the brain that determines our behaviour and how we react to our experiences. It is this area that psychology is interested in. Learning

The brain's capability for learning is enormous. We have an amazing ability to learn right from earliest childhood. This ability becomes dulled later in life for many people. However, there is nothing physical, which says that this must be so. The change is mainly psychological. As we grow up we learn not only useful skills, but also things that hinder further learning. In some cases we even learn to shut down our own learning capability. Physiologically speaking, our mental abilities do not begin to decline before they are 60 years old. We become, in other words, worse at learning when we become older, not more forgetful. Instead of the essential repetition that characterises children's learning we think we are able to do something straight away and so will be able to do it next time we try as well. Look at small children: When they learn something new, they repeat it again and again, and eventually they master the new skill. Children do the same thing with computers, repeating the same process again and again. We adults do something once or possibly twice then believe that we have learnt it. We should not just tell children to practice, we must practice ourselves. This also applies to map reading, whether sitting at the desk or running at high speed in the forest. We do not become good orienteers just because we learnt the theory once in the classroom. We must repeat the procedure often and learn the right system. Each learning opportunity is about getting feedback that something is right or that we have succeeded. When we are actually competing and working as hard as we can, and at the same time facing stress from both inner expectations and external factors, it is exceptionally important to apply the right behaviour to the job. If our learning has been positive, offensive, and creative it is easier for the brain to choose the right system to quickly solve the task. And it is up to us to decide what information we wanted to programme in during our learning.

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Nervous system

Our actions are either inherited, for example breathing and swallowing, or learned, such as running, cycling and swimming. Learning movements like this involves the nervous system learning to send impulses and make the right connections. The nervous system learns to activate the right muscle fibres and fire the right number of fibres so that movements can be carried out gently and with precision. The more a movement is trained, the more securely impulse traffic is transmitted to the right muscles so that they work in synergy. Fewer and fewer signals go to the wrong receptors or antagonists. We learn to work or run efficiently without opposing muscles working and hindering the intended movement. We learn to work at the right effort, the correct amount of motor units fire straight away. In this way we get develop a better running economy. We are able to carry out more work for the same energy expenditure. For this reason training at competition speed, with elements of technique included, is essential to performance. The task of the nervous system is to accept impulses from the different senses – sight, hearing, feeling, balance and pain, and to transfer these signals to the brain. The brain collects these impressions, evaluates and then sends signals out to the different organs in the body, such as the muscles. Impulses are sent to the muscles with the help of electrical signals. The speed of impulses can be as high as 100-200 m/sec in the larger nerves. The nerve connects to the muscle via a so-called motor end plate. Transfer of the impulse to the muscle is carried out chemically here via a synapse and with the help of a transmitter substance containing charged part. This nerve and muscle unit is called a motor unit (see figure).

In the eye, where movements are often extremely fine, the number of muscle fibres in a motor unit only is only about ten. In the thigh muscle, in contrast, such a unit can consist of hundreds of fibres. Motor units work on an “all or nothing” principle. This means that when a motor unit is reached by an impulse it will either activated fully or not at all. There is no middle ground. This is why the principle “we become good at what we train” applies. It

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means that competition relevant training should be frequently included in whole or parts of training sessions. Muscle fibres are recruited in a certain, determined and rule governed order, depending on their properties. Low threshold fibres (type I) are recruited first, and in the electrical current required to activate the fibre can be relatively low. Fibres in different work groups are then activated, based on the power that is required (see figure).

This is an extremely important core in all training, both physiologically and psychologically or technically. If all training during the build up period is carried out at low speed only certain groups of fibres, a part of the whole muscle, will be activated. These groups will develop an increased capacity. When we run an orienteering competition, the pace will be considerably higher and the nervous system will also activate high threshold fibres. However, we have not stimulated these during training and they are therefore totally untrained. For this reason, we need to train at a variety of speeds so that different fibre types are forced to work in a way that reflects an orienteering competition.

Thoughts becomes actions

We believe that we think in words and language. Researchers believe on the other hand that we think in images. Words come from these images and we can then use these words to transmit ideas to others. If we are lucky the other person will imagine the same picture from the words we supply and we will be in agreement. If we do not imagine the same picture problems begin. Take a simple word like “run”. For an orienteer this means running in the forest, for a marathon runner it means running on a hard asphalt road surface. Two totally different nerve and muscle recruitment systems are at work. For a sailor the word “run” has a totally different meaning - sailing with the wind behind the boat. Another issue which thinking

in pictures raises is the difficulty of negotiation. It is difficult to get the brain to work in a “not” environment. Try to draw “not fast”, “not fun”, “not strong”, “not fit”, “not miss”, “not push it”, “not twist my ankle”. We would have to draw a picture with a cross over it or something similar. We would start with a picture, but then add a language symbol, a cross, to make the idea negative. There is no such thing as a crossed-out

Light load

High load

Moderate load

Power

Number of motor units

Type I

Type IIa

Type IIb

Slow fibres

Fast fibres

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picture in the brain. We can only create this type of picture by trying to combine two different pictures. The consequence of this is that we end up describing the things we are trying to avoid. The word “not” should be taken away from our language. In English we use “not” far too much, while in Japanese, for example, there is no word for “not”. What are the consequences then if we talk about how we took the “wrong” route choice, or how much time we wasted when we “missed” a control? The more we use the right picture for the thing we want to achieve, the easier is for the brain to send the right signals to the different organs of the body. If we want to be relaxed we need to create pictures of a calm environment and not pictures of “do not stress now”. When we start with pictures we end up with hormonal neuromuscular actions and effects. Hormones are both affected by and stimulate or feelings, mood an orienteering technique, while the neuromuscular system controls our running technique.

Self-image and self-confidence

Much research, into both our performance at work and in competitive sport, shows that people with a positive image of themselves are able to perform near the roof that is set by their physical and technical capabilities. These individuals have a better capacity to “make the best” of the present situation and perform to the “best of their ability”. People who have a negative image of themselves on the other hand, perform far worse than they are physically and technically capable of doing. Self-image is really the sum of three parts: the ideal self-images, the real self-image, and reflected self-image. The sign of a mentally strong and stable person is that these three aspects are close to each other, or are the same. If there are significant differences between these different aspects there will be conflict between them and this affects performance negatively. The ideal self-image contains short and long term goals – in sport, studies, work, love or life. It also contains our ideals concerning appearance, behaviour, relationships, and how good we want to be. The real self-image is harsh reality, that is how good we really are at different things – a strong hill runner, good at fine orienteering, good at man-to-man running and so on. The reflected self-image consists of what the individual believes that others think about him or her. How internal and external factors affect self-confidence

Children learn a lot about themselves through reactions from their environment and the people around them. These external factors affect a child’s self-awareness. This awareness or self-image is the basis for self-confidence. When the child becomes older and can choose

his or her own activities and how they are carried out he or she has the opportunity to influence this internally. By carrying out an individual, self-chosen task the child has the chance to achieve his or own result. This result, from an individual activity, forms the basis of a developing inner self-confidence. Individual activity is the basis of inner self-confidence. The results that we experience are the inner factors which build inner self-confidence, the real self-confidence. With a true inner self-confidence an individual can act in unknown situations in a creative, dynamic and effective way. The individual has experience – feelings – as reference points, not things that are able to be measured in reality. The advantage of having

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feelings as reference points is that works every time and everywhere, in all situations. In addition you take your emotions and feelings with you everywhere and it always works.

Motivation

Motivation is the inner fuel that lets us do things. There are of course also external motivations such as reward or avoiding punishment but these are far less powerful than inner motivation. Inner motivation helps us to get from where we are now to where we want to be, be that a vision or a finish line in a race. Motivation is about inner power, wanting to do something, being inspired, being offensive and seeing possibilities. A performance-focused orienteer who enjoys the sport will act in a different way to an orienteering who also enjoys the sport but is less interested in winning and more interested in socialising with like-minded people. The orienteer who cares about his or her performance will train harder, in a way that is relevant to competition, put more time into pre-race preparation, look for any little “nugget” of information that can help to be the best when it counts. Motivation comes from inside. It is an expression of the will of the individual. At the same time external factors can help to “pep up” the individual and give a boost to the individual’s training or performance capacity. Coaches and leaders can help to create opportunities in the training environment, so that runners feel that orienteering is a fun sport to be involved in, rich in experiences, is socially rewarding and that competing is exciting and challenging. If motivation is to work well, we need goals. Goals are visions and visions are pictures: We have motivation when we see ourselves in a situation which is currently unachievable but measurable. By measurable we mean that at least one person, you yourself, can judge if you have achieved your goal or not. Often others can also judge if you have succeeded. There is nothing wrong with others seeing our goals, but remember that there are goals that only we can see – goals that are based on inner feelings! A goal based on feelings can be a perfect run. By perfect we mean that it has been carried out exactly as we wanted – controls, running, map reading and so on. When we get to the finish we feel satisfied – I did that well, that was right, I have achieved my goal. Motivation can be created through money, threats, punishment or even war, and these methods will have a certain amount of success. Motivations which come from inside and which is handled in the right way is always stronger and more effective than outer motivations. That is why my starting point must be individual when I create the images that will motivate me and not based on what others think of me. This can lead to uncertainty in the beginning – how should I do this, what is right? As in many other cases this is a learning process and we have to let ourselves make mistakes.

Will power

Will is really a type of motivation. Often it is a negative form of motivation. Will power is a kind of desire to fight which usually burns up too much of our motivation fuel. Will power is only positive if we have an excess of this fuel or if it is focused on a very limited activity.

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When coaches shout “Come on!”, “Stick in there!”, “Push it now!” to athletes, they do so with the best of intentions. The problem is that this enthusiastic encouragement often makes the individuals performance worse when they use their will power to try to fight harder and push themselves to the limit. Nerve impulses are sent to antagonists and the muscles stop working in relaxed synergy. Will power is usually focused on something external, while motivation comes from inside a person. If the object of our efforts is changed, we must also re-focus our will power on the new goal – we start over again. As motivation comes from inside it continues to work even if the object is changed – our inner goal remains, regardless of external changes, until we decide to change it from within. Will power is more easily influenced by our surroundings in the form of pep talks for example – go, push, fight. Will power is often used to achieve a goal that someone else has created.

Concentration

During an orienteering competition it is absolutely essential that you are concentrated. Most mistakes and misses that we make during a competition are due to difficulties with concentration. The course planner tries to force us to vary both our speed and the brainwork we need to navigate efficiently on different parts of the course. When we have several controls which sit close together it is often relatively easy to stay concentrated on the task. But as soon as the orienteering becomes a little easier and our speed increases, our thoughts tend to wander. How many of us for example have begun to hum the last song we heard on the radio as we stand on the start line? Being concentrated means that our attention is tightly focused on a very narrow field. We live in the present and are so focused on the task that we shut out time and space from our consciousness. Concentration is in other words increased attention on a reduced area. We have all experienced this sensation before, for example when reading a gripping book. We lose ourselves in the content, perhaps we even become the person in the book so that we forget everything and shut our surroundings out of our consciousness. We wake suddenly from this dream after a while and become aware of ourselves and our surroundings again. Often we have no idea how long we have been in this concentrated state. This feeling of focus can be created because competition is something we long for, something we think is fun, exciting and challenging, and something in which we can succeed. If our feelings about the competition are negative so that it is something we do not enjoy, or we are afraid of making mistakes, or afraid of the unknown, we become defensive and cautious. There is a significant risk that our thoughts will focus on the wrong things. We may catch sight of other runners, think about what we will be doing after the competition, begin to grumble about the map or that the terrain does not suit us. In short we lose focus on the task, and we cease to think in the present. This is what concentration is - thinking in the present. It is only the present time that we can influence, not the past or the future.

Performance-focused or concentrated?

Being performance-focused could be said to be nearly the same thing as being concentrated. The difference is that being performance-focused is a state before performance while concentration is what we are during competition.

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We are performance-focused when we turn our attention to the task and shut out everything else apart from the job we are going to do. We are concentrated during the activity and non-essential things do not disturb our performance. We become very selective and register only things that are interesting for our performance. We hear our coach among 1000 screaming voices, as if he or she stood alone. Some people become irritable when they are trying to concentrate. Studies of the influence of distractions have shown that three distractions

in a short period mean that concentration must start again. It is therefore not so remarkable that a person reacts when distractions come frequently. We can develop systems, teach ourselves, so that we become less sensitive to distractions and the time that is required to be concentrated reduces. If this process becomes shorter it is easier to be left in peace long enough. On the other hand, by showing consideration for a person who is trying to become concentrated we help them both towards better concentration and make it easier for them to get on with the people around them.

Attitude

We think in pictures. It is important to remember this when we analyse different attitudes. Simply speaking, an attitude is the way we feel about something – a person, an event, or a competition. Attitudes arise when we get a picture of something, a situation or a person and begin to live according to that picture. We react in a planned way when we meet a situation like the picture we have. If we say to ourselves that we are “useless”, “weak”, “a loser”, “can’t run in Scottish terrain” or “that control looks hard, I’d better not miss that one” what will the result be? As the brain prefers to think in terms of positive pictures, energetic work starts going on behind the scenes. The brain must swim against the current, first drawing up a positive picture, “I hit the control” and then crossing out that picture. What is left afterwards? Our picture of doing it right consists of “not” doing it wrong! The image is based on failure. Is that what we are trying to achieve – making a mistake and not hitting the control? Why is this then the dominant picture in the process? Our starting point must be positive and offensive. The word “not” does not exist in Japanese, and from now the word is removed from the orienteering dictionary. Try to attack instead, try to do the thing you want to do, the thing you want to achieve. We must dare to succeed. In “Winnie the Pooh” two extremes are described, the donkey Eor and the tiger Tigger. Eor is a pessimist and only sees problems, while Tigger is offensive, optimistic, positive and only sees opportunities. Choose yourself! Who do you want to be? It is much easier for the brain to think in a positive, realistic way.

Tigger! Eor...

Tiggers like orienteering! I can! I will! I feel bouncy and energetic!

What a thick forest... I can’t be bothered today... I might... I’ll just plod round like usual...

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Yippee! Home again soon... The important issue is getting a positive picture of what we want to happen. We should look for pictures which reflect our attitude and feeling for the situation. When the situation is unknown we must create a picture then test it reality. When we have a picture we can begin to change it by trying to influence it in advance.

- I run easily, I run more easily
 - I’ve done this before, I can do this
 - I am strong on hills, I am stronger than my rivals
 - I enjoy running in tough terrain, I can run well in this type of terrain
 - I am nervous and become calm, I become calmer
 - It is cold and pouring with rain, I have already beaten half of the other runners
- The words in *italics* are the ones that the brain works with. The attitude we take is totally decisive in the process of freeing up energy for the job we face now. We cannot change the weather, the length of the course, the start list, the steepness of the terrain, or any other external factor. We can only try to like the situation and change our attitude to the external factors. Let them be my friends during the competition. They are positive and help my result.

Plus and minus psyche

The difference between a plus and minus psyche lies in the pictures we send to the brain from our language. A plus psyche sends clear, realistic pictures and creates a positive, realistic attitude. Minus psyche creates unclear and unrealistic pictures and our attitude reflects this.

Vision and goals

When you were a child you maybe talked about the future, “I want to be a train driver”, “I want to be a professional football player”, “I want to be in the squad”, “I want to own a Porsche”. A vision is a dream or desire about something we want to become, have, or how we would like to be. It could be about possessions or a job that wakes feelings of excitement and experience. It could be about life itself, what I want to do with my life, what I want to leave behind me. A vision can also be a dream within sport. All successful sportsmen and sportswomen have or have had a vision which helped them find the passion and commitment to train to reach the top. Visions or dreams release the power and desire in us to train, and also creates the creativity needed find new solutions to achieve success. In children’s games visions are important both to carry out the game and for enthusiasm and enjoyment in the game. As a child we often imagined that we were someone else. Today’s children play that they are Michael Owen or Ryan Giggs on the football pitch. A vision can be said to be about seeing a goal. A goal is therefore when we see ourselves in a situation where we do, carry out, perform in the task which is our goal. By making the picture clear and repeating it time after time we internalise it. If the picture is fully internalised the brain can easily recall it and repetition becomes easier. At this stage we begin to have a type of pre-muscular learning.

Practically this means that our muscles learn to carry out a task

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which they have not yet actually tried! If we use visions and goals in the right way we can save many hours of running in our training. Visions create goals, goals create sub goals, sub goals create the content of our training. To find the thread in this process we can make a comparison with golf:

“In golf we have placed the individual and her life as a whole at the centre of the process. Our vision is that “we as players and leaders achieve our potential as individuals through golf”. With this attitude we believe that we can be a leading nation in the golf world and that the things we learn through golf give us joy in other parts of our lives. A great deal of our success is based on heart and head working together. It is through inner motivation that we can perform optimally. One of my most important tasks as leader is to help players get to know themselves and find their own way to the top, so that they can produce a maximal performance on a special occasion and to ensure that players have the tools they need to take full responsibility for their own performance. It takes time to find your own style so we must allow time for that process. Everything is based on the long-term view. Every player has their own way of doing things, whether we are talking about playing golf or their life as a whole. It is important to find an individual rhythm that suits each player. We have become better than before at finding different ways of achieving our goals. We want each golfer to find her own way to a winning performance. Success is about being prepared for new situations and requirements, listening to signals from players and creating different levels of performance peaking. Each competition, regardless of its importance has its own level of performance peaking.” Pia Nilsson, head coach of the Swedish golf team

Within golf players have worked with a very concrete vision for a long time – “the 54” – to show how important the philosophy above is. If a player completes each hole on stroke under par, with a so-called birdie, at each hole, on an 18-hole course, then he or she has a turned

this vision into a reality. The par for most courses is 72 strokes. If a player birdies every hole they will go round in 54 strokes, 72 minus 18. The vision of "the 54" creates motivation, concentration, creativity, energy and not least joy in being able to slowly but surely achieve the impossible. As a coach the vision of "the 54" can be used to create a starting point to make the impossible possible. Visions or dreams are important for everyone within all areas of work. Visions make people think in new ways and find different solutions. The Fosbury flop and Bjorn Borg's double-handed grip in tennis were invented thanks to the desire to jump higher in the high jump and hit harder in tennis. Visions and goals share a lot in common. The difference is that a vision deals with the future in general, whereas a goal is something that will be achieved at a chosen point in time.

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"23-0"

This can be a vision for orienteers. "23-0" means that the orienteer won the match, won the orienteering competition against herself. 23-0 means in other words that the orienteer spiked all the controls in the competition. 23-0 also creates a focus on the task: go clean, flow, all the controls are just sitting there, spike each one. One can also break a race up into different phases. We know that as soon as we lose concentration a mistake is just around the corner. The beginning of a course, points where the tempo or terrain changes, and the last few controls are several areas particularly prone to mistakes. One could theoretically split the match into the following periods: "3-0", "5-0", "8-0", "4-0" and "3-0". This is just playing with numbers of course, but nevertheless it is a way to try and create a vision of the perfect run. When our technique is so good that this is a realistic goal, we can begin to carry it out faster and more securely in stressful conditions.

Harmony, enjoyment and security

If self-confidence is built on inner factors then we also have an inner security. If we feel secure we are able to enjoy things more. When we feel secure and enjoy what we do we experience harmony. Concentration is natural, we work in a systematic way, and our ability to take responsibility, for ourselves and together with others, increases so that our results become optimal. An optimal result is when we have had the opportunity to perform to our absolute best, regardless of whether we actually came first or last in the race.

Feedback

Is knowledge of how we learn important for orienteers? What do we talk about after a competition? Of course, we talk about which controls we missed and how we made our mistakes! We can sit for hours comparing notes on our favourite mistakes. What we are actually doing here is programming in the mistake over and over again. If we consider the principles of learning we have discussed it is a wonder that any orienteer ever finds a single control in competition. Of course we must analyse what went wrong and try to correct it. But is going through a mistake many times really the right way to make something better next time? Would it not be better to go through the things that were good? Should we train to stumble through the forest or should we train to bound lightly over rocks and brushings? Do we practise taking the wrong route choice, or do we train ourselves to find the smartest way between controls? Do we rush into the circle carelessly or do we train so that we spike every control safely and confidently? We should perhaps think a bit more about how we analyse and evaluate our performance from the point of view of feedback and learning. What we learn during orienteering training will be used fully during a competition later. The brain works to store everything that it has learnt, right or wrong, and then applies it later at the right moment. The brain always tries to find a system to solve the task in the best way it can. When we have plenty of time and we are in calm surroundings, then the brain has plenty of time

to work out the right solution logically. When time is limited and the situation is becoming more and more stressful, the brain must work more and more quickly until we begin to react instinctively. In this stressful situation it is easy for the brain to choose from the store of “right and wrong” behaviours we have programmed in through our training. If we have stored

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many mistakes or strengthened incorrect behaviours there is a very great risk that the brain will make an incorrect decision. This is why learning must be strengthened with the correct behaviour and feelings. Give praise to yourself! Emphasise what you have done well and remember the successful things. Note mistakes and turn them into concrete and positive events. The competition analysis form at the end of the chapter can be a useful tool for ofensive and positive but critical evaluation of performance in competitions.

Mental training

By studying the results from the Mental Profile and Competition Analysis we can get a good picture of where we are at the moment mentally and in our performance level. From this analysis we can get a performance profile for ourselves. This picture looks backwards in time and describes how our performance was. This can now be a basis for a future performance profile. The main task should be to find those things which we believe are good at. We build our ability to perform around these things. The things that you think you are good at give you the framework to develop further. The things you think you are not so good at are the things you will work with. You will later tie these two aspects together as you increase the number of things you believe you are good at.

Mental training models

There are many different models and systems which are referred to as mental training. The better and more effective mental training models build on visualisation. This means that we create pictures of the thing we want to do and what we want to achieve! By repeatedly seeing ourselves carry out that thing we want to do we are eventually able to do the same in reality. This requires that the new picture is realistic. There are many different ways to create the pictures we want to visualise. The simplest way is to make use of a model or example that exists in reality of the things we want to learn. We look at someone who is good at the thing we want to do. We copy his or her way of doing things in our own picture, but adapted to our own level. If we watch a basketball player who is good at getting the ball into the basket, we can copy her in our picture but adapt her style to a level that is achievable for us. We should preferably not copy someone totally, but try to capture the essence of how they act and recreate that feeling. In that way we can do the thing she is good at but in our own way – an excellent solution. A person who tries to copy someone else 100 percent must be exactly like the original or else they will not succeed – and no person is exactly the same as any other. Through letters or personal conversations coaches can also strengthen the athlete's feelings and visions. Here is an example of a letter that gets runners to imagine the right pictures before an important competition, in this case the big Swedish relay competition Tiomila. Imagine if everyone is fit and healthy... Imagine if everyone is well prepared...

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Imagine if everyone longs for the race to start... Imagine if everyone is fired up... Imagine if everyone is motivated... Imagine if everyone is in harmony... Imagine if everyone fights... Imagine if everyone tries to do his best... Imagine if everyone gives 100 percent... Imagine if everyone is thinking, “I’m going to perform well...” Imagine if everyone takes responsibility for his job... Imagine if everyone is focused on the job... Imagine if everyone is thinking in the present... Imagine if everyone sees the opportunities his capabilities give him... Imagine if everyone runs as well as they usually do... Imagine if everyone makes the right route

choices... Imagine if everyone “spikes” every control... Imagine if everyone enjoys “shoulder-to-shoulder” running... Imagine if everyone runs for each other... Imagine if everyone supports each other... Imagine if everyone believes that we can do it... Imagine if everyone thinks that Tiomila is really cool... Imagine if everyone thinks that this party is the best ever... and Imagine if everyone was thinking...

- we want to
- we can
- we dare

...then the first weekend in May could be a really good one! Turn these thoughts over in your head as the days pass and we get closer to Tiomila!

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Performance psychology questions to work with

The questions below with example answers can be used to find your own model for mental training. Try to solve each question with yourself as the starting point, or together with your coach, and then transfer these ideas into reality.

1. What is inner motivation? Answer: When the sport and competing are sufficient reward in themselves and when the enthusiasm and desire to have a perfect run mean more than money or other forms of material reward.
2. What is outer motivation? Answer: Money is a good example of outer motivation. When we need other reasons to compete than enjoyment of the sport itself. Discuss the difference in the power to perform that comes from inner compared to outer motivation. Which gives most power?
3. How can we go about trying to influence unwanted behaviours? Answer: One way of working to influence our behaviour during competition is to work with a model such as “inform-accept-change”. The idea is to first inform yourself on where you stand at the moment, accept that and then try to change your behaviour in the desired direction.
4. How can you as an athlete optimise the present moment? Answer: It is important to remember that you cannot change history. What you have done is done. Do not spend too much time pondering on the future either. You can only very slightly affect what will happen in the future. Your task is to concentrate on what is happening now. It is the present time that you can influence most.
5. Give some examples of the form psychological training can take. Answer:
 - Monthly Letter: Every month the athlete gets a letter from her coach or leader. The letter contains a theoretical and practical section on performance psychology. The athlete reads the letter and tries the practical exercises, and can then evaluate the contents and her experience with the methods. The idea with this monthly letter is to help bring psychological training continually into focus.
 - Meetings: It can be useful to have regular contact with someone outside the club or team who can give everyone in the training group the same basic education or further education in the area of performance psychology.
 - Personal discussions: A chance to discuss psychological problems and opportunities for development with someone who is not involved in selecting the team.
6. Why can visualisation be effective? Answer: The human brain has difficulty in distinguishing between real and imagined events. By visualising a picture of the desired action internally the neuromuscular system is programmed in advance for different actions and movements. This is the most intense and effective type of technique training, at home on your most comfortable sofa!
7. What can you do during a competition to positively influence your performance? Answer:
 - Mental time-outs: The brain must be allowed to relax a little during long, demanding performances.
 - Work in the present: You are always best now. Never show negative body

language: Björn Borg won many matches thanks to his calm and secure outward appearance

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while he fought with everything he had under the surface. Never reveal your thoughts to your rivals. Physical activity reduces tension: When you feel nervous, get up and move about a bit. Rituals and routine: Many athletes use rituals and routine to help them feel secure.

Secure through a habit that when it is carried out or repeated, lets them know “now I am ready to perform.” Focus your concentration on the right things: Never on the actual parts of your body that take you round the course, never of the automatic details of your technique, but rather on the task which is to be carried out, such as spiking the next control. Always use the same routine in training and competition. Quick reminder: Imagine a picture of yourself when you are performing at you absolute best. 8. What is a thought stop and when can we use it? Answer: When an athlete realises that he or she is thinking a lot of negative thoughts, a “thought stop” can be a good strategy to try. The idea with a thought stop is to put a stop to all these negative thoughts and replace them with positive ones instead. Each time a negative thought crops up it is replaced with a positive one. So that a thought stop is as effective as possible the athlete should follow these six steps:

1. Learn to identify the things that bring on negative thoughts. 2. Realise that these thoughts prevent you from performing at your best. 3. Choose a clear signal that it going to mark your thought stop. It could be you saying

“STOP” to yourself, clench your hand or that you imagine a red light or flag. 4. Replace the negative thoughts with constructive thoughts that you have practised in advance. 5. Practice your thought stop during training. 6. Use the technique during competition.

9. How can positive self-talk and encouragement strengthen performance? Answer: Positive encouragement builds a good feeling in the team: “Nobody can touch us. We are unbeatable!” It also sends a message to other teams: “They are so good – we are going to have a hard time to beat them.” By using positive encouragement or self-talk we can create an ideal psychological environment for performance, an environment where we are continually reminded that our team can compete with the best. This influences performance, as an idea thought in a secure environment many times, tends to become reality.

10. Why can an imagined headline about a future success increase our chances of winning when it matters? Answer: A thought, turned over in the mind again and again in a secure environment, has a tendency to become reality. An authentic looking headline which athletes see often helps to remind them of opportunities for success: “We can win! We can perform well!”

BRITS

FIND

GOLD IN

FINLAND

Chapter 6 – Food and performance

Food and

performance

Energy balance

Orienteers have a high-energy requirement as they continually work their bodies hard, both in training and competition. It is very important that our diet, the fuel and building material for our body, is of a high quality.

In orienteering both aerobic and anaerobic metabolism contribute to energy production in the muscles. The size of the muscle and liver glycogen reserves is therefore extremely important so that the brain can be supplied with blood sugar and make the right decisions while the body continues to work at a high intensity. Large amounts of carbohydrate in our diet before training and competition are therefore absolutely essential.

The intensity of our training is the main factor that determines energy expenditure. The more intensively we train the more energy is taken from the glycogen reserves, while at lower intensities both fat and glycogen are used as fuel. The energy turnover during an hour of training lies between 1000-1500 kcal. The factors that determine how much energy is used are the training intensity, sex, age and weight. As general rules the following are true: High intensity requires more energy than low intensity training, men require more than women - men have more muscle mass, young people (15-25 years old) require more energy than older people and people with a higher body weight require more energy than those with lower body weight.

How much energy an orienteer requires is in other words totally individual. For this reason advice on nutrition for orienteers must start with the individual. What is sensible for one person can be totally crazy for another. But to give a basic idea, for a competitive junior or senior 4000 kcal for women and 5000 kcal for men are reasonable daily figures. However, how we train and what we do with the rest of the time during the day are crucial factors in the energy balance equation. It is important when giving nutrition advice for sport to consider how different the requirements of an active sports person can be from the average man on the street. The most important thing to remember is that we should always start with the individual.

Many people train hard to try to improve their performance. Technique must be continually polished and the right muscles developed. But to get the most benefit out of this training there must be a suitable balance between three crucial components: food, rest and training. When the energy in our food is less than that we use during training and competition, there is no energy left to build new muscles or for recovery. Training becomes wearing and breaks the body down instead of encouraging development.

A simple rule to follow is that food and recovery must also be increased when we increase training volume. The right balance between these three factors can be compared with an equilateral triangle. If one side of the triangle is changed then

Food Training

Recovery

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the other two must follow if the triangle is to remain balanced (see diagram). Food groups

The food we eat consists of carbohydrates, fat, proteins, vitamins, minerals and water.

Carbohydrate, fat and protein are energy sources while vitamins, minerals and water have other important jobs such as protecting the body against infection. Protein

Protein is the body's most important building material. Muscles and other tissues need protein to build up and repair the body. Protein consists of around 20 different amino acids, which are formed into different chains, with different appearances. Eight of these are essential and cannot be synthesized by the body. If we eat a balanced diet we will get the protein that the body requires without problems. Protein should count for 10-15 percent of our total energy intake, which means that if we are in energy balance a well-trained orienteer will easily meet her requirement for protein. The following foods contain the proteins we need: Animal based products:

- Meat
- Fish
- Poultry
- Eggs
- Milk products Combinations of vegetable products:
- Corn + peas
- Lentils + bread
- Corn/rice + beans Fat

Fat is a food group with many important roles. Unfortunately fat does not enjoy good press among sportspeople or the general public. Yet fat is very important part of an orienteer's diet, as it is indeed for all endurance athletes and especially for women. Fat is a large source of energy for all types of activity. The lower the intensity, the more energy is taken from the "fat tank". Fat works as a support for the inner organs and helps to keep body temperature constant. Fat also provides energy for the hormone system, surrounds each nerve fibre as an insulation material, like an electrical cable and is present as a building material in cell membranes. We get several important fatty acids and the fat-soluble vitamins A, D, E and K through the fat in our diet. As there is a strong link between the female hormone oestrogen and percentage body fat for women, menstruation is a signal of whether there is balance between food, training and recovery. If menstruation ceases or is irregular then a woman has too little fat in the body and is unable to produce oestrogen. This leads to serious problems with calcium absorption, which is essential for maintaining the bones. Too little fat can actually lead to low bone density or eventual osteoporosis and there is a greatly increased risk for injuries such as stress fractures. Fat should make up about 25-30 percent of our total daily energy intake, but for an orienteer who has an extremely high energy requirement it is sensible to lie on the upper limit, and

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indeed perhaps a bit over, 35 percent, so that the food we eat is not so bulky. If we compare fat with carbohydrate, it provides the same energy in half the volume, which is perfect for a sportsperson who requires a lot of energy. If food contains some fat, less food is needed than if we only to eat carbohydrate based food. Fat is present in the following foods:

- Dairy products
- Vegetable oils, margarine
- Fatty fish: Salmon, mackerel, herring
- Mayonnaise
- Nuts Carbohydrate

Carbohydrates are the body's primary source of energy during training and competition. Carbohydrate is the chemical name for the food group consisting of sugars, starch and dietary fibre. Carbohydrate is stored in the muscles and liver in the form of glycogen reserves and is also in the blood in the form of blood sugar. When the body requires energy, glycogen is broken down into glucose. A simple way of looking at the process is that muscle glycogen is used for muscle work and blood sugar is the fuel for mental work. The main part of our diet should consist of carbohydrates – about 55-65 percent of the total energy requirement. This requirement can also be calculated as an amount per kilogram of body weight per day. Suitable amounts range from 6 to 10 grams per kilogram per day. The upper

limit would be suitable for someone training twice a day. Carbohydrates can be found in the following foods:

- Pasta, rice, potatoes
- Bread
- Oats, porridge, muesli
- Fruit
- Vegetables and root vegetables Glycaemic Index - Slow and fast carbohydrates

Certain carbohydrates are broken down more quickly than others and are taken more quickly into the blood stream. Carbohydrates that are broken down slowly are said to have a low glycaemic index and those that are broken down quickly have a high glycaemic index. The rate of breakdown depends on fibre content, carbohydrate structure and chemical makeup and the amount and type of cooking the food has undergone. A high intake of just fast carbohydrates with a high glycaemic index will give a rapid rise in blood sugar levels, which can lead to a significant insulin reaction. Insulin is responsible for transporting glucose from the blood vessels into the cells. The result can be that you will feel hungry and tired again shortly after this type of food. Slower carbohydrates with a lower glycaemic index will give a slower rise in blood sugar levels and a more even insulin reaction. Somewhat simplified, we should eat slow carbohydrates before training and competition, while during and directly after we should eat fast carbohydrates, and then slow carbohydrates again some hours after. During training and competition insulin is not activated in the same

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way, in fact the muscle cells stand more or less wide open for carbohydrate metabolism and storage. If we on the other hand only eat fast carbohydrates during the day when we are working or studying there is a risk that the blood sugar will rise and fall in a series of peaks and troughs and we will feel energetic for a while before becoming rapidly tired and losing concentration (see diagrams). Slow carbohydrates give a slow and controlled rise in blood sugar.

Fast carbohydrate foods can lead to peaks and troughs in blood sugar levels.

Examples of slow carbohydrates

- Couscous
- Baked beans
- Split peas and other pulses
- Lentils Pasta, fluffy rice (lightly cooked)
- Wholegrain bread
- Milk, yoghurt
- Apple, banana
- Juice

Examples of fast carbohydrates

- Puffed rice
- Cornflakes
- Sports drinks
- White bread
- Brown bread

- Porridge
- Sticky rice (well cooked)
- Potatoes

Energy reserves in the body

Energy in the body

Blood sugar level

Normal level

Time

Blood sugar level

Normal level

Time

Low blood sugar: Tiredness, difficulties concentrating.

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Liver 50-100 grams (liver glycogen) Muscle 350-1000 grams (muscle glycogen) Fat > 10 000 grams Blood glucose 5 gram Competition Percent of Energy requirement duration VO2 max for that pace < 1 min >> 100 percent > 15 g/min (muscle glycogen) 1-5 > 100 ≈ 10 6-15 ≈ 100 ≈ 7 16-60 98-94 ≈ 6 61-120 ≈ 90 ≈ 5 > 120 89-75 ≈ 4 For an orienteer this means that the “fuel tank” in the muscles that are used for running, the legs, may be nearly empty after 75-80 minutes running. It is therefore important to consider how we can fill this tank before, during and directly after training and competition. Vitamins and minerals

Several vitamins and minerals are referred to as antioxidants due to their capability to mop up so called free radicals in the body. Free radicals are very unstable, highly reactive elements and compounds that, among other routes, are sometimes formed naturally in the body during aerobic metabolism. Free radicals can, if we are not protected from them, damage cells by oxidising fatty acids in the cell walls, which can lead to illness. During periods of hard training, such as training camps or when competing in multi day competitions or the big Scandinavian relays where you may have to run through the night, the production of free radicals will increase. In these cases antioxidants may be particularly important. In other words, it is important to eat a balanced and varied diet and make sure that the food contains products rich in antioxidants such as vitamin C, E, A and selenium. Vitamins have an important role in controlling several processes in the body. They are essential and we need to eat them daily. An orienteer has a high energy requirement and therefore needs a lot of food. If we eat a varied and balanced diet during the day we will get the vitamins that we need. It is not necessary to supplement this with countless different vitamin pills. In certain extreme cases during very hard training, or when visiting countries where food is a limiting factor it can be useful to take vitamin supplements, but not otherwise. Vitamins can either be fat or water-soluble. Vitamin A, D, E, and K are fat soluble while vitamin B and C are water-soluble.

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Fat-soluble vitamins

Vitamin A is important for vision in the dark, the skin, mucous membranes, growth and resistance to illnesses and infections. Vitamin A is present in liver, milk products, margarine, cheese, fatty fish, carrots, peppers, and apricots. Vitamin D regulates the calcium balance in the skeleton and teeth which means that it is important in ensuring that enough calcium is absorbed. Vitamin D is found in fatty fish, egg yolks, milk products and margarine and is also synthesised in the skin when we are exposed to sunlight. Vitamin E protects vitamin A and the unsaturated fatty acids against oxidation. Vitamin E is thought to contribute to the

stability of the cell membrane by protecting the polyunsaturated fatty acids in the membrane from attack by free radicals and oxidation. Vitamin E is found in cereals, especially in wheat germ, margarine, oils, fish, almonds, nuts, beans, liver, eggs, vegetables and root vegetables. Vitamin K is necessary for blood coagulation and is also important for nerves, muscles protein and energy metabolism, the liver and the formation of blood. About half of our vitamin K requirement comes from food; the rest is produced by bacteria in the intestine. Vitamin K is found in vegetables such as broccoli, spinach, sprouts and also in kidney and liver.

Water-soluble vitamins

Vitamin B exists in different forms: B1, B2, B6 and B12. Vitamin B is necessary to convert carbohydrate, protein and fat into energy. They are also necessary for muscle and nerve function. Vitamin B deficiency affects several metabolic processes and leads to an increase in pyruvate and lactic acid in the blood among other problems. Vitamin B is also important in the formation of red blood cells. Orienteering is essentially an endurance sport where athletes work both aerobically and anaerobically. This means that our requirement for vitamin B is greater as our glycogen stores must be broken down and converted to energy. Vitamin B is found in flour, oats, bread, pasta, wheat germ, potatoes, meat, fish, chicken, shellfish, vegetables, fruit and milk products. Vitamin C is necessary for the formation of connective tissue in the skin, blood vessels, skeleton and the teeth. It is also beneficial for healing of cuts and bruises and protects vitamin A and E and fatty acids against oxidation. Vitamin C is found in fruit and berries, citrus juices, vegetables, potatoes and root vegetables. It is beneficial to include a food or drink rich in vitamin C at mealtimes as this helps the body to absorb iron from food. Minerals

Minerals account for only 5 percent of our body weight but are very important for both structure and function. Approximately 2 percent of the body's weight is calcium and most of this is found in the bones. Other elements include phosphorus, potassium, sulphur, sodium, chlorine and magnesium in decreasing order, all the way down to the small amounts of iron, selenium, fluoride and zinc, which nevertheless have an important role to play, regulating the many chemical reactions that take place inside our bodies. If we eat a well balanced and varied diet we will get enough of all the minerals we need. As far as sport and particularly endurance sports are concerned, however, iron and calcium can

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be exceptions. Women who train very hard sometimes have difficulties in meeting their iron requirements. Iron is often in the food we eat but our ability to absorb it depends on the other foods that make up the meal. Vegetarians should also be observant as most of our selenium requirement comes from animal products. Iron is found in the haemoglobin in red blood cells, in the myoglobin in the muscles and in iron-dependant enzymes. Iron deficiency, can lead to reduced resistance to infection and anaemia, which leads to reduced performance. One of the important jobs iron has in the body is to transport oxygen from the lungs to the working muscle cells. Iron deficiency can give a low Hb value which means that we have a reduced capacity to transport oxygen. High Hb values indicate a higher capacity for oxygen transportation and this is naturally a very important physiological factor for an elite orienteer. Iron is found in abundance in animal products such as meat, liver and fish but also occurs in non-organic form in fruit, beans, bread, oats and pasta. Absorption of iron is significantly affected by the food and drink served with the iron rich food. Foods rich in vitamin C such as fruit, berries, juice and vegetables encourage iron absorption. Caffeine in tea and coffee, and milk chocolate, significantly reduces iron uptake. Calcium Calcium is found in our bones, soft tissues and in bodily fluids. Calcium is very important for the bone formation and for muscle contractions. Calcium deficiency can lead to low bone density and osteoporosis. If this

happens before puberty or during puberty then the spine and pelvis are worst affected. The bones in the limbs will be affected if this process takes place later. According to current research we can store calcium and build up the skeleton until we are 25 or perhaps 30 years old. After this, the storage process declines and our bones become weaker with age. It is therefore very important to ensure that we have strong bones during our younger years, both for our sporting career and for our later years. Milk and cheese are the best calcium sources, without these products it is difficult to meet our calcium requirements. Calcium is also found in broccoli, cauliflower, sardines, mussels, prawns, and eggs. Selenium is a trace element which helps prevent cell damage. It protects the body together with vitamin E by mopping up free radicals. Hard training and poor recovery leads to an increase in free radicals in the body. Selenium is mainly found in animal products such as fish, meat and milk products. Water

Water is an essential substance for life. Fluid turnover in normal climatic conditions is around 3 litres per day. When humidity is higher, as it was at the 1996 summer Olympics in Atlanta for example, a turnover of 6-8 litres is normal. About half of our fluid requirement comes from what we drink and the rest comes from water in our food. About 60 percent of our body weight is water. Men have more than women, due to their greater muscle mass. Water is found in and outside all the cells of the body, but also in the various body cavities. It can pass freely through all our tissues and cell walls. Water equalizes differences in salt content Chapter 6 – Food and performance

between different tissues in the body and keeps our salt levels in balance in this way. Water also helps to clear away waste products which are formed during different metabolic processes. Correct fluid balance is easy to check daily. Urine should be colourless or light yellow in appearance: If it is darker, drink more water! When training and competing or carrying out other physical activities we lose a lot of water. Increased metabolism to provide energy for the work results in heat production and to prevent overheating we begin to sweat. Fluid lost through sweating must be replaced. If we lose as little as 2 percent of our body weight, or about 1.4 litres for a person who weighs 70 kg, performance is reduced to 80-85 percent of optimum. Fluid reserves must be filled before training and competition and kept topped up during activity. During training and competition it is recommended that you drink 200-250 ml every 15-20 minutes. If the session is less than 45 minutes long water is suitable. When training or competing for a longer time you should drink a weak carbohydrate mixture, such as a sports drink.

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40 60

80 10

0

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Fluid Loss: % body weight

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After training and competition it is important to restore energy and fluid balance in the body. Suitable foods include bananas, white bread, raisins and sugary sweets such as Jelly Babies. Drinks such as water, sports drink, squash or juice are suitable. It is easiest to replenish the glycogen stores directly after hard activity. Recovery will be considerably quicker if you remember this. Mealtimes

In order for training to be effective we should plan for breakfast, lunch, dinner and 1-2 snacks. Meals do not necessarily need to come in this order. For example, if training is in the early evening when you would have eaten dinner, you could have a light supper before, then take a proper meal for dinner afterwards. The important thing to remember is that there must be a balance between “energy in” and “energy out” – eat enough to cover your energy requirements. Meals should be at times that fit in with your routine.

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A simple rule of thumb can help to eat appropriate amounts of different food groups. A good meal for an orienteer who has high-energy requirements might be as follows: $\frac{1}{2}$ plate of potatoes, rice or pasta $\frac{1}{4}$ plate of root vegetables and vegetables $\frac{1}{4}$ with meat, fish, chicken or egg. Bread is a good addition to the meal, with or without a spread. Milk or another similar drink is ideal and you can finish with a simple pudding such as fruit or berries. If you need more energy it is easy to increase the size or number of portions and add snacks in between meals. When you are training at your absolute hardest it is probably easiest and best to increase the potato or pasta portion in the main meal and perhaps add a little sauce to make the food moister and creamier.

Chapter 7 – Sports anorexia

Eating disorders

and sports anorexia

New perspectives on sports anorexia

In a study Diagnosis and Treatment of Anorexia in Elite Sport (1995), which the Anorexia Centre at Huddinge Hospital in Stockholm carried out with help from the Swedish Sports Federation, it was shown that the main causes of anorexia in elite sport were high levels of physical activity, including heavy training, in combination with difficulties in eating enough food to meet the energy requirement of this activity. The body did not get enough energy to replace the energy it expended. The research also showed that there were differences between sports anorexia and “normal” anorexia. In comparison with “normal” anorexia it is clear that sports anorexia is less severe. Weight loss is not as great in sports anorexics as it is in classic anorexia cases. Distorted body image was not noted in sports anorexics. The typical picture of a waif-like girl who nevertheless sees herself as fat in the mirror does not apply. Physical activity combined with restricted eating and the fact that boys and girls involved in sport are extremely focused on performance and results are the main factors that make them susceptible to sports anorexia. When an athlete develops anorexia they often also have a feeling of failure. Many blame themselves, which means that they place higher, sometimes unrealistically high, demands on themselves and their performance. Many of the girls in the study were also vegetarians who ate a lot of plain and very low energy food, lacking adequate knowledge about nutrition. This can lead to significant problems with heavy training. It was also found that menstruation had stopped in all sports anorexics. If the menstrual cycle is very irregular or has ceased, it is a sign that fat reserves are far too low. Production of the female hormone oestrogen ceases and the absorption of calcium decreases, leading to a risk for low bone-density and osteoporosis. Sleeplessness,

concentration difficulties and problems with tiredness were also very common. As far as mental problems are concerned, the researchers at the Anorexia Centre believed that these came as a product of sports anorexia rather than being the root of the problem. If mental problems lay at the root of sports anorexia, such problems would be 10-15 times more common in the sports world than the population at large, as anorexia itself is 10-15 times more common in sport. Those mental problems that were noted in the research, such as anxiety, obsession with weight and thoughts of suicide, can be seen as a result of sports anorexia. However, this interpretation should not reduce the significance of those psychological problems. What we can say is that when diagnosing anorexia, including sports anorexia, it is important to not just look at the psychiatric perspective. In earlier Swedish and Norwegian studies, it was found that eating disorders are more common in elite athletes than the general population, but also that certain sports were more affected than others. They found that one in seven girls in specialist sports schools showed symptoms of anorexia or anorexic tendencies. Anorexia is also known to be the third most common illness in girls aged 14-19.

Chapter 7 – Sports anorexia

Endurance and aesthetic sports had more problems with anorexia than other sports, but it was noted that all sports had or have had problems with anorexia. It was also noted that coaches played an important role in detecting anorexic tendencies in athletes. As training is a very big part of an athlete's life and too much training is one of the main causes of anorexia within sport, it is clear that we need to work with this variable when treating patients. Moreover, training can in itself be a way to increase the effectiveness of the rehabilitation process. This requires a lot of work with the athlete to develop an activity and training plan that is appropriate for the individual during the different phases of treatment. The role of training in the treatment of both sports anorexics and other people with eating disorders is described in a report entitled *Physical Training in the Rehabilitation of Patients with Eating Disorders* (RF 1997). Training was shown both to make treatment more effective and to speed the process up. This treatment has been carried out together with the Anorexia Centre. Difficulties with eating enough

Studying boys and girls eating habits is of course central to the prevention and treatment of anorexia. Men and women can eat different amounts of food. From a biological perspective humans have not changed for several thousand years. Men, who were out for extended periods to hunt, ate large amounts of food when he got the chance. Women, who stayed at home and looked after the children, ate smaller amounts of food more frequently. There is therefore a historical basis for men and women's different eating habits. Women find it difficult to eat as much as men. A study was carried out where a number of men and women were given breakfast, lunch and dinner every day for one week. On the sixth day dinner was taken away, and they were not given any breakfast on the seventh day. The following lunchtime the men ate enough food to "fill the gap" from the missed dinner and breakfast. However, the women only ate as much as they had eaten at previous lunchtimes. It is probable that we have a genetically inherited mechanism from the Stone Age that controls this behaviour. With this in mind we can say that the traditional pattern of three larger meals per day may not be ideal for women. It is probably better to eat "little but often", for example "breakfast-snack-lunch-snack-dinner-supper". As women find it difficult to eat large quantities of food long endurance training should be treated with great respect. The energy requirements for his type of training are large and diet is essential in ensuring that these requirements are covered. Early signals

Coaches should be observant for signs which point towards eating disorders in athletes. These signs are clearest during the phases referred to as “restrictive eating” and the “positive phase”. It is much easier to detect anorexic tendencies during training camps and competition tours. It is of course mainly at mealtimes that we can detect these tendencies. If as a coach you become aware of one of the signs listed below it is time to be extra-observant. If you note more than one of the signs you must, as a responsible adult, react and take positive action to help. Show you care! Act when you have a chance to help. It will only be more difficult later.

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- I have already eaten.
- Playing with food. Moving food around plate.
- Cutting food into small pieces.
- Using small plates.
- Eating a lot of low-energy, plain food or salads.
- Eating alone.
- Changing eating habits.

These signals are clear signs of worrying eating habits. Nobody can be a good sportsman or woman by eating like this. If any of the factors below also come into the equation, then you have got a person with a clear anorexic in your group. If you do not react now you are not doing your job as a responsible coach, or even as a person who should be there to look out for others and care. It is our duty to react and take positive action.

- Increased training volume, most often in the form of more or longer sessions.
- Menstruation ceases.
- Increased activity in other areas.
- Unexplainable mood swings and unhappiness. Feeling down.

If we do not react at this stage and some of the following factors arise...

- Thin and undernourished appearance
- Feels cold. Prefers to wear bulky clothes.
- Obsessed with food and weight.
- Headache: Often takes aspirin, paracetamol, Neurofen or similar.
- Problems with sleep.
- Dizziness and fainting.
- Bloodshot eyes.
- Dry skin and hair.
- Long, frequent visits to the toilet.
- Constipation.

...then medical help must be sought. At this stage it is difficult to get through to the affected athlete. Professional support is required. Body Mass Index, BMI

As a coach it can be useful to have some simple tools on which to base discussion of suspected cases. Vegetarians who eat a limited diet, high training volume, no or irregular menstruation, thin and frozen appearance and BMI are extremely useful indicators. BMI is a calculation where weight in kilograms is divided by height squared, in meters. For example: A woman weighs 55kg and is 1.65m tall: $55 / (1.65 \times 1.65) = 20.2$ The same woman later

weighs 47kg: $47 (1.65 \times 1.65) = 17.3$ The standard values for normal BMI, which are used in the medical profession are:

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Underweight Normal Overweight Women less than 19 19-24 more than 24 Men less than 20 20-25 more than 25

Women in the Swedish orienteering squad have BMI values of 19-20, in the lower region of normal weight. If an orienteer went down to a BMI of 18 it would be time to consider what that athlete was doing. If the BMI fell to 17 or 16 it is time to take more immediate action.

Many elite sportswomen have BMI values in the range 18-20, or even move around in this range. This in itself need not be a negative sign, but the risk that an athlete can lose weight and fall down to 16 or 17 is always there. Training

It is very important that training is reduced when a person is suffering from sports anorexia, but at the same time contact with club mates and sports friends needs to be maintained.

Training volume can be reduced to a quarter and should consist of short sessions of 20-30 minutes fast walking or light jogging, 2-3 times per week. This speed corresponds to 50-60 percent of VO₂ max and could also be called “chatting pace.” Competition should be avoided until a healthy balance is restored. By healthy balance we mean that the athlete has normal eating habits, weight is normal and menstruation has returned.

Food

Puberty, the time when a girl develops into a woman, is a particularly sensitive period for the development of sports anorexia. A great deal is happening during the teenage years, both biologically and in the social and psychological spheres. When the athlete is in addition training and competing intensively during this period very high demands are placed on what she eats. Energy and nourishment is required in relatively large quantities. Normal energy requirements for a young person who trains daily are between 3000-5000 kcal per day. The “normal” energy intake for a sports anorexic can be between 800-1500 kcal per day. This means that she is eating less than her basic requirement: The minimum amount of energy which is needed to keep the body functioning is not being provided.

Together with a low training volume, or no training at all, and progressively increasing amounts of food we can turn the trend around very quickly and increase weight. The important factor is that we divide the daily food requirement over several mealtimes according to the following table:

kcal	kcal	kcal	kcal	Breakfast, 25 percent	500	625	750	875	
Snack, 5 percent	100	125	150	175	Lunch, 30 percent	600	750	900	1050
Snack, 5 percent	100	125	150	175	Dinner, 25 percent	500	625	750	875
Supper, 10 percent	200	250	300	350	Total, 100 percent	2000	2500	3000	3500

It is valuable to find out how much the athlete is eating before implementing the plan above. There are now several good diet analysis programs available. If we start with too much food in the first week the risk for a negative reaction is greater. Food can become the enemy, not a friend in the recovery process. It is probably better to start with 2000 kcal per day or sometimes even less so that the athlete can adapt to the new eating routine. It is also easier to

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eat if the athlete has food that she likes. A mealtime should be an enjoyable time: It is important to think about appearance, taste and aroma when preparing food. Anorexia in orienteering

At the end of the eighties it was noted in Sweden that there were girls with anorexic tendencies in training groups. The federation doctor was brought in to try and find suitable solutions to the problems. Everyone was in agreement that it was not acceptable to tolerate anorexic tendencies in a national coaching system. Ignoring the problem let down those girls

who were affected and was contrary to the whole philosophy of the sport. The signals which a coach gives to athletes about what is right and wrong are extremely important. Turning a blind eye can be seen as tacit approval of this type of behaviour. Orienteering is an extremely demanding and physically tough sport at international level and there is a risk that the wrong signals from leaders actually create and encourage inappropriate training and eating habits. We must always be observant for worrying trends in the development of training and think of the possible consequences for current athletes, and also the next generation of runners. The role of the coach

A coach has a very important role, both in the manner in which they lead a group during training and competition and in the way that they work to develop the athlete as a person through sport. This is especially true for male coaches working with female athletes. As leaders we must also understand that everyone is unique: We all have different needs for attention, support, enthusiasm, inspiration and feedback. A coach needs to make training fun and create enthusiasm. The following points are useful to remember when working with female athletes:

- Be aware that there are likely to be many individuals in the group who could easily develop eating disorders.
- Be careful with seemingly innocent comments about weight or physical appearance. At the wrong moment, these can have dramatic consequences.
- Coaches, leaders and parents must never demand that young people who are still growing lose weight.
- Emphasise the importance of regular and nutritious food. Fat is necessary, especially for growing young people.
- Learn to spot early signals of anorexic tendencies.
- Set goals that are shared and realistic.
- Make sure that communication between athlete and coach is open and effective.
- Athlete and coach must have trust for each other.

Human Physiology

The muscles

Our muscles make up about 40 percent of our bodyweight. For a man of average build this could be 30-35 kg and 20-25 kg for a woman. There are over 300 muscles in the body with different sizes and functions (see diagram). The smallest muscles in our fingers only weigh a couple of grams while the largest muscles in our thighs can weigh several kilos. Skeletal muscles have tendons at each end which are attached to two parts of the skeleton. When the muscle works and makes a contraction, the two parts move nearer each other (see diagram). The tendon at the end of the muscle turns into a thin membrane called a fascia, which surrounds the whole muscle. This means that the muscle can glide smoothly against other muscles and organs and work independently.

The functional unit in a muscle is called a muscle fibre. These fibres stretch along the whole length of the muscle like thin threads. They are often somewhat thicker in the middle. A fibre consists in its turn of smaller units called myofibrils. The smallest units are the myofilaments, which lie linked into long rows in the fibrils. These consist of two proteins, actin and myosin. These proteins are the tools that can transform stored energy into muscle movements. When the muscle works the two proteins move into each other to shorten the filaments. The Z-band works like a concertina that moves in and out (see diagram).

The main function of the muscles is to turn energy into movement. In orienteering, which is an endurance sport, this takes place with the help of oxygen. The higher the oxygen turnover, the greater the work done by the muscle can be.

Muscle fibril

Muscle fibre

Muscle sheath

Muscle

Bundle of muscle fibres

Connective tissue membrane

Muscle fibres have different properties and are significantly influenced by training. Different types of training will influence fibres in different ways. Training that places a significant load on the muscles such as strength training will make the fibres increase in volume, while general aerobic fitness training will improve the systems around the muscle that supply it with energy and oxygen during long periods of activity. There are two main types of muscle fibre, slow twitch (Type I) and fast twitch (Type IIa and IIb). The names refer to their different maximum contraction speeds. The different properties of the fibres also make them suitable for different types of work.

Properties of the different fibre types

Property	Type I	Type IIa	Type IIb
Contraction speed	Low	High	High
Time to develop maximum power	Slow	Fast	Fast
Power	Moderate	High	High
Lactate formation	Moderate	High	High
Capillaries	Many	Few	Few
Mitochondrial density	High	Low	Low
Aerobic properties	Good	Moderate	Moderate
Fat-burning capacity	High	Moderate	Poor
Resistance to tiredness	Good	Poor	Poor

In humans both types of fibre are often found in the same muscle. Each person has a different amount of each type of fibre, based on his or her genetic makeup. A potential long distance runner might be born with 80 percent slow twitch fibres while a potential sprinter could have 70 percent fast fibres. Our dominant fibre type gives us different capabilities for different types of physical work. However, an interesting feature of muscle fibres is that they are influenced by training. A muscle fibre that repeatedly undergoes a certain type of training will

Muscle fascia

Muscle fibre

Myofibril

Myofilament

Actin

Myosin

Muscle

Sarcomer

Z band

change slowly and develop the properties which are required by the activity carried out.

Once again, the importance of training that is relevant to competition is apparent. We become good at the things we train!

Metabolism

The term metabolism covers all the chemical reactions which take place in our cells. This includes for example the breakdown of compounds from food which are needed for the body to function, and the creation of new cells. Metabolism is regulated by a very complicated chemical and hormonal system involving the different organs of the body. To be able to run, the muscles need energy. This is supplied in the form of carbohydrates and or fat which are broken down and transformed into fuel for muscle work. This fuel is an energy-packed

phosphate compound, ATP, which is the only form of energy that the muscle can use directly for physical activity, such as running or cycling. Depending on the demands of the activity carried out, we use different systems for breaking down carbohydrate and fat. This is why the intensity of our training is so important. In order to get a useful training effect we must activate the right systems, so that our metabolic processes adapt to the new load and intensity. Training breaks the body down, while rest and refuelling with food allow us to recover and benefit from that training. The training process also breaks down cells, which must then be replaced. In this way training stimulates tissue formation and repair in the muscles, tendons and skeleton. The body uses proteins for this repair process. Different types of food provide different amounts of energy. The table below shows the relationship between the amount of food and the energy it can release when metabolised:

Fuel Energy from complete metabolism
1 gram carbohydrate 4 kcal (17 kJ)
1 gram fat 9 kcal (39 kJ)
1 gram protein 4 kcal (17 kJ)

As a store for energy, fat is a clear winner. On long walks in the mountains or long training sessions on a bike fat is the best source of energy. However, carbohydrate is a richer fuel in some ways. It burns more efficiently, like high-octane petrol, requiring less oxygen to release the same amount of energy. Given the same amount of oxygen, carbohydrate supplies 6-8 percent more energy than fat. Energy production

In order to work, muscles need access to ATP (adenosine triphosphate). There is always a small amount of ATP stored in the muscles. This energy source can only fuel a few seconds of work. If activity continues more ATP must be produced. This can either be done aerobically, with the help of oxygen, or anaerobically, without oxygen. As running speed and terrain are always changing during a race, orienteering has a continually changing balance between aerobic and anaerobic work. Energy for our muscles can be produced in the following ways:

- 1 ATP breakdown Maximal effort 3-6 sec

- 2 Carbohydrate breakdown Anaerobic glycolysis 10 sec-2 min, lactate produced

- 3 Carbohydrate burning Aerobic glycolysis 1-2 hr, until carbohydrate runs out
- 4 Fat burning Aerobic process 2-10 hr, but at a lower pace

Steps 1 and 2 take place without oxygen while steps 3 and 4 require a continual supply of oxygen. Glycolysis means breakdown of glycogen - the compound that the body uses to store carbohydrate in the muscles.

Carbohydrate is also stored in the liver. These two sources of fuel are called muscle glycogen and liver glycogen respectively. Fat cannot be broken down directly in the same way as carbohydrates and is therefore not able to be used for short, maximal efforts. The muscle itself determines which of the above processes will provide energy. If the load varies, for example an increase in speed, decrease in speed, uphill or downhill running, sprinting, a change from path to forest running the muscle immediately changes to another process or uses two process simultaneously. ATP production is two times greater in the anaerobic processes, but as these processes involve breakdown without oxygen, lactate is produced. As the muscle is provided with oxygen, step 3 and 4 will come into action. Different effort levels activate different energy systems, resulting in increased capacity and effectiveness. This is the basis for understanding why training must be varied. We become good at the things we train. Lactate

In short, maximal efforts such as 200 m canoeing, 400 m running or 100 m swimming, the energy requirements are very high. The muscles are unable to get enough oxygen. Step 1 and 2 produce the initial energy required. Carbohydrate is broken down directly, without oxygen, and lactate is produced as a waste product. Lactic acid is produced inside the muscle and is actually a rich source of energy in itself. But, lactic acid alters the pH value in the muscle's internal environment and leads to less effective muscle function. Some of the lactic

acid is transported away from the muscle by the smallest blood vessels, the capillaries. The lactic acid that is transported away is processed by the liver and rebuilt into new fuel for the muscles. However, the process of transporting and rebuilding the lactic acid demands a lot of energy. The lactic acid which remains in the muscle can actually also be used directly as a fuel.

Glycogen

Glucose

Pyruvate

Lactic Acid $H_2O + CO_2$

Glycogen Glycogen

Liver Heart muscle

Capillary

O_2 Glucose Lactate

CO_2

Muscle fibre

Whether lactic acid is transported away from the muscle or is recycled directly in the muscle energy system depends on the way that the energy system has been adapted through training. The processes which transport lactate away or use it in the muscle are controlled by an enzyme called LDH, lactate dehydrogenase, which exists in five different forms. If your training has been mainly based on sprinting and short intervals the enzyme which transports lactate out of the muscle will dominate. If your training has included more endurance work then more lactic acid will be recycled directly in the muscle. In other words, the content of your training indirectly determines the pathways that lactic acid will follow, by influencing the LDH enzymes in your muscles. If you have developed a rich capillary system through endurance training you will also have a much better capacity to transport lactate away from the muscle. You will be much better at clearing lactate. Oxygen pathways to the muscles

A national team orienteer has a very high capacity in the organs responsible for transporting oxygen. On the way from the air to the working muscle cell, oxygen passes through several limiting links. These links can be improved, but also made worse, by training, nutrition and rest. The air we breathe contains about 21 percent oxygen gas, O_2 . If we are at altitude the available oxygen reduces, which also reduces our capacity for physical activity. It is possible to notice a difference even at 1000 m. We become breathless more easily, have a higher pulse and our legs become rapidly heavy with lactic acid. At extreme altitude, 7000 – 8000 m, each step is an exhausting effort.

The Lungs

The lungs Ventilation Diffusion

The heart Haemoglobin, blood-volume Volume per minute (stroke volume x heart rate)

Arterial blood pressure

Peripheral circulation Circulation in organs, tissues, and non-active muscles. Muscle blood flow Blood vessel dimensions and the size of the capillary network.

Metabolism Supply of energy-rich fuel – sugars and fat. Muscle mass Myoglobin

Mitochondria Enzyme activity

At rest lung ventilation is 5-6 litre/min and this value increases with increased activity. Each person has a theoretical ceiling which limits ventilation. Maximum ventilation is around 200-220 litre/min for men and 125-150 litre/min for women. Lung volume is not influenced significantly by training, but the muscles involved in breathing become stronger and develop better endurance. Oxygen reaches small cavities in the lungs, the alveoli, via the windpipe and bronchi. There are millions of alveoli in the lungs. This means that the total area

available for oxygen transfer is very large: 70-100 m². Around each lung cavity there is a very fine network of thin blood vessels. Oxygen passes through the walls by diffusion and attaches to the red blood cells in the blood. This oxygenated blood is then transported to internal organs and muscles, with the help of the heart. The Blood

Blood volume for an adult man is approximately 4-5 litres. Women have a lower volume, around 3-4 litres. Blood consists of red blood cells and plasma. Red blood cells make up approximately 40-45 percent of the total volume. This percentage is called the haematocrit value. It is usually higher in men than women. Red blood cells contain an iron based compound called haemoglobin, which is able to bind with oxygen and enables red blood cells to transport oxygen around the body. Each gram of haemoglobin can hold 1.34 ml of oxygen. A litre of blood contains 140-160 grams haemoglobin in men and 120-140 grams in women. Haemoglobin concentration, the Hb value, determines how much oxygen can be bound to the red blood cells, and therefore how much oxygen can be pumped by the heart. At the same work level a woman's heart must pump around 10 percent more blood than a man's to compensate for her lower Hb value.

Haemoglobin Concentration

100

110

120

130

140

150

160

170

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

Age

g /li

tr e

Blood volume increases with training and can be as much as 6-7 litres in well-trained athletes. However training does not increase the Hb value, indeed it actually reduces it. The food that we eat is the most important factor in maintaining or increasing the Hb value in blood.

Men

Women

Blood also has other important jobs, such as providing the body with nutrients and being a transport system for hormones. Blood also helps to regulate body temperature. The heart and circulatory system

The heart is the body's most central organ. The heart is muscle which tirelessly pumps blood around the circulatory system. There are in fact two systems – one works to collect oxygen from the lungs, and a larger one pumps oxygenated blood round the body to the muscles.

Oxygenated blood from the lungs fills the heart. When the heart muscle contracts blood is pumped out into the main artery, the aorta. Each beat of the heart contains 1-2 dl blood. This amount, the stroke volume, varies significantly between trained and untrained people. A well-trained heart can pump as much as 40 litre/min. A person who exercises lightly might be able to pump 20 – 25 litre/min. The pump capacity of the heart is an important limiting factor in oxygen transport, and can be improved with training. Resting pulse

Increased aerobic training makes the heart increase in both size and effectiveness. Resting heart rate reduces over time and this is a sign that fitness is increasing. Normal resting pulse

is around 60-70 beats/min, while the resting pulse for an elite orienteer can be around 40-45 and in certain extreme cases around 30 beats/min. The explanation for this is that the body requires the same amount of oxygen at rest, regardless of whether it we are trained or untrained. As the same volume of oxygen is required and each beat from a trained heart provides more oxygenated blood due to its higher stroke volume, the pulse will fall. The relationship can be expressed as a volume per minute: $\text{Volume per minute} = \text{Stroke volume} \times \text{Pulse}$ It can be useful to know your resting pulse. An increased resting pulse can be a sign of infections or the onset of overtraining. The pulse will typically increase by 5-20 beats per minute in these cases. Maximum pulse

As our pace increases the amount of oxygen required in the muscles also increases. Work rate and pulse increase linearly. The pulse increases to an upper limit, the maximum heart rate (Max HR). This generally lies between 170-220 for adults and is somewhat higher for children. A high or low maximum pulse is not an indicator of fitness or training, but depends more on inherited factors which determine the maximum heart rate for each different person. It can be useful to know what your maximum heart rate is, as a guide for training intensity later. When describing training intensity we often refer to percentage of maximum pulse: 70, 80, 90 percent for example. Measuring maximum heart rate can actually be difficult. A heart rate monitor is required. In the lab, an ECG machine may be used. It is possible to carry out the measurement running outside or on a treadmill inside. The procedure involves a thorough warm up, as if for a competition, and then to run for at least 4-5 minutes at a continually increasing work rate. On a treadmill you would run until you have to jump off due to tiredness.

Measurement of maximum heart rate outdoors can be done as follows:

1. Thorough competition warm-up
2. Run for 2 minutes at a continuous fast pace.
3. Run for 2 minutes at a slightly higher pace.
4. Run as hard as you can up a long hill for at least 1 minute, or until you are not able to carry on.

Ideally, you should repeat the whole procedure a few times during a week to get a reliable value for maximum pulse. The variation should not be more than 2-3 beats per minute.

Maximum pulse testing should only be undertaken by healthy and well-trained orienteers!
Oxygen use in the muscles

The largest arteries nearest the heart branch into smaller arteries. The blood rushes down these channels to the organs which have the greatest requirement for oxygen. After eating the stomach and intestine require oxygen and during physical work it is the muscles. The brain and nervous system always require a certain amount of oxygen regardless of how hard the body is working elsewhere. The arteries divide into smaller and smaller blood vessels which supply the smallest vessels at the end of the system, the capillaries. These small tubes surround the actual muscle cells and it is here that oxygen leaves the red blood cells and is transported by diffusion into the muscle cells. A compound related to haemoglobin, myoglobin, plays an important role in this final transport process, where oxygen is taken to its destination, the mitochondria. Metabolism of carbohydrates and fat takes place in the mitochondria and it is here the ATP is formed, providing the processed fuel that the muscles need. The distance between the capillaries and the mitochondria is a very important factor in determining the oxygen transport rate. As the slow twitch fibres (Type I) are somewhat thinner, the distance is shorter than in the fast twitch fibres. The myoglobin content in these fibres is also greater and the capacity for aerobic work is therefore better in these muscle fibres.

Mitochondria

CO₂ O₂ Fatty acids Glucose

Contractive muscle fibres (actin and myosin)

ADP ATP

Arterioles Pre-capillary sphincter Capillaries

ARTERY VEIN

Apart from energy and heat, both carbon dioxide and water are formed. These “waste products” are transported away by the blood to the lungs where they are breathed out. Deoxygenated blood is transferred via capillaries to larger blood vessels and is pumped back to the heart through the veins. The veins join together and take the blood back to the heart where it is pumped round the smaller circulatory system connecting the heart and lungs. Mitochondria – the muscle cell’s power station

The mitochondria are the final link in the oxygen transport chain. It is in the mitochondria that power for muscle contractions is created. The more mitochondria we have, the higher our capacity for providing the energy necessary for muscle work. Training at intensities lower than the “lactate threshold” encourages the body to form more mitochondria. Too much hard training, where you are working in the “lactate zone” can both damage the mitochondria and worsen their function. Capillaries – the body’s postmen

The more postmen who work to deliver out the post, the quicker letters and packages will get to recipients. The capillary system works in a similar manner. Through aerobic fitness training, mainly at low and moderate intensities, it is possible to double the number of capillaries with time. This means that the mitochondria are provided with more oxygen and the oxygen supply during hard work will be much better. This capacity must be continually maintained by orienteers. If training ceases both mitochondria and capillaries will decrease to a normal level.

Central and Local Capacity

The idea of “central” and “local” capacity is an essential concept in all physical training. Local refers to the capacity we have in the parts of the body that are most involved in the movements specific to our sport. For an orienteer this means the muscles used when running in the forest, and expressed simply our training should be in the forest, not on the road. The stride required on the road is very different to the running style needed in the forest. On the other hand road running, mountain biking, swimming, cross-country skiing and a whole variety of aerobic sports can be excellent training for our central capacity. Central capacity

The heart’s pump capacity, the capacity of the lungs and the capability of the blood to transport oxygen are referred to as our central capacity. This capacity is the basis for all aerobic training. The stroke volume of the heart and the blood’s haemoglobin value are two important factors in the central capacity. The stroke volume can be improved through training at different intensities, both low and high intensities in sports which recruit large muscle groups, such as the aerobic sports named above. The haemoglobin value can be maintained through nutrition, recovery and rest. Building the central capacity up to a high level takes many years, but when we have developed a high central capacity it is possible to have short breaks or even train less for longer periods without decreasing this capacity significantly. When building up the central capacity is a good idea to vary the training methods and sports we use, so that the heart and lungs can be worked without repeated stress on the same muscles and joints. Local capacity

Local capacity is directly related to the movement that we actually carry out. For an orienteer it is important that we try to transfer the important aspects of competitive orienteering into our training. Similar intensity, speed, effort and similar underfoot conditions are important to be

able to train local capacity optimally. Orienteers need muscle mass to move themselves over rough terrain, but also need a rich network of capillaries round these muscles. The muscles must be well coordinated with the right nerve impulses providing a smooth, efficient movement. In the muscle cells there must be a high density of mitochondria and a high enzyme activity. The type of energy metabolism that the muscle will perform in competition must also be recreated in training. There is only one way that an orienteer can maintain and improve local capacity: Train in the same way that you compete.

Oxygen uptake

Oxygen is essential to the body, whether it is resting or working. At rest oxygen is about 0.2 – 0.3 litres per minute and at maximal work level for an orienteer, oxygen uptake is about 4-4.5 litres per minute for women and 5-6 litres per minute for men. At rest oxygen is used to supply the brain, the nervous system and the internal organs, and at work it is used to create energy in the muscle cells, together with glycogen. At rest the body uses about 1 kcal per kilogram of body weight: Around 70 kcal for a man who weighs 70 kg. Energy consumption during a competition can increase to approximately 14-15 kcal per kilogram and hour, or about 1000 kcal for the same man.

0

1

2

3

Effort (km/h)

Oxygen

uptake

(l/min)

Oxygen uptake and work load increase linearly until oxygen uptake reaches the individual's maximum level (see diagram). The work level cannot be increased any more at this point and the muscles rely on anaerobic, and produce lactic acid. A high maximal oxygen uptake is one of the most important factors to be a good elite orienteer - see more information in the Profile of physiological demands. Compared with other sports orienteering has one of the highest maximal oxygen uptake value (see table). The test value for an orienteer or runner, who carries his or her body weight during activity, is often expressed as millilitres per kilogram of body weight per minute: ml/kg x min.

Rest Walking 5 km/h

Jogging 9 km/h

Running at increasing speed 12 km/h 15 km/h

VO2 Max ml/kg x min

0 10 20 30 40 50 60 70 80 90

Cross country skiing

Cycling

Orienteering

Triathlon

Running

Walking

Swimming

Kayaking

Mountain biking
Indoor hockey
Alpine skiing
Football
Table tennis
Ice hockey
Water skiing
Basketball
Wrestling
Untrained, healthy

The data in the above table comes from male Swedish national teams.

Training at the
right intensity

There are several ways to measure effort and check that you are training at the right speed or intensity for different types of endurance training. We have already mentioned percentage of maximum heart rate and perceived exertion, in your breathing or your legs. There are also other ways to measure intensity, including measuring lactate during a session or using the speed in minutes per kilometre or mile as a guide. For practical reasons we have chosen the simplest measures of effort, heart rate and perceived exertion to control intensity and give feedback during training. Perceived effort is really the only method which can both provide reliable feedback and help control training pace. Estimating training intensity with the Borg Scale

Perceived effort is actually an old method for describing training intensity. Gunnar Borg, professor in psychology at Stockholm University, developed this method nearly 30 years ago. The scale below is often used now when testing athletes, but very seldom used as a way of guiding training. The method is under-rated in endurance sport, but when you learn to use the system, it is far better than all other methods. By using the Borg Scale, or the RPE scale (Rating of Perceived Exertion) as it is actually called, it is possible to rate your effort level both for your breathing and your legs. Within orienteering the method is being used more and more and is often used by elite runners now. The feeling associated with different levels of effort is directly linked to the work you are carrying out that moment. If you measured heart rate or lactate level at the same moment you would get a result based on what was happening before. Heart rate and lactate give secondary feedback. If you increase your effort, increase speed, more lactate will be produced and this in turn means that the body must compensate with increased breathing and raised heart rate. However, these responses are delayed in comparison with the differences in effort that you can immediately feel yourself. Lactic acid measurements are usually taken from blood samples from the tip of the finger, and this is a long way from the legs which are actually producing the lactate. However, you will feel lactic acid in your legs as soon as it is produced and this means that your own perceived effort in your leg muscles is a simple and reliable indicator of how intensive your training is at that particular time. We talk about central and local effort. Local effort includes, in the case of an orienteer, the legs while central effort is linked to breathing and heart rate. Gunnar Borg developed two scales, the "lactate scale", 0-10, and the "heart rate scale", 6-20. It is simpler to use just one scale during training. Experience has shown that the pulse scale seems to be the most relevant for endurance sports, both for the legs and breathing. When you are training it is important to consider what you are measuring. When running in the forest you might rate the heaviness of your breathing at 17 while the

effort in your legs is 13. This is why we must always consider effort level for central and local effort separately.

It can be useful to have the Borg Scale with you during training, for example on the back of a map, so that you can learn how different effort levels feel and to help guide training intensity in different types of training. The Borg Scale is especially useful for holding a steady pace during a long run or running at the right speed during continuous fast, interval and lactate training. Borg's heart rate scale is based on percentage of maximum pulse. If the effort level is 9 this corresponds to a very easy effort and the pulse may be just 70. Training which is perceived as hard is rated at 15 and the pulse could be around 150. Remember that maximum pulse is very individual and can vary from 170-220. This is why it is not possible to give absolute heart rates to the different points on the scale, and we work with percentages instead. Comparing absolute heart rate with effort level works well when external factors such as undulation and terrain are constant. This is very unusual in orienteering training. Heart rate is often raised and remains at a high level during a whole run. During uphill running oxygen is needed to create energy in the muscles and when running downhill, oxygen is required to help eliminate and neutralise lactic acid in the muscle cells. Heart rate remains high throughout the whole session, while lactate levels rise and fall in relation to the effort level. See also the "Dalarna Study" in the analysis of the demands of the sport. The Borg Scale is an excellent tool to use for guiding training intensity, as it is possible to judge actual central and local effort levels immediately.

BORG-RPE

Using heart rate to get training feedback

Heart rate and lactate level are both good methods to use if we want to know what happens during training. Measuring pulse and lactate level gives excellent feedback and helps us to describe and learn the different intensity levels. Unfortunately it is relatively difficult and complicated for a coach or athlete to carry out lactate measurements in the field. If it is possible to gain access to the equipment and competence needed, lactate measurement can be a good way of describing what happens in training or competition, especially from a learning perspective. On the other hand for the orienteer heart rate and lactate are less useful tools for controlling training. Heart rate is a useful tool to teach yourself about the type of levels you work at when training and competing. If you work out what percentage of your maximum heart rate you are aiming for during training and try to subjectively judge your effort level at the same time it is easier to find the right pace. However, you should remember that heart rate is a "blunt" tool in comparison to the Borg Scale. If you are going to use heart rate measurement, you need to know your maximum pulse for reference.

Perceived effort Corresponding heart rate

6 7 Extremely easy < 60 percent of max HR 8 9 Very easy 10 60-75 percent of max HR 11 Fairly easy 12

13 Somewhat hard 75-85 percent of max HR 14

15 Hard 16 85-95 percent of max HR 17 Very hard 18 95-100 percent of max HR 19

Extremely hard 20 Maximum effort 100 percent of max HR

Planning Training

Training or competition planning?

For an elite orienteer all training must build on previous analysis – of the sport and its requirements, capacity and past competitions – and also on the coming competition program. In the first chapter, under the heading "Strategic Thinking", we pointed out that "the competition will dictate the content of the plan" and "every training session has relevance

and meaning for the coming competition.” We should start with our goals, develop a competition plan and then work on a training plan. This chapter presumes that you have carried out the first two steps. We now want to combine the different aspects of physiology, psychology, orienteering technique and other factors into a complete training program that suits the individual. Model for planning training and competition: Vision/Goals

Competition Plan

Year Plan Training Plan Training/Competition

Change

Analysis Evaluation Vision and goals

Coach and athlete develop goals based on the athlete’s own dream or vision for their orienteering. These visions and goals help to release the energy needed to train more and create the inner motivation to become better. Planning competition and training

We should plan out training so that we are able to perform well in the competitions we have identified as important goals. These important races are the basis of a well thought out plan. The competitions guide our training and not vice-versa. Consider first which competitions are important – the ones where you really want to succeed. These competitions require physical, mental and technical preparation. Examples of such competitions could be BOC, selection races or international races. At these important competitions you need to be prepared and focused on success. If you compete in a big race like this without appropriate preparation you are effectively saying that this is not an important competition for you. “Unimportant” competitions can be planned into your schedule next. By unimportant we mean competitions that you regard as preparation for the bigger races. These competitions do not require any special preparation; they are in fact a part of your training. The third step is to plan your training. Training needs to be planned on several different levels: Period plan Time of year - winter or summer, autumn or spring. Each period has a different type of training associated with it. In orienteering we talk about build-up, conversion, specific, competition and recovery periods. Week Plan Pulsed training between and within weeks. A week with seven days is perhaps too short to fit everything you want to train into. Why not make your week plan longer - 10 or 14 days? Training Session One or more sessions during the day, aerobic or anaerobic, central or local, general or specific, warm up and jog down, maintenance or developmental? There are many questions that must be answered before you can step out of the door. Carrying out competition and training

Of course, carrying out the actual session is the most important step in the model. With the help of your enthusiasm and motivation you are able to carry out what you have planned. During the actual session you try to find the right intensity and do things in the right way so that you are really training what you want to be good at. The clearer you are on what type of training you need to do, the more you think training is fun and the more stimulating your training environment is, the easier it is to train. Analysis

Competition results and testing (see Tests of Physical Capacity) can show if your plan and its execution have been successful. A training diary, competition analyses, comments and discussion with a coach are all good ways of getting feedback on success or making small changes in your future plans. Evaluation

Did you achieve all your goals this season so that you are 100 percent satisfied? The goals that you set can be both long term and short term. Winning BOC, getting into the WOC team or making the first team for Tiomila are all examples of result based goals. In comparison, running a time-trial loop 30 seconds faster, be a better forest runner especially uphill, spike 90 percent of all controls in important races or win “23-0” in the “match” against the course planner are performance goals. The difference between result and performance goals is that

you can achieve your performance goals under your own steam, whereas results goals are dependant on the competition. Even if you performed extremely well, there was perhaps someone who had a good day and performed even better. Performance goals are more important than results goals when evaluating training. Individualised training

The aim when planning your training is to develop a training program tailored for you as an individual based on the analysis you have done, and the future competitions where you want to succeed. It is very difficult for a coach to develop a training program for all the athletes he or she works with and it is also wrong to believe that he or she should do so. The job of the coach is to give the athlete advice that helps them to be their own coach. It is usually the case that the athlete knows what sort of training he or she should do, and even how it should be done, but a coach can help avoid unnecessary pitfalls. The coach can help supply the athlete with the tools and knowledge they need to plan their own training and succeed.

When a coach asks, "Why, when, where, what, how are you going to train?", he or she inspires a much more valuable process of reflection in the athlete than if he or she had just said, "This is what you need to do!" It is difficult to follow any plan 100 percent, so you must build some flexibility into your schedule. Illness and injury can crop up when you least expect it. "Think long term" is a theme running through the whole planning process that can be useful to remember. The general long-term plan is the most important aspect of your training. Detailed planning is more important within each individual session. Your weekly plan or two-week plan should contain all the aspects you want to train and how the sessions fit together. However, each period needs to have real training content. Balanced training
The weakest link in a chain determines its limit and breaking point. Orienteering training is no different. Brain and brawn must be in balance. You cannot just train aerobic fitness, run on paths the whole time, or jog round technical training. Every training period in the year has its own focus and during a whole a year all aspects must be trained and improved - basic fitness, running strength, mental skills, technique, speed and flow.

Training for development and maintenance
over several years

All competence and skill has to be developed before it can be maintained. General training comes before specialization, speed comes before endurance, technique comes before strength, and quality comes before quantity. As a junior all-round training, speed, technique and quality are the priorities. In practice this means the following for an orienteer: All-round training

Take part in several sports, train shorter, faster sessions and interval training. Training should be fun, like a game, especially for younger juniors. Coordination develops from running in the forest, work on map and compass skills so that harder courses can be tackled. Every training session is a quality session. As an older junior and young senior the athlete can begin to specialise in their training. Endurance, strength and to some extent quantity. For an orienteer this means:

Specialised training

Become more and more focused on orienteering, spending less time on other sports, run longer and longer sessions and increase training load both in terms of volume and content. This relationship between all-round training and specialised training is an idea that athlete and coach should discuss carefully. The diagram below shows how all-round training is developmental under the first part of a sports career and then later becomes a type of maintenance training. Specialised training has the opposite pattern. All-round training is also the single most important factor during the junior years for later success as a senior orienteer.

Performance Level

Training

in winter

and summer

and autumn

and spring

Specialised training

All-round training

Development and maintenance training in a training year

During winter and summer training should be similar in principal, whereas during spring and autumn when there are more competitions training should be identical. Types of training which develop fitness and technique during winter and summer are included as maintenance training in spring and autumn. Training that offers further development during spring and autumn is included as maintenance in winter and summer. The balance between different types of training during the year might look like this:

Regular training throughout the whole year It is unfortunately the case that all physical capacity must first be built up, then maintained. If you do not maintain existing fitness and strength they disappear relatively quickly. Aerobic capacity has a longer "survival period", than both aerobic capacity and strength. It is therefore pointless to work hard at developing strength during the winter, and then not bother maintaining it during the competition periods. Remember also that giving the body time to adapt to regular training helps to prevent injuries and improves performance.

Training method Winter/summer Spring/autumn

Steady running, low intensity ↗ → Steady running, moderate intensity ↗ →

Steady running, high intensity → ↗

Long intervals ↗ → Short intervals → ↗ Natural intervals → ↗ Fartlek → ↗

Hill sprints → ↗ Longer fast intervals → ↗

General strength ↗ → Specific strength → ↗

General flexibility ↗ → Specific strength → ↗

Orienteering technique → ↗ Mental training ↗ →

↗ = training for development

→ = training for maintenance

Orienteering technique and mental skills are totally different in this respect. If you have laid down the foundations for good technique, self-confidence and the right attitude then these qualities remain stored in your subconscious. Even if you have a long break from competitive orienteering it often only takes a few races to shake off your rustiness. It is important to remember that mental skills and your orienteering technical abilities are closely linked to each other and strengthen, or weaken, each other. Increasing training

If the training load is very small the effect can be negligible. On the other hand, injuries and overtraining can be the result of a training load that is too high. There is, for each individual and situation, an upper and a lower limit for training load and its effect. Both these limits move successively upwards with increased training. Increased training load is a requirement if you want to continue improving your performance capability. Training load can of course be increased with small or large steps, but it is always wise to consider previous training and not increase too quickly or dramatically. Many juniors dramatically increase their training, by 50-100 percent, in the later junior years as they become inspired by the chance to run at JWOC or make the national team. Girls especially often get problems with knee and hip injury

es due to the suddenly increased training load. The most important tool for planning a sensible increase in training load over several years is a training diary. The risk for injuries and overtraining is significant with increases in training of more than 25-30 percent per year for juniors, or 15-20 percent for seniors. Training load or training volume during a given period is the result of the duration of sessions, intensity and the number of sessions carried out. Training + recovery + food = training effect

Recovery or rest and food are really the most important part of a training session. All physical training breaks the body down, whereas food and rest build the body up again. If the relationship between training and recovery is optimal, then the training will have a positive effect on the body. The body adapts to the increased training load. This process is called super-compensation:

Training load

Recovery

Compensation

Time

Performance capability

The compensation phase can take different amounts of time, anything from a couple of days to a week, depending on the amount and type of training. The shortest recovery periods in your training are the rests between efforts in short interval training while the longest can be a whole week of easier training after a hard week. Recovery starts immediately after training with a warm down jog and refuelling with fast carbohydrates and fluid – see Nutrition. Next comes a shower and a proper meal, such as lunch or dinner. These steps are essential if your body is to benefit from the training you have carried out. The routine you have directly after training and competition can dramatically affect recovery time. The table below shows how different types of training and intensity can affect recovery in the muscles active during training:

Type of training Aerobic Aerobic/ Anaerobic Anaerobic Strength

Intensity Low (60-75 percent of max HR)

Moderate (75-85 percent of max HR)

High (85-95 percent of max HR)

Maximal (95-100 percent of max HR)

Specific strength training

Recovery level Minimal recovery Immediate approx. 1 hr 1-2 hr approx. 2 hr 2-3 hr Near full recovery 2-4 hr 6-12 hr approx. 12 hr 12-36 hr 12-36 hr Full recovery 6-36 hr 12-48 hr 24-72 hr 24-96 hr 24-96 hr Full recovery means that the muscle groups which have been worked are 100% refuelled with glycogen, fluid balance is restored, protein structures are repaired, hormone and enzyme activity is normal and vitamin and mineral levels are correct. This process takes between 1-4 days depending on the type of training, how hard it was and how well trained the athlete is. With near full recovery, 90-95 percent, performance capability is good again and a well-trained orienteer can train in this condition and derive benefit from the training. But after about a week, complete recovery should be allowed to take place, to minimise the risk for overuse and overtraining.

Time

Performance capability

1 Performance curve without refuelling 2 Refuelling straight after training 3 Optimal diet and routine during whole recovery period

2

3

Pulsed training volume and intensity

Physical training and recovery, rest, work together to give a good training effect. Training can vary from week to week, from day to day, or from session to session. A normal pattern is 1:3 – that is an easy week, moderate week, hard week, easy week again and so on. Another is a 1:2 pattern, with alternate easy and hard weeks. During an easy week recovery is the most important aim, both physically and mentally. These weeks can include different sports to help recovery – see Cross Training. An easy week might contain more aerobic training and general strength training. A moderate week may contain more continuous fast running and some anaerobic training. A hard week should probably be mostly aerobic training, long aerobic sessions and long intervals, with little anaerobic training. In that example both the moderate and hard weeks are actually quite hard, but emphasise different types of training. The moderate week has more hard anaerobic training, while the hard week is mainly aerobic. Easy and hard weeks during the basic training period might look like the plan below. The alternative sports could be altered depending on local conditions, for example mountain biking instead of skiing if you live somewhere without snow.

Easy week	Hard week
Session 1	Session 2
Steady run, easy	Circuits
Circuits	Steady run, easy

Tuesday Long intervals Long intervals General strength

Wednesday MTB + swim, moderate Ski + run + swim, moderate

Thursday Indoor intervals General strength Indoor intervals General strength

Friday MTB + swimming, low

Saturday Long run, moderate Light strength – core stability

Long intervals Ski, moderate

Sunday Long ski, moderate Long ski, mod Ball game

Differences between men and women

The cycles of our biological clock are of course, much more significant and noticeable for women than they are for men. This cycle is around 28 days for women. Many coaches working with women in other sports have begun to consider the effect of the menstrual cycle on training patterns. Some recommend hard training, long aerobic sessions, hard strength training and anaerobic training after menstruation. Between ovulation and menstruation training should be lighter with lower quantity, more aerobic interval training, and less anaerobic work. Research in this area is still in its infancy, but studies so far show that there is a link between training effect and the menstrual cycle. Listen to your body

Your training plan is a guide which helps you direct your training, but your form and how you feel on the day are the most important factors for how and when you train. One way of guiding your training is to let your own biological clock decide when you feel in good shape and train hard then. At other times you feel tired or heavy and easier training is more appropriate, letting your feeling and form guide the pace. Sometimes rest feels like a must and then you should definitely take a rest day, with a good conscience. Signals from your body must always be allowed to decide the content of your training, not a written plan with theoretical calculations on how your training should develop. Listen to your body!

Periodisation

How you divide the year up into different training periods depends partly on where you live, and partly what goals you have. Here is an example of a young senior who plans to compete in the British domestic season in the spring and early summer, then hopefully gain selection to a late international such as Euromeeting. A 6/11-19/11 Recovery period 1 The season is over and new competition goals feel a long way off. Mind and body need a contrast to the

previous weeks of racing and training. This recovery period allows rest after earlier training and races and creates new energy for the coming winter training. The body needs a chance to recharge its batteries: During this short period it is best to train in ways that you enjoy, or through other sports. Training volume should be considerably less, although keeping a little bit of quality training in your schedule will help you maintain earlier form. Training in itself should not be your main focus - these weeks are more about enjoying life and doing something different. It is also a time to reflect and evaluate the past year and to begin planning new goals for next year.

B 19/11-18/3 Basic training period 1 Improve central capacity and local aerobic capacity. Cross training in other sports helps to maintain a higher training volume without overloading joints and muscles too much – mountain biking, running in water and swimming can all contribute to training. There is however a core of running, which includes intervals and longer runs, often in terrain. Strength is built up through general strength training, circuits and resistance running, such as hill running or bounding. Work at your core stability and general flexibility, if necessary working on specific strength or flexibility in problem areas. Do not just train long slow sessions. Interval training, fartlek and perhaps a few fell or cross-country races help to vary training, maintain your speed and give a better training benefit than endless plodding. Orienteering technique training should not be forgotten either. Now is a good time to analyse strengths and weaknesses from previous races and work on the basics. Any competitions should be regarded as training and you should focus on doing things in the right way, not the final result. Improve your mental skills, work on your attitude and the psychological techniques that you can use later in stressful situations to improve concentration. Visualisation and goal setting help to create the energy needed to carry out the tough winter training in this period.

C 16/3-15/4 Conversion period Train in much the same way as during the basic period, but begin to focus your training more on the coming competitions. For example, gradually introduce more sessions in the forest and terrain relevant to coming races. More of your training percentage wise may be interval training and faster running. Technique sessions may be carried out at competition speed; you begin to put the separate techniques together and try to develop a good flow and rhythm in your orienteering. Early races such as UK Cup races provide good opportunities for practice. You may include more resistance running such as hill intervals in your training. You do less alternative training now – training volume is not so important and you are becoming better adapted to hard running training in the terrain.

D 16/4-31/5 Competition period 1 Training volume reduces significantly and interval training, fartlek and pace running in the forest accounts for a significant part of training. This training mainly improves local capacity, both aerobic and anaerobic. Few long steady runs are needed - competitions in themselves may be quite long. Specific strength comes from running in the forest at competition speed, but a few sessions to maintain general strength is time well spent. Technique training between competitions is best carried out as short courses, control picking and similar. Its aim is to strengthen the right behaviour in the races and help you orienteer cleanly and confidently and at high speed. Mental training is important in this period to prepare for coming races. Look out old maps of competition areas. There may be 1-2 competitions per weekend in this period, including important domestic races like the JK, British Champs and Scottish Champs. International races in this period include Tiomila.

E 1/6-15/6 Recovery period 2 Shorter steady runs, and cross training such as mountain biking, general strength and other sports. No special focus on orienteering training. Evaluate the spring races. What went well? What can you do better? In what way and with whose help?

F 16/6-31/7 Basic training period 2 Building up central capacity again - long runs in the fells or in the forest, mountain biking and running in the water or swimming again. Training also includes intervals and

fartlek, general strength and flexibility work. Improve orienteering technique, going through competition analyses from spring races. Work on psychological skills again. Improve stress tolerance through fast orienteering intervals sessions, chasing tart training and similar. Training camp with friends or team in a new environment – perhaps a camp abroad for an important future international competition. G 1/8-15/8 Specific training period Training for a physical, technical and mental peak. Learning how to perform on a chosen occasion in this case a B-international in Sweden, after good performance at WOC selection races. Taper down training volume so that quality dominates – hill sprints, intervals, running at competition speed on technical courses. Time trial loop at full speed confirms good shape. Use competitions to develop the flow and rhythm in your orienteering you need for good performance. H 16/8-30/9 Competition period 2 Your first senior international race is getting closer. Study any maps of relevant terrain that you can find. Your training is very similar to the previous competition period and you continue working towards peak form around the competition days. Quantity is reduced and quality is predominant, with only occasional maintenance sessions for general strength. Try to train in more technical terrain to prepare you for the coming race. Short orienteering courses at race speed help to sharpen technique. I 1/10-15/10 Recovery period 3 Last recovery period before Euromeeeting. Recharging mental energies after summer racing, and resting for the last intensive sessions before Eu omeeeting. Shorter runs, mountain biking, and general strength training. A little break from rienteering technique – analyse instead what went well and what can be better after the summer races. How can you improve in your last races this year? J 15/10-4/11 Competition period 3

Training is now totally focused on Euromeeeting. The race is in Switzerland where the World Champs will take place in two years. Quality training s predominant, and you therefore cut down on long runs. Everything is focused on increasing your speed in the forest and increasing local aerobic and anaerobic local capacity. Most training is short and intensive. Technique sessions are aimed at making you technique sharp: Short, fast courses and control picking at high speed.

Sketch of a training year

The diagram below shows how training volume and intensity can vary during a year in relation to desired form:

Year planner

This competition and training plan is a suggestion for how part of your schedule for a year might look. The training plan is guided by the competitions where you want to get good results. In this case the athlete wants to have a good spring season and show selectors that she is a likely candidate for the national team. She plans to get selected for the Euromeeeting in November, which takes place in the same country as the next World Champs.

Week plan

A training diary is an essential tool for planning, carrying out and evaluating training. The following extract comes from a week in “Competition Period 2”, the week before the main race.

Intensity

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ec h
R P
S -p
hy si
ca l
R P
S -t
ec hn
ic al

Comment

M 24 45 Run 3.5 30 75 Jogging with Tobbe, core stability. Tu 25 15 25 Interval 6.0 40 8.5 9.0
50 x 15-15 intervals in fast terrain. W 26 45 Run 4.0 Steady running in forest. Th 27 45
O-tech 5.6 15 60 45 Test course in relevant terrain. Strength. F 28 Travelled to SM, cinema
in evening. Sa 29 15 65 O-race 6.0 80 65 9.0 9.0 Qualification, no problems. Felt great. Su
30 15 90 O-race 6.0 115 90 9.0 9.5 Perfect run – really pleased! 3rd

Total 135 225 45 360 200

Planned training

75 Easy running, core stability.

45 Short intervals in terrain.

45 Easy running.

60 Fartlek with map, core stability. Rest 50 SM Qualifier

Felt good right from the beginning of the week. Really motivated for the weekend – harmony
in other parts of life. Totally involved in the orienteering, concentrating on running a clean
race.

90 SM Classic

Summary

We can summarise training planning in the following way:

- Competition goals determine training content
- Training should be

-individualised -aimed at development or maintenance -regular -progressive -pulsed volume and intensity -planned

- Training effect is the sum of training, recovery and food.

INJURIES

Injuries in orienteering

During the 1980s there was a change in philosophy in the treatment of injuries in the Swedish orienteering team. Earlier doctors had worked with injuries after they happened. Their job was to “fix” the problem. However, the medical team began to change this policy under the enthusiastic leadership of team doctor Christer Rolf. The new way of working focused on educating the athletes how to treat simple injuries that occur during training and competition themselves. For example each athlete was encouraged to carry an elastic bandage in their pocket so that they could treat a twisted ankle in the forest immediately. The doctor’s job became educating athletes in how they could prevent and minimise injuries, giving general health and medical advice, and treating acute injuries that arose during camps and competition trips. The Swedish policy was based on the fact that the most common injuries in orienteering are minor injuries. These injuries can usually be treated by the athlete. Blisters, bruised nails, cuts, skin irritation from taping, chapped skin, warts and athlete’s foot are some of the commonest minor injuries which can hinder training or stop an orienteer performing optimally during competition. Every orienteer should have a first aid bag containing medical equipment to treat these problems. Such a bag could contain, tape, plasters, elastic bandages, blister pads, foam pads, scissors, nail clippers, and a foot file. Other common injuries in orienteering are overuse and acute injuries. Around 80 percent of all injuries occur below the knee.

Reasons for injuries

Every new activity requires time for adaption. The different structures in the body adapt at different rates. Muscle strength and aerobic fitness is relatively easy to build up, while resistance to impact and loading in bones, connective tissues and tendons takes considerably longer to develop. It is always the weakest link that will develop problems when stressed. Most overuse injuries are due to lack of variation in training, increased pace or effort, high impact, poor warming up, poor recovery, change of training content or training environment. Change in training environment could be from road to forest, home training to training camp, cycling to running, or when using new footwear or other equipment. The diagram below shows risk periods for orienteers during a training year. A Change of surface, climate and pace on a training camp in Spain. Sudden increase in both volume and intensity. Risk for overuse injuries and colds. B Changeover from winter training to more running in the terrain and faster paced running as the spring season starts. Risk for overuse injuries. C Training camp during spring where you train twice a day, compete in several competitions and stay in a group in a cottage. Risk for overuse injuries and colds. D Beginning of the main competition season. Not used to running fast, and change to competition shoes. Risk for overuse injuries, blisters, bruised nails. E The end of the competition season. Tired and worn down. Risk for overuse

injuries and colds. F Changeover to basic training again, longer runs, often on the roads due to the

dark. Risk for overuse injuries. G Heavy training during training camps. Risk for overuse injuries. I Whole competition season: Risk for acute injuries, twists and knocks, especially to the lower leg.

Overuse injuries

A basic difference between overuse and acute injuries is that the symptoms of an overuse injury are not necessarily located in the same place as the cause of the injury. If you run into a stone you will get a bruise on your leg where you hit the stone. If on the other hand you have a wart under your right foot and cannot take your full weight on your right leg, you may end up with a sore left knee because you have run out of balance. The overuse injury that you have developed in your left knee has arisen because of the wart on your right foot. The knee injury may improve if you treat the wart on your left foot. It is important to try and work out such connections when overuse injuries arise. A training diary is a useful tool here, containing a record of training and also comments on how well recovered you feel or special notes about things like the wart in the above example. Overuse injuries in orienteering are most often seen in the knee and below. Common overuse injuries are Achilles tendonitis, shin splints, and ITB tightness leading to runner's knee problems.

Listen to your body

Most overuse injuries develop gradually and it is easy to ignore the first signals. When you have had pain for some time, and come to the stage where you are unable to train the symptoms have often become more diffuse. It is easy to forget how the problem started, identify the probable cause and find a solution to the problem. It is important to always be alert and listen to your body. The definition of an overuse injury is locally reduced function, often combined with a local inflammation and pain, which often arises due to a new or unaccustomed movement, increased load or too little variation in training. Inflammation is the body's signal that something is wrong. There will be pain when you try to work the tendon or joint, the skin may be reddish and the injured area may feel warm. It is important that you learn to recognise and react to these signals. Continued training can both make the injury worse and give rise to another injury as you will work your muscles and joints in a different way due to the tenderness.

There is often a simple and logical explanation to why the inflammation arose. Perhaps you have started using new shoes which are not worn in yet – having two pairs of shoes for training and competition in use at a time is sensible. Perhaps you have not bothered with a proper warm-up and got a muscle strain or forgotten to warm down after a race and been stiff and sore the day after. Maybe you have not slept or eaten properly in the days leading up to the race, especially at races like Tiomila or Jukola, and you have twisted a foot on a slippery stone due to pure tiredness or lack of coordination. You may also catch a cold in the days after a tiring competition like this if you are not careful with diet and sleep. Overuse injury

Inflammation Symptom: Heat Redness Swelling Stiffness Pain The diagram shows common signs of inflammation. Inflammation is the body's reaction to overuse. Treatment: Stop training at the first signs, consider the reasons, treat the problem, and carry out alternative training until the injury is better. Alternative training means any training that does not stress the injured area.

Infections

Another common cause of enforced training breaks is infections. Most colds are relatively minor and will get better within a week. It is, of course, not as easy to prevent an infection as an injury. But there are many small points worth remembering, that can help reduce the risk

for complications. Wearing the right clothes for the weather conditions, keeping your head and feet warm, changing into dry clothes quickly, and drinking and eating fast carbohydrates straight after training are simple measures which we often forget. Infections lead to an imbalance in the body's immune system, when bacteria, a virus, or other foreign bodies invade the body. The infection places the whole body in a type of overuse condition.

Never train or compete when you have an infection

The risk for serious complications, such as inflammation of the heart muscle, is far too serious to take a chance with. Signs of infection include raised pulse, raised temperature, and local symptoms such as a sore throat, runny nose or headache. To know if your pulse or temperature is higher than normal, you need to measure your

normal resting pulse and temperature several times when healthy, so that you have a

reference value. Virus, bacteria, Micro-organisms Infektion Symptom: Raised resting HR

Raised temperature Sore throat Runny nose Headache The diagram shows signs of

infection. Infection is caused by external factors such as viruses and bacteria. Treatment:

Stop all training, rest, drink plenty of water and eat so that the body can begin to fight the

infection. If you have a cold for longer than a week you should visit a doctor. Your immune system has not been able to cope in this case, and the infection has been able to take hold.

It may be that you require antibiotics, if the infection is bacterial. It is important to remember that your body needs a relatively long time to recover after you have had an infection before you can train fully again. Reckon with at least as long after the infection has gone as the symptoms themselves actually lasted.

Acute injuries

The most common acute injury in orienteering is a twisted ankle. The injury can happen through pure bad luck, but often it is to do with tiredness, lack of coordination, not being used to running in the forest, stiffness or poor general flexibility. The immediate treatment of acute injuries is very important and can dramatically affect the time that it takes to come over the injury. If you can treat a twisted ankle effectively straight away, it is often possible to start training again a few days after the injury. If not such an injury can easily lead to an enforced break as long as two to three weeks. A twisted ankle usually leads to injury on the outside of the foot. At the moment that the foot is twisted, small blood vessels in the area are torn. The bleeding which follows gives rise to a lot of the problems that follow. The coagulated blood encourages more fluid from the blood vessels out into the injured tissue. The result is a swelling, which can increase in the days immediately after the injury. This swelling increases the pressure on the structures in the foot. The ankle becomes stiff and is sore when you try to move the joint, or bear your full weight. After a few days, the ankle may also be visibly bruised and discoloured due to the breakdown of blood. If you can limit the initial bleeding in the acute stage of the injury you can significantly shorten the healing process, by reducing the subsequent swelling and associated problems.

Measures to take after acute injuries:

- Stop all training and competition.
- Use the elastic bandage which you should carry with you during all training

sessions. Minimise bleeding by compressing the injured area with the bandage. Bandage the injury from below (see picture).

- Elevate the injured area to minimise swelling.
- Keep the compression bandage on for about 30 minutes. Remove bandage and to

ensure adequate blood circulation. Repeat 2-3 times.

- Cool the injured area to reduce pain. Cooling does not affect bleeding, but helps to

minimise pain.

- Use an elastic bandage such as a tubigrip in the days following the injury.
- After a few days it is possible to begin working the ankle gently again, through

alternative training and possible with a warming bandage, such as neoprene support. This gentle activity encourages the body to transport away waste products from the injured area, and warmth can help this process.

- After a few days it should be possible to run again, without the ankle swelling up. Taping for support can be very useful to prevent re-injury.

Alternative training

When injured it is often possible to train in other ways. Training which does not cause pain, and does not strain the injured area can be carried out. Swimming, running in water, cycling and spinning, strength training, walking or running on soft surfaces are all good examples of alternative training. The list below gives several good ideas for alternative training sessions. Many runners have had an injury, but then actually become better and stronger by being forced to train in different ways to traditional running training. Examples of sessions:

- Running intervals in water.
- Continuous fast cycling.
- Running intervals in a soft marsh.
- Mountainbiking in hilly terrain to give natural intervals.
- Strength training and circuit training.
- Walking in terrain, with a high knee-lift.
- Aerobics with hops and jumps on a crash mat. Prevention of injuries It is possible to avoid almost all injuries if you train "smart" and apply common sense. The advice below can help to avoid injuries.
- Remember that new training methods require time for adaption.
- Plan your training for the long term.
- Vary your training.

Stop training - Compression - Elevation - Cool - Rest - Alternative training - Warmth

- Be careful when making changes to your training.
- Eat and drink enough.
- Increase training slowly.
- Always warm up and jog down.
- Work on equal strength and flexibility in your left and right legs.
- Listen to your body and the warning signals it gives you.
- Rest adequately between sessions.
- Include alternative training when you are injured.
- Never train when suffering from an infection.

Chapter 8 – Physiological testing

In this chapter we describe the physiological tests that have been developed and used with BOF teams. The tests have been developed with the following aims: 1. Provide information on which types of

aerobic training should be prioritised. 2. Give a basis for optimising training with respect to volume and intensity. 3. Investigate where improvements due to training have taken place. Maximal aerobic capacity ($\text{VO}_{2\text{max}}$) is mainly a product of the heart ability to supply the musculature with oxygenated blood. This capacity is often referred to as a central capacity. $\text{VO}_{2\text{max}}$ is tested on a treadmill with steadily increasing speed and gradient during 6-8 minutes. Lung ventilation and the amount of oxygen taken up is measured every 10 seconds during the test. These values give the volume (V) of oxygen (O_2) taken up per minute (VO_2). When VO_2 no longer increases despite increased gradient and/or speed $\text{VO}_{2\text{max}}$ has been reached (Fig 8). The validity of $\text{VO}_{2\text{max}}$ as a single variable for explanation of running performance for orienteering is however low.

Fig 8. Increase in VO_2 with time on a treadmill for two individuals with different $\text{VO}_{2\text{max}}$ values. Individual number 1 reached a max of 74.9 ml/kg x min while individual number 2 reached a max of 80.2 ml/kg x min, which meant that individual 2 could keep running at a higher power output and kept going for approx 70 seconds longer.

During the BOF project the relationship between $\text{VO}_{2\text{max}}$ and average running speed on a marked test circuit was established in three different terrain types and for three different courses. In fig 9 the result from a representative standard course in Portugal is shown. The result shows that there is only a weak relationship and, consequently, a low explanation (r^2) for the $\text{VO}_{2\text{max}}$ on the mean running competition speed. This means that in this group with runners having relatively high $\text{VO}_{2\text{max}}$ values, $\text{VO}_{2\text{max}}$ can only explain about 10% of the differences in running performance between the male runners and about 27% of the differences for the female runners. (Fig 9).

Fig 9. The relationship between $\text{VO}_{2\text{max}}$ and average running speed during a simulated competition on a marked course for 15 runners. Red dots denotes female runners and blue dots male runners. Running performance with the two groups was therefore respectively 90% and 78% unexplained when $\text{VO}_{2\text{max}}$ was used as a single explanation variable (Fig 10). Expressed in running speed the $\text{VO}_{2\text{max}}$ test could only explain the forest running performance with an average error of ± 44 sec/km. The validity of the $\text{VO}_{2\text{max}}$ can, consequently, be concluded to be far too low to use as a single variable for training evaluation and training prescription in elite athletes.

Fig 10. Description of the level of explanation that $\text{VO}_{2\text{max}}$ gives for running performance during a simulated competition on a marked course in terrain. Aerobic running economy/effectiveness, describes the ability of the muscles to use oxygen in an economical way. This capacity is often referred to as a local capacity (3). Running economy is tested during an incremental speed test with 4-5 stages each of 8 minutes. The speed within each stage is constant. Individuals with poor running economy use more oxygen at a given running speed than individuals with good running economy (Fig 11).

Fig 11. The difference in VO_2 consumption at the same running speed for two subjects. Subject no:1 (blue) showed poor running economy, requiring a VO_2 of about 70 ml/kgxmin at 3:30 min/km speed compared with subject no:2 (red) who showed good running economy and required a VO_2 of 62 ml/kgxmin. The difference in running economy meant that subject no:2 could run about 15 seconds faster per km at the same VO_2 level.

Running economy can be tested on treadmill or in the terrain. The treadmill running test gave, independent of sex, a similar level of explanation as VO_2max gave for running performance in the terrain (Fig 12).

Fig 12. The relationship between running economy on treadmill at 4 min/km pace and average running speed during a simulated competition run on a marked terrain course for 15 runners. An interesting secondary discovery was that the women tested appear to have poorer running economy than the men at 4 min/km pace (Fig 12). This difference seems to be due to the use of too high test speed rather than to a true gender difference because the 4 min/km testing pace resulted in an accumulation of lactate for the women, thus causing a higher oxygen consumption, which was not the case for the men. In addition the relationship suggests that the fastest runners in terrain also showed the best developed running economy. When recalculated as sec/km, treadmill running economy could only be used to predict running performance with an error margin of ± 34 sec/km. Despite this being an improvement of ± 10 sec/km compared with the VO_2max test, it is still too imprecise to be used for intensity settings at elite level (Fig 13).

Fig 13. The relationship between running economy in terrain at 5 min/km pace and average running speed during a simulated competition run on a marked terrain course for 15 runners. However, in terrain a different pattern emerged. It appeared that the women tested had better running economy in the terrain than the men, in contrast to the results from the treadmill. In addition it appears that the fastest runners in terrain also have the poorest running economy in terrain, contradicting the results from the treadmill. (Fig 13). When recalculated in sec/km, the running economy test in terrain predicted running performance with a ± 30 sec/km error margin. This result suggests that there is no correlation between running economy in terrain and treadmill running economy. In addition the correlation is too weak to justify using running economy on its own to plan training, as about 75% of the differences in running performance cannot be explained by differences in running economy regardless of how running economy is measured (Fig 14). When comparing treadmill running economy with running economy in terrain it was not possible to demonstrate any relationship at all ($r^2=0.08$) for the male runners, whilst a moderate relationship ($r^2=0.77$) existed for the women. (Fig 15).

Fig 14. Description of the level of explanation that running economy in terrain gives for running performance during a simulated competition on a marked course in terrain. Comparison of running economy in terrain and on the treadmill again indicates a gender difference.

Fig 15. The relationship between running economy in terrain at 5 min/km pace and treadmill running economy at 4 min/km pace for 15 runners. While there is no correlation at all for the male runners, the values for female runners appear to show a correlation between good running economy in the terrain and poor running economy on the treadmill. The group is however too small to draw any definitive conclusions from.

From the training perspective these results show that running economy tests should be carried out in terrain and not on the treadmill as treadmill tests can lead to incorrect conclusions about good running economy, which is in reality poor in the terrain and vice versa. If we instead investigate the relationship between VO_2max and treadmill running economy in combination, that is, the running speed reached at VO_2max ($v\text{VO}_2\text{max}$) and running performance the degree of explanation doubled from 20-25% to 40-45% (Fig 16).

Fig 16. The relationship between treadmill running speed at VO_2max and average running speed during a simulated competition run on a marked course for 15 runners. The increase in the degree of explanation meant that the margin of error for prediction was more than

halved, from ± 30 sec/km to ± 13 sec/km. Despite this, the margin of error was still too large for the test to be used for intensity settings on elite level. Finally, if the running economy value from the terrain test was used in combination with the VO₂max value the degree of explanation nearly doubled, from 40%-45%, to 74%-85%, and the error margin reduced to ± 9 sec/km (Fig 17 and 18).

Fig 17. The relationship between running speed in the terrain at the VO₂max and average running speed during a simulated competition run on a marked course for 15 runners.

Fig 18. Description of the level of explanation that the velocity at VO₂max in terrain gives for running performance during a simulated competition on a marked course in terrain. An interesting and unexpected secondary discovery was the significant relationship between treadmill running economy and VO₂max (Fig 19 and 20).

Fig 19. The relationship between treadmill running economy and VO₂max for 15 male seniors and juniors. These comparisons are based on the runners who were tested at the camp in Portugal and also on runners who only were tested on treadmill in the laboratory.

Fig 20. The relationship between treadmill running economy and VO₂max for 15 female seniors and juniors. A further analysis of those runners who had taken part in camp in Portugal showed an even higher significant relationship between running economy in the forest and VO₂max compared with treadmill running. The values for the men showed a degree of explanation as high as 90% for running economy as and explanation for differences in VO₂max (Fig 21).

Fig 21. The relationship between running economy in terrain and VO₂max for 9 male runners. The corresponding figure for the women was 85% (Fig 22).

Fig 22. The relationship between running economy in terrain and VO₂max for 6 female runners. This relationship suggests that VO₂max does not only reflect central capacity and general fitness level. It also suggests that there is a strong connection between how efficient oxygen is used in the muscles and the adaptation of the heart to this level of effectiveness. This can in the worst case mean that the effect of a high VO₂max is cancelled and maximum running speed is the same, or even lower, compared with an athlete who has a low VO₂max in combination with a good running economy (Fig 23).

Fig 23. Differences in maximal running speed for two individuals with different VO₂max and different running economies. Individual 1 could run 6 sec faster per km compared with individual 2 because of his better running economy, despite the fact that individual 2 had a significantly higher VO₂max. Further evidence for this is that if running economy is improved by specific running economy training then VO₂max might fall and stabilise at a level that meets the oxygen requirement at the new level of muscle effectiveness. This has been demonstrated in three cases during one training year where the main aim of winter training has been to improve running economy. In these cases VO₂max reduced by an average of 1.8 ml/kgxmin during the year while running economy improved by about 3.2 ml/kgxmin at the same running speed (Fig 24).

Fig 24. The average net effect of an improved running economy on treadmill on the theoretical VO₂max speed for two occasions – before and after one years running economy training for three runners.

The theoretical net effect of improved running economy with lower VO₂max was in the above case increased running speed during the treadmill test, with an average of 5 sec/km for the three runners. In reality the average improvement was as much as 14 sec/km in terrain, as running economy in terrain improved by 6.6 ml/kgxmin at the same oxygen consumption level as the treadmill test (Fig 25).

Fig 25. The average net effect of an improved running economy in the terrain on the theoretical VO₂max speed on two occasions – before and after one years running economy training for three runners. The difference in improvement between the test on the treadmill and in the terrain again shows the importance of testing the individual in the right training and competition environment. Measurement of running economy must therefore take place in terrain and in combination with measurement of VO₂max in the lab to give a valid basis for evaluation of changes in both running economy and VO₂max. The validity for running economy in combination with VO₂max, with a method error ± 9 sec/km, is high enough to be a suitable method for evaluation of training. However, this error is still too large for the method to be used as a guide for training focus, volume and intensity. We have shown that muscle endurance gives further information as far as validity and carrying out training in practice is concerned. The last muscle capacity which we have measured during the project is covered in the following section.

3. Aerobic endurance: This describes the capacity of the muscles to work as near to VO₂max as possible for a period of at least 30 min, without relying substantially on anaerobic energy systems (4). The degree of endurance is limited by; 1. fiber composition between slow (type 1) and fast (type 2a and 2b) twitch fibers, the greater the proportion of slow type 1 fibers the higher endurance capacity will be developed (4), 2. The development of the slow twitch fibers, which is dependent on their capillary density and mitochondria mass (5), 3. The energy effectiveness of the fat liberation which depresses the formation of lactate production (6), and, 4. The elimination of produced lactic acid out of the muscle (7) and the oxidation of produced lactate in heart and other muscle tissues (8). If running speed is too high, certain muscle cells will form so much lactic acid that the pH falls. Lactate will be formed and is transported out of the cell to the surrounding tissue. When pH falls a reduction of the muscle contraction power and co-ordination occurs, and the individual will experience increasing tiredness. Eventually the individual must reduce running speed to a level where lactate production rate does not exceed the elimination capacity rate and instead reaches a maximal steady state level (MaxLass) at which the subject can work for at least 30 min (10). MaxLass is defined as the highest possible intensity which can be maintained during at least 30 min at the same time as blood lactate can be held at a steady state level (Fig 26).

Fig 26. The difference in blood lactate concentration for five different test runs with different running speeds for one runner. Maximal lactate steady state pace was determined in this case to be 3:30 min/km which corresponded to a MaxLass intensity of 82.7% of VO₂max at a lactate concentration of 3 mmol/L.

The most accurate method for determination of MaxLass is for the individual to carry out 4-5 test runs each of 50 min on a treadmill at different steady speeds. The lactate concentration is measured at regular intervals and a steady state concentration is determined for different running speeds (Fig 26) (10). When a steady state concentration is no longer present the test is interrupted and the MaxLass can be determined (10). However, this method, with five days of testing, is very expensive and impractical for the individual. A number of simplified prediction tests for MaxLass have been developed. One of the best known and most often used is the so called Onset of Blood Lactate Accumulation (OBLA) test. This test is also sometimes referred to as an Anaerobic Threshold test (AT). The test is carried out once with a steadily increasing running pace and based on an assumed level for lactate accumulation of 4.0 mmol/L regardless of the individual's level of training and muscle fiber composition (Fig 27).

Fig 27. The effect of increased running speed on the blood lactate concentration for one runner during a OBLA or AT test on treadmill. The vOBLA/AT denotes the critical running

speed at an assumed concentration accumulation of 4.0 mmol/l. These tests only give a reliable result at a so called group level (10). This means that only the average value for a group of individuals will be reliable, because of the large individual differences (2-7 mmol/L) in concentration from 4.0 mmol/L which can be observed (10).

The explanation degree for OBLA/AT for running performance was 67% for the men and 62% for the women, and the error margin increased to ± 14 sec/km which is worse compared to the VO₂max in combination with the running economy test. Therefore the OBLA test is not precise enough to be able to use for evaluation and planning of training at elite level (Fig 28).

Fig 28. Relationship between treadmill running speed at OBLA and average running speed during a simulated competition run on a marked course for 15 runners. The relationship that appears in our results is backed up by results from earlier publications covering endurance work for runners who have carried out several year's endurance training (12). The treadmill lactate test in combination with an assumed MaxLass value of 4 mmol/L does not actually give better validity for predicting endurance capacity compared with running economy test in forest in combination with VO₂max. A new treadmill test, the Fox-Lass test, was therefore developed. The definition of MaxLass in the new Fox-Lass test is; the exercise intensity at which the lactate concentration is 0.7 mmol/L higher than the value after 4 min of exercise on the same intensity (fig 29).

Fig 29. Differences in lactate concentration after 4 and 8 mins running at different running speeds for one runner during the new Fox-Lass test. When the Fox-Lass intensity was used instead of the assumed OBLA value, the level of explanation for running performance increased from 62% to 87% for the men and from 67% to 92% for the women (fig 30).

Fig 30. Relationship between treadmill MaxLass pace and average running speed during a simulated competition run on a marked course for 15 runners. Moreover, the error margin fell from ± 9 sec/km to ± 6 sec/km which is sufficient to be able to use the test results for evaluation and planning of training because most runners can't set their training intensity in forest with higher precision in running speed than within 6 sec/km.

However, in its current form, the treadmill Fox-Lass test overestimated running speed by an average of approximately 1min/km for both groups. Currently athletes run on the flat at increasing speed, without any fixed or increasing gradient. This overestimation corresponded to a running speed of 3:50 min/km on MaxLass average for treadmill running, whereas testing in terrain showed the correct MaxLass speed to be 4:55 min/km. Moreover, the relative endurance intensity level tested on treadmill (Fox-Lass%VO₂max) was underestimated by an average of 4%, 80.2% vs 84.2%, regardless of sex, compared with the correct average endurance level observed in the terrain test (Fig 31).

Fig 31. The relationship between relative Fox-Lass (%VO₂max) intensity on treadmill and mean relative intensity (%VO₂/VO₂max) during a simulated competition run on a marked course for 15 runners. The difference in running speed for achievement of Fox-Lass on the treadmill, with very little vertical exercise, indicates a higher recruitment degree of fast type 2 muscle fibres compared to when running in terrain on the same level of VO₂ consumption. Such a recruitment will result in a higher production of lactate at a lower oxygen consumption level compared to when only type 1 muscle fibers are used for the exercise. This theory can be demonstrated by a reduction in lactate concentration of 70% when using a slower running speed of 1 min/km in combination with an inclination angle of 3-4° when running on the treadmill compared to when running on a horizontal level for the same relative oxygen consumption level (Fig 32).

Fig 32. The difference in lactate response to different inclination protocols and to horizontal and forest running at the same relative level of VO₂ consumption for one runner on four different occasions. A similar result was seen for two other cases which strongly indicates that future tests on the treadmill should be carried out with an inclination angle between 3-4° in combination with a 1 min/km slower running speed of compared to the running speed used for horizontal tests. When comparing the average running speed during the forest test with the Fox-Lass predicted running speed during the same forest test (vFox-Lass) the degree of explanation for forest running capacity increased further from 87% to 90% for the men and from 92% to 94% for the women (Fig 33). At the same time the margin of error was reduced from ± 6 sec/km to ± 4 sec/km on an individual level which is an equivalent precision to which the speed the runners can set their training intensity.

Fig 33. The relationship between forest Fox-lass velocity and average running speed during a simulated competition run on a marked course for 15 runners. The high degree of explanation for the vFox-Lass speed on the running performance in forest, independent of sex, means that only about 6% of the variation in running performance between the runners is dependent of other variables that are unknown such as fatigue tolerance and motivation (Fig 34).

Fig 34. Description of the level of explanation that the velocity at Fox-Lass in terrain gives for running performance during a simulated competition on a marked course in terrain.

The highest validity of the investigated test methods was achieved when carrying out the forest test and is consequently recommended for testing athletes of an elite level. Especially if the aim of the test is to recommend training intensities the forest test gives more valid information about the local muscle efficiency and endurance capacity. In the following section the Fox-Lass forest test and the perceived exertion of the runners during the test is described. The Fox-Lass test involves the runner completing 35-40 min of interval running wearing mobile analysis equipment (0.7kg) which can measure oxygen consumption, carbon dioxide production, and lung ventilation (Fig 35).

Fig 35. The portable oxygen analyser "Metamax 1" which allows measurement of oxygen consumption and carbon dioxide production in forest every 10:sec during 70-80 min. Due to the big variation in vertical exercise during forest running the intensity can't be controlled or set by using a km speed. A more relevant way to set training intensity is to use a standardized rating scale of perceived exertion which has been developed earlier for sports testing (14) (Fig 36). The Fox-Lass forest test consists of eight lap intervals, each of at least 900-1000m for the men and of 800-850m for the women, performed on a marked course. The two first laps are carried out at an RPE-level corresponding to the normal fatigue level during continuous/steady running. The next two laps are carried out at an RPE-level corresponding to the normal fatigue level during long interval (3-5 min) training.

Fig 36. The rating scale of perceived exertion (RPE). The next two laps are carried out at an RPE-level corresponding to the normal fatigue level during short interval (45-90sec) training. The last two laps are carried out at an RPE-level corresponding to the normal fatigue level during a 70 min competition for men and a 50 min competition for women. After each lap the runner is stopped and a capillary blood sample from a finger tip is taken at the same time as the average RPE level is noted together with the elapsed running time. After the test the collected values of oxygen consumption are analyzed and related to the relative intensity and to the earlier measured VO₂max (Fig 37).

Fig 37. One example of oxygen consumption from a forest Fox-Lass test. The average value for each RPE level is denoted with striped lines and relative figures to VO₂max. The lactate

concentration for each interval is also related to the running speed (Fig 38) and to the RPE level (Fig 39).

The MaxLass intensity during forest running was assessed as either one of the three following definitions: 1) the running intensity that corresponded to an increase in lactate concentration of at least 0.7 mmol/L between the two last interval levels in combination with a speed increase of no more than 3 sec/km (Fig 38).

Fig 38. One representative example of the relationship between produced running speed and lactate response during the Fox-Lass test in forest for one runner. 2) the intensity that resulted in a speed increase of no more than 3 sec/km between the two last interval levels despite an increase in RPE of at least 0.7 units (Fig 39).

Fig 39. One example of the relationship between produced running speed and lactate response during the Fox-Lass test in forest for one runner. 3) the running intensity that corresponded to an increase in lactate concentration of 0.7 mmol/L in combination with a speed decrease between the two last interval levels of no more than 3sec/km (Fig 40).

Fig 40. One example of the relationship between produced running speed and lactate response during the Fox-Lass test in forest for one runner. If none of these definitions is fulfilled the MaxLass intensity can't be predicted and the test has to be repeated another day. The use of three different definitions of the MaxLass during forest running is necessary because the individual variation in experience of setting running intensity in combination with the variation in relationship between the RPE and the lactate accumulation. These variations are not a problem when performing the test on a treadmill because the running speeds used are standardized and controlled. The result from the forest test can be used in the following models: 1.a The test is carried out on the same terrain course which makes it possible to compare a change in running speed from time to time. In this case the effect of a training programme in terms of change in running performance speed and different aerobic capacities are possible to compare (Fig 41).

Fig 41. The difference in oxygen consumption and running speeds on the same terrain course for four different RPE levels before (blue color) and after (green color) one year of training on an optimized programme for one subject.

1.b A more interesting comparison for established senior athletes is the change in running economy that might occur when a change in VO₂max no longer is possible (Fig 42).

Fig 42. The difference in running economy on the same terrain course for four different RPE levels before (blue color) and after (green color) one year of training on an optimized programme for one subject. 1.c Moreover, it is also possible to detect a difference in lactate accumulation due to an increased endurance capacity in the muscles despite no change in VO₂max (Fig 43).

Fig 43. The difference in lactate accumulation on the same terrain course for four different RPE levels before (blue color) and after (green color) one year of training on an optimized programme for one subject. The second way to use forest tests is when the test cannot be carried out on the same course. This can often be the case when competition preparations from a technical point of view has to be done in new terrain and the tests have to be carried out during the same week in that terrain. In these cases the running speed cannot be used due to different vertical work conditions.

Instead relative capacities have to be used as in the following models: 2.a The first model looks into a difference in relative oxygen consumption (Fig 44) which can reflect a higher endurance capacity or a higher fatigue tolerance. These two capacities are changed by different training models and are therefore important to investigate for the future training set up.

Fig 44. The difference in relative oxygen consumption on two different terrain courses for four different RPE levels before (blue color) and after (green color) one year of training on an optimized programme for one subject. 2b. The second model looks into the relative lactate accumulation response to the running performance in order to investigate if the lactate metabolism (muscle endurance) has changed (Fig 45).

Fig 45. The difference in relative lactate accumulation and MaxLass on two different terrain courses for four different RPE levels before (blue color) and after (green color) one year of training on an optimized programme for one subject. 2c. The third model investigates the fatigue tolerance and the RPE figures are compared which can show an increased MaxLass RPE level which allows the athlete to run faster without an increased muscle endurance or running economy capacity (Fig 46).

Fig 46. The difference in relative RPE level on two different terrain courses for four different RPE levels before (blue color) and after (green color) one year of training on an optimized programme for one subject. These models allow a thorough and valid evaluation of the individual aerobic capacity independent of whether it has been tested in the lab or in the forest. In conclusion this chapter can be summarized as follows: 1. Aerobic performance during forest running

can be explained to a degree between 90-94% by using the Maximal lactate steady state “MaxLass” concept. The “MaxLass” concept was used together with a new test protocol, the “Fox-Lass” protocol in combination with a new mobile oxygen analyzer, the “Metamax II” and with measurement of finger capillary blood lactate.

2. The new “Fox-Lass” field test also allows a low method error for setting of optimal training intensities using a rating scale of perceived exertion, “RPE”, at an individual level. An error of only ± 4 sec/km between 6 and 4 min/km running speeds was found which probably will not have any significance on the oxygen consumed or lactate concentration.

3. If the “Fox-Lass” test for any reason cannot be carried out in the forest a similar test on the treadmill in the lab has been developed. However, the treadmill test was found to increase the method error from ± 4 to ± 6 sec/km which might cause unwanted effects for those athletes who are able to control their running speed with a higher precision.

4. The increase in method error was mainly due to the fact that running economy on the treadmill is not always a reliable predictor of running economy in forest. Athletes with poor economy in forest can have a good economy on treadmill giving false information to the coach and the athlete about the need for specific economy training.

5. Moreover, the endurance capacity in the forest was underestimated on treadmill by 4% mainly caused by high running speeds with no inclination when testing on the treadmill. Therefore an angle between 3-4° is recommended with a decreased speed of 1 min/km compared to the speed used at horizontal level earlier.

6. The use of a rating scale for perceived exertion has been shown to be valuable in combination with oxygen and lactate measurements on different terrain courses. This is because running speeds from different courses cannot be used but the athletes rating figures can be used as a valid variable for increased aerobic capacity at submaximal level. In the following chapter we will describe the training concept we have used and how the interpretation of the test results has been the base for an optimization training concept.

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Physical Training

How should we train?

The starting point of all training is a thorough analysis of the sport, its demands, and our capabilities. When we have analysed these, it is much easier to plan the content of our training. Our competition goals should always be at the centre of our training. By that we mean that we train in order to be well prepared for competition and perform as well as we can. This is why it is so important to remember that we “become good at the things we train.” It has been claimed before that orienteers “run too much and train too little.” This has probably improved during the last few years thanks to better coaching work, at least at national team level. All success is built on a lot of work and simple ideas, or put another way, a lot of perspiration and little inspiration. There are no short cuts to the top. The way there is training, and training again, backed up with a healthy lifestyle. It is therefore sensible to think a little bit before starting on your training. Every orienteer must decide for him or herself, or together with a coach, what his or her optimal training solution is. Everyone has a different starting point, strengths and weaknesses and other factors which may influence training.

All-round training

“You cannot be stronger than your weakest link.” You need to train the whole body if you want to be able to do specific training. Your coordination, motor skills, awareness of your body and strength are an important base for intensive and frequent training. In addition, all-round training helps prevent injury, as stress on each separate part of the body is less if training is varied. The terms central and local or general and specific are useful when talking about training. The question is which capacity or capacities you want to improve or maintain and which training methods you can use to do so.

Focus on quality in training

If you have a plan or goal for your training, it is easier to get more out of each training session. A plan helps you to improve your weaknesses, and at the same time maintain or even improve your strengths. To be a good runner in the forest, to run faster and further in forested terrain, you should prioritise the following types of training:

- General strength training such as circuit training or calisthenics.
- Long runs in terrain.
- Fast runs in varying terrain.
- Intervals - different types in varying terrain.
- Specific strength training and running against resistance, such as hill running.
- Flexibility and coordination training. Map and compass should be part of training as often as possible.

Aerobic endurance - central and local

Orienteering is essentially an aerobic sport, which means that oxygen is the rate-determining step in energy turnover. When we train endurance aerobically the body is able to eliminate lactic acid. If we begin to accumulate lactic acid when we are training aerobically our speed is

either too high or external factors such as steepness or underfoot conditions are making us work harder. Our so-called central aerobic endurance can be trained effectively through other sports. The important factor in making the heart, lungs and circulation work is using large muscle groups. Cross-country skiing is often quoted as the best sport for this type of training, but in a British context lack of snow probably rules this out! Cycling, especially mountain biking, is also good training, as are swimming, rowing, kayaking, and of course running. If you train using other sports, you also have the benefit of variety in your training, less wear and tear on the "orienteering muscles" and it will be easier to increase your total training volume. When training local aerobic endurance capacity it is important that we train in the forest in a way that is as similar to competition as possible. Local aerobic endurance is often the decisive link in our ability to perform. For example a cross-country skier who trained only skiing would have a high central capacity and a high local capacity in his own sport. But the same skier, if running were not included in his training, would not be able to take advantage of his central capacity using the running muscles. Once again, we become good at the things we train. The table below gives a systematic overview of different types of training:

Percent of max Borg Lactate Time

Aerobic training		heart frequency (mmol/l)
Continuous low intensity	60-75 10-13 2-3 90+ minutes	Continuous moderate 75-85 12-15 2-3 1-2 hours
intensity	Continuous fast running 85-95 15-17 3-4 30-60 minutes	Long intervals 85-95 15-17 3-4 20-60 minutes
Short intervals	85-95 15-17 3-4 10-45 minutes	Natural intervals 75-95 13-17 2-4 20-60 minutes
Fartlek	60-95 10-17 2-4 20-60 minutes	Anaerobic training Hill sprints 95-100 17-19 4+ 20-45 minutes
Fast intervals	90-100 17-19 4+ 20-45 minutes	Continuous training

Continuous training involves moving from A to B at a more or less steady speed. The speed or intensity of the training can vary, depending on the aim of the session. For example, the aim of low or moderate intensity continuous training is to adapt the body to working for long periods of time, by improving the central and local endurance capacities. If we carry out technique training in the forest at this intensity we will make improvements in both central and local capacity for orienteering. Training through cycling, cross-country skiing or canoeing will also increase the orienteer's central capacity. Variation is the key, and a combination of running in the forest and other training such as cycling, swimming and paddling is ideal. Combining different types of training in a single session, for example running-cycling-swimming, allows a greater training volume which has a positive effect on the central capacity while placing less stress and strain on the different muscles and joints. A two hour combination session like this is indeed better training than a comparable session which only includes running.

The main aim of continuous running in the forest with low or moderate intensity is to improve the ability of the muscles to use oxygen. The effects of continuous training are increased mitochondrial density, through increased number and size, an increased number of capillaries and an increased capacity to use fat as a fuel for exercise. At low intensity slow twitch fibres (Type I) are primarily recruited and fat burning is dominant. The skeletal structure, joints and tendons become stronger and abrasion of the skin improves resistance to blisters and similar problems. Gentle to moderate undulation is ideal for low intensity

training but steeply undulating terrain can be suitable for moderate intensity training. Our own subjective rating of effort is often the best way to set the speed during a session. The Borg Scale can be a useful tool for this job. Low intensity training can also be described as “conversation pace” whereas at moderate intensity we can only manage 4-6 words per breath. Heart rate can also be a good way of judging the right pace so long as we also consider how a particular effort level feels as well. A heart rate monitor can be a good learning tool and give feedback about our training. But for an orienteer it is a less useful tool for guiding and controlling training pace. Continuous training improves aerobic fitness and provides a base for more intensive training.

- Continuous training should be present all year.
- Continuous training dominates during winter and summer – the basic training periods.
- Continuous training can beneficially be carried out in other sports or through combination

sessions where both running and other activities are mixed together. Continuous fast training
Fast training for orienteers means running at speeds approaching competition speed. The athlete might run for 30-60 minutes at a higher speed, but so that her running style still feels controlled and coordinated. The body should feel light and the athlete floats over the ground. Arms and legs are coordinated and move powerfully together in the right rhythm. The body manages to eliminate any lactic acid that is produced, that is the athlete is running at her threshold or slightly under. Flat or moderately undulating terrain is ideal for this type of training, and steeper runnable terrain can also be suitable. The aim of the training is to develop both central and local aerobic capacity and neuromuscular function for orienteering running. By having a steady high speed during the session we also improve running economy. This type of training can be included now and then during the build-up period but should mainly be included in the specific training and competition periods. Continuous fast training can also be carried out as a type of long interval to allow a slightly higher speed. The distance can be split up into a number of natural sections, for example 2-4 km with about 5 minutes of active recovery such as jogging. The speed can then be raised slightly so that you feel lactate accumulating in the last half-minute or so. As the effort is followed by active rest for about 5 minutes, the body has a chance to recover relatively quickly for the next section of fast running. This type of training is best carried out in the specific training and competition training periods. The important factor with these types of training, is that you let your subjective rating of effort guide your speed. You should feel that your breathing effort is “hard to very hard”, 15-

17 on the Borg Scale, and the effort in the legs should feel “moderate to hard”, 10-15 on the Borg Scale. At this level there is a considerable load on the body, but lactic acid is still eliminated. Long interval training

There are several advantages to splitting a high-intensity session into several sections of work and rest. Through this type of interval training the athlete can work near their maximal oxygen uptake capacity for a long total time without building up lactic acid. The training is carried out, as with continuous fast running at competition speed. The aim of interval training is to improve central aerobic capacity, heart and lungs, and local aerobic capacity in the orienteering muscles for running in the forest. The speed is high in each interval, the heart works at 10-15 beats under maximum pulse, and producing more lactate than the body can eliminate is inevitable in the later section of each interval. The rest period between each

interval is therefore very important. As heart rate will reach the planned level during each interval anyway the actual length of each rest period is not especially important. The guiding principal should be that you must rest long enough to allow you to carry out the next interval at the same speed as the earlier ones. Long interval training typically involves 2-10 minutes work alternated with half as much rest. A common interval is 4 minutes running and 2 minutes active rest. The number of repetitions depends on previous training levels. A normal number of repetitions might be 3-6 but some elite athletes will train many more. Each interval should be carried out for the same amount of time and at the same speed. Long interval training is included mainly in basic training, but can also be adapted to the specific training and competition periods. The important thing with this training form is again that your own rating of effort determines your speed. You should find the training "hard to very hard", 15-17 on the Borg Scale with respect to your breathing and your legs should be working "fairly hard to very hard", 13-17 on the Borg Scale. At this level the body is working very hard during each interval but has a chance to neutralise lactic acid during each rest period. Interval training is especially suitable for orienters in the later years of puberty and into young adulthood, around 20-22 years old. During this time the central capacity is most receptive to this type of training. It is during this time that we can work to develop the aerobic capacity for the longer term.

Short Interval Training

Interval training can also be carried out with short rest periods. The most common work-rest times are "90-30", "70-20", "45-15" or "15-15". The last interval type can also be called a short-short interval. As each interval in itself does not give a high heart frequency, it is important that the rest period is kept short. After a few intervals the pulse will reach the right level. The aim of short interval training is as with long intervals, to raise the central capacity, that is, to increase the heart's pumping capacity. As the rest periods are frequent the muscles manage to eliminate lactic acid to a greater or lesser extent. During the rest the muscles can also store some oxygen for the next work period. This means that despite the fast pace relatively little lactic acid is accumulated, if the work period is not too long. These intervals can be beneficially carried out in the forest, but even indoors on an obstacle course or as part of a circuits session. If the intervals are carried out in the forest both slow and fast twitch muscle fibres are used. This means that more muscle fibres are involved in running and local capacity in the orienteering muscles is improved. The speed during each interval should be competition speed or faster and this high speed has a particularly beneficial effect on running technique. Sometimes this type of training can feel tough and should therefore mainly be used when you are trying to achieve a performance peak. Short intervals should mainly be carried out during the specific training and competition period. The important point with this training type is that subjective effort level determines the speed. Your breathing effort should be "hard to very hard", 15-17, while the effort level in your legs should be "somewhat hard to hard", 13-15. At this level the body is working hard, but still manages to eliminate any lactic acid formation during the rest period. Including short interval running on an obstacle course, circuit training or in general strength training is an excellent training method. Each exercise is carried out several times, and the rest between each exercise is short. After each rest you move onto a new exercise. Each exercise is carried out intensively during the work periods – read more under "General Strength". This type of training is best carried out during the build-up period.

Natural Intervals

During the above types of training the work period is controlled by time, whereas the terrain determines the effort level and length of each effort in natural interval training. By choosing terrain which is steeply undulating or soft ground, such as marshy or heather-covered areas the load placed on the heart and lungs varies naturally during the session. Changing

runnability and steepness also mean that your stride is continually varying. The idea with natural interval training is to increase your effort and speed on for example hills, or tough sections, and then to reduce it again in the runnable or downhill sections. The terrain guides your speed and the length of each interval. It is also possible to carry this type of training out in "reverse" so that you take it easy uphill then increase your speed on the downhill sections. A so-called Murder Loop is a good example of how one can train natural intervals, continuous fast running and competition like forest running. Such a course can be planned in varied terrain to include for example short steep hills, long gentle hills, marshy ground, tough areas of felling and runnable terrain. It can be in the form of an easy orienteering course, or line orienteering. The course can be divided into sections and perhaps marked to show where to run intervals and where the easier rest sections are. Fartlek

Fartlek is a Swedish work which literally means "speed-play". The aim of fartlek is mainly to improve running technique for orienteering. At the same time both central and local aerobic capacity are trained. By "running technique" we mean moving over the obstacles that rough terrain presents as quickly as possible and with the minimum possible energy expenditure. The aim is to be able to run in a relaxed way even in rough terrain. The more efficiently the right muscles work together to give a strong stride the easier it is to run quickly. If running feels like a "battle" against the terrain, then it is very likely that the wrong muscles, antagonists, are also being used. It is therefore important for the orienteer to train at competition speed and even faster during fartlek in varied terrain so that the right running style for forest terrain is learnt and maintained. Orienteers should run fartlek training in the type of terrain that we compete in. The unique factor running with orienteering running is that no one stride is like another. Stony ground, thick forest are steep descents are some of the areas that require "relaxed strength" from the runner. If you dare to run fast in tough, demanding terrain you give yourself the opportunity to develop your running technique. A series of running technique loops would be a good way to concentrate on high speed with a relaxed running style.

Anaerobic endurance

The aim with anaerobic training is to accustom the body to running with accumulated lactic acid, at the same time as maintaining running technique and encouraging the development of the chemical processes which deal with an "acid environment" in the muscles. By this we mean that teaching ourselves to tolerate lactate, increase our capacity to buffer or neutralise lactate and improve our capacity to transport lactic acid away from the muscles. Studies from recent years show that anaerobic capacity can be influenced positively by short training periods at high altitude. In the so-called Kenya study, carried out by the Swedish Athletics Federation, it was shown that the muscles developed an increased buffer capacity, meaning that the muscle cells can neutralise lactic acid more effectively and also regulate the acidity (pH value) for a longer time. There was also found to be a reduction in lactic acid forming enzymes and an increase in enzymes which broke down lactic acid. Heart rate and the subjective effort reported in a treadmill test reduced on return to sea level. This points to an increased anaerobic capacity. Similar results were experienced in an altitude-training project undertaken by the Swedish Orienteering Federation. Both short periods at altitude and high-intensity lactate training probably have the same effect on the anaerobic capacity. The difference lies in the method. Lactate training must be carried out in a sport specific way. The training effect gained is very specific and only affects the muscle fibres which are involved in the particular training action. So for orienteers the best method involves running in the forest with resistance such as hills or rough, marshy ground. Hill sprints

The simplest form of lactate training is running up a slope at maximum speed for 30-90 seconds. At the end of each effort you should feel tired and heavy in your legs. A long rest is required before starting the next effort – at least 3-4 minutes. If you jog easily during the rest period it is much easier to clear the lactate. Each series should contain 3-5 repetitions and you can carry out 1-2 series. The training should feel “very hard to extremely hard”, 17-19 on the Borg Scale, both in your breathing and in your legs. At this effort level there is a very high load on the body, and lactic acid will be partially transported away during the rest period. This type of training should only be used in the last 4-8 weeks before important competitions where you want to be in form. If you have a break from lactate training any effect it has had will reduce quickly. Aerobic capacity has a short lifetime and within a month the effect of

lactate training will have totally disappeared. It is however possible to maintain anaerobic capacity through competition and occasional lactate sessions. The training is carried out mainly when trying to peak and as maintenance training during the competition period.

Lactate interval

Another form of lactate training is to reverse the sort interval idea, so that you have a longer rest and shorter work period: “30-90”, “20-70” and “15-45” are some possible ideas, which can either be carried out on their own or mixed. Short hill sprints are also a possible lactate training method, with for example a 20 second sprint uphill followed by 70 seconds easy jogging back down. The easiest form is perhaps to spontaneously vary the length of each effort, let the terrain determine how long each interval is. In this case, one idea could be to run a Murder Loop as hard as possible on the hills and then jog on the downhill and easier sections. As with the hill sprints above, the training should feel “very hard to extremely hard”, 17-19 on the Borg Scale both in your breathing and your legs. At this level of effort the body is working very hard and manages to clear some lactate in each rest period. Lactate training is an extreme type of training. You need to prepare yourself very well mentally and physically before a session of this type. The training is demanding and requires a much longer rest period afterwards than aerobic training. The injury risk is also greater and you are working with maximum muscle power. Muscle strains or twisted ankles are common injuries associated with lactate training. The reasons can be increase in training load, poor basic training, too little recovery time between sessions or inadequate warming up.

Running Technique

In all the descriptions of different types of training above we have pointed out again and again that we only become good at the things we train. Orienteering running technique is very specific as the terrain and load on the body vary during the competition and from competition to competition. When we run in terrain our knees and hips are slightly bent and work to keep our centre of gravity low and help us maintain balance. The ability to run fast on rough ground without looking where you are putting your feet is probably not just dependant on strong ankles. When we study the progress of elite runners in the forest, it appears that the runner can smooth out the uneven ground, by shifting their centre of balance and coordinating their movements with active upper body work. The signals for the rapid and subtle movements of centre of gravity probably come from the actual foot strike with each stride. The feet probably work like “sensors” responding to the uneven ground under the runner. The more used you are to running in forest terrain the more relaxed you can be as you run in terrain with varying underfoot conditions. In a final year project at the Sports High School in Stockholm, carried out in 1989 by Kurt Svensson, Erik Svensson and Jan Sundbue, it was shown that running in the forest had very few similarities to running on the road (see figure). The only similarity in technique was the link: “Tilt of the upper body when

running downhill in forest terrain". This table underlines very clearly the importance of analysing the actual requirements of the sport when thinking about how training should be carried out.

Forest terrain

Forest uphill

Forest downhill

Felled ground

Marshy ground

Stony ground

Stride length (cm)

- - - - -

Stride frequency (double strides/min)

- - - - - +

Foot plant (contact time)

+ + + - + + + + +

Arm (difference from a symmetric pattern)

+ + + + + + + + + + +

Knee lift + + + - + + + + + Hip (flexion) + + + + - + + + + + +

Upper body (forward lean)

+ + + + 0 + + + +

The table shows the results of research which analysed many runners in the forest, using a video camera. The biomechanical measurements compare running in a variety of forest terrain to running at a steady pace on a flat surfaced road. The symbols show the range of differences: --- = much less than 0 = the same as +++ = much greater than.

Cross Training

Cross training means training in a sport other than your main sport. The aim of cross training or combination sessions is all round fitness and strength and to train the central aerobic capacity. As orienteering is an aerobic sport, requiring a lot of aerobic training, it is ideal to be able to vary training as much as possible when training the central capacity. The benefit of combination training is that you work your heart and lungs for a long time but are able to use different parts of your body in the session. Another benefit is that you "empty" one muscle group at a time, and that tendons, joints and ligaments are stressed less as the training is varied. Triathlon (swimming, cycling and running) is a variety of cross training. The athlete uses her upper body in swimming, thigh and hip muscles when cycling and the calf muscles when running. This shows that it is possible to keep going for a very long time with a high speed. The following combination sessions are possible variations for an orienteer, depending on the time of year, and where you live: XC skiing – running – swimming XC skiing – running XC skiing – running – general strength XC skiing - cycling XC skiing – general strength - swimming XC skiing – aerobics - swimming XC skiing – ice-skating – swimming XC skiing – cycling - swimming Running – cycling – swimming Running – general strength - swimming

Running – aerobics Running – aerobics - swimming Cycling – running – cycling – swimming Cycling – general strength - swimming General strength – running in water - swimming The possible combinations are endless – the limiting factor is your own imagination. Inline skating, a rowing machine or climbing for strength are possible alternatives for snow and ice free climates! Cross training is a form of training which is used more and more by today's elite runners and the advantages are summarised below:

- The whole body can be trained
- The aerobic central capacity can be trained for long periods of time
- Over use injuries are rare
- It is varied and fun

Warming Up

The aim of warming up is to prepare the body for hard exertion. Warming up should take 15 – 20 minutes and raise the muscle temperature during this time. The increased temperature means that oxygen is transported to the muscle cells more easily, metabolism increases and that nerve impulses from the brain reach the muscle cells more quickly. Muscles sense position and react to impulses more quickly, which makes it easier to correct small stumbles. Example of a warm-up routine:

- Jogging for 5-10 minutes
- Stretching 5 minutes, mainly for leg muscles
- Movements to raise the pulse: high knee raises, kicking your backside, ankle work,

skipping and bounding or similar, 3-5 minutes

- Finish by running 50-60 meters at a steadily increasing pace 3-4 times
- Rest and collect your thoughts for a few minutes before the start of the training session or

competition Warming up improves performance and prevents injuries.

Warming down

A warm down jog is the first part of the next training session. When the body has undertaken hard physical work, there will be waste products left in the muscles. Warming down is especially important after competitions, when the muscles have produced a lot of lactic acid. Lactic acid remains in the muscles for a longer time and recovery is slower if you do not include light activity after the session. By warming down lactic acid and other waste products are processed more quickly and transported away by blood circulation. Example of a warm down routine:

- Light jogging 5-15 mins, straight after the competition or training session
- Stretching approx. 5 mins, mainly for the leg muscle
- Relaxation, sitting or lying, tense and relax the main muscle groups from head to toe.

Warming down improves performance and prevents injuries.

Strength

To be efficient in the forest we need to be strong without being “muscle bound”. The orienteer should be strong but light. In our sport we propel ourselves over the different obstacles presented by the terrain. Strength for an orienteer is therefore related to body weight. This means that an orienteer should have good strength endurance and be able to support his own weight. Orienteers also need to be especially strong in the feet, calves, thighs and hips as these parts of the body do most of the work when running in the forest. There is no point trying to develop bulging muscles, as these will only be like a heavy rucksack when you run. The aim of strength training is to develop the muscles so that they can work longer and have better endurance or produce more power and thereby be stronger. The training effect of strength training is often rapid, but in the beginning this usually takes time to do.

with improved neuromuscular coordination and the muscles actually become stronger later. The strength of tendons, ligaments and cartilage develops more slowly than muscle strength, which can lead to injuries if training is too intensive in the beginning. It is very important that movements in strength training are carried out with good form and in a controlled way from the beginning. In the beginning it is also wise to carry exercises out slowly. As training progresses the load can be increased. The training effect depends on the type of load that is chosen. If a high load is chosen, in the form of a heavier weight perhaps, fewer repetitions can be carried out. If the load is reduced it is possible to continue working at the same exercise for a longer time. Training with heavy weights and few repetitions gives increased muscle strength. The training effect of strength training with few repetitions and heavy weights is very noticeable. The muscles will increase in volume. Each individual muscle fibre will be thicker, but no new muscle fibres will develop. During endurance training on the other hand, with many repetitions and less weight, local blood supply increases and the muscles' ability to work at higher lactate concentrations is improved. Muscles can either work statically or dynamically. Dynamic strength means that power is developed while the muscle length is changing. If the muscle is shortened we refer to this as concentric work, for example the quadriceps on the front side of the thigh or the calf when running uphill. Static strength means that the muscle works without any change in length for example the abdominal muscles when lifting the knee. General strength training

In the 1970s in Sweden many of the best elite orienteers got a lot of strength training "for free" through their work. Many worked in the forest or with another similar physical job. They would carry out many repetitions with low weight naturally during the course of the day. The body became strong through this physical work. Today there are very few elite orienteers who have manual jobs. Most have a more or less sedentary job which means that general strength training is a very important part of a training programme.

In orienteering, as in all sports, we need good general strength to ensure balance between the different muscle groups in the body. It is particularly important to build up the strength of muscles around the torso and back. In orienteering, all our movements come from the hip, which means that we must have excellent stability so that the legs and to some extent the arms have a strong base to work from. Forest running places specially high demands on this core stability as the hip flexors work intensively with each stride. The more knee lift required the more important core stability is. If the muscles around the torso are weak, the runner will soon slip into a "sitting" running position and loses form and speed. The aim with strength training for orienteers is to work all the muscles in the body with lighter strength exercises that can be repeated often. The orienteers own body weight is enough load for most exercises. However, light free weights can also be used and fixed weights can also be used occasionally. Aerobics is a very suitable type of strength training for orienteering. Circuit training, where different exercise "stations" are planned out to provide a varied strength workout is also suitable for general strength training. General strength should be carried out throughout the year, but is especially important in the build up period and as maintenance training in the competition period. Note: Different examples of general strength training can be found at the back of the book. Specific Strength Training

For most of the year training should be coupled to the individual and reflect what they will face in competition. If an orienteer plans to carry out specific strength training, the movements must be relevant to the "orienteering stride" that he needs in the forest. Specific strength training can be divided into two - muscle specific training and movement specific training. Muscle specific training can be included in the form of maximum strength training, with few repetitions and heavy load. It is not certain that orienteers gain any benefit from this

type of strength training if the body already has good muscle balance. But after injury, when certain muscles have been forced to rest, for example after broken leg this type of training can be very useful. It can also help when there is imbalance between the left and right sides or other opposing muscle groups. This type of training can be beneficially carried out with the help of fixed equipment. The aim with the training is to build up muscle mass in a particular muscle. Movement specific strength training should be trained by running against resistance. The training is carried out with light resistance and as fast as the resistance allows. Carrying out movement specific strength training using fixed equipment would be pointless. Instead the orienteer should use hills, downhills, marshes or snow during the winter as the resistance for this type of training. These conditions act as a brake against which the orienteer works. It is also possible to work with a partner and run against the resistance of an elastic band or even drag a tyre or small parachute. Movement specific strength training is very similar to anaerobic training and should be treated with the same respect as far as intensity and recovery are concerned.

In addition, orienteers are going to be faced with greater stretch-shortening demands if the sport begins to move out of the forest and becomes more park-like. Stretch-shortening means that the muscle and tendon are stretched and then make a quick, powerful contraction. An example of this is when we jump down from a platform, then bound up onto another platform immediately. Stretch-shortening can help us to run faster and improves elasticity so that we get more from each stride without expending more energy. If we look at modern middle distance and long distance runners it is no coincidence that they move very economically and quickly, running on their toes. For the orienteer the best way of including this type of training is probably uphill bounding, with different hops and jumps. This would increase the elasticity of the foot and calf. This type of training can also be included as a part of other training sessions such as fartlek or hill sprints. Movement specific strength training can be included throughout the year, particularly in the specific training and competition periods. Muscle specific training should only be carried out the build up period or when it is required for a special reason such as after injury.

Flexibility

It is important to maintain the natural suppleness which we have as children. Most young people have adequate flexibility, but muscles soon become tighter when we train intensely. Some people are even born hyper-flexible, especially in the joints. This type of flexibility problem should not be tackled with more flexibility training, but strength training instead. There is an important difference between flexibility in a joint and flexibility in a muscle. In general, when we increase training volume muscles and other structures round the joints need to be stretched so that they do not become tight and stiff. General flexibility training is required to improve the range of movement in many of the body's joints to ensure good general flexibility. Extreme flexibility is not the aim. Specific flexibility training involves improving flexibility in joints which can be limiting factors for performance in a specific sport, orienteering running. By carrying this type of training out we try to improve our forest running performance. In orienteering it is primarily the hip flexors, the deep calf muscle and the thigh muscles that are important for running technique. All training shortens the muscles somewhat, especially intense specific training. In addition, age and increased muscle mass negatively influence flexibility. Women are naturally more flexible than men. There are many reasons to make flexibility training an important part of your training programme all year round. Most importantly, good flexibility means that you will suffer fewer injuries. Good flexibility in the muscles that are important for running technique increases your performance capability. The connective tissues around the muscles become more flexible which can

increase stride length and coordination. By improving this, your running speed will also increase. Certain muscles easily become stiff or shortened when the muscle changes volume, which is a common effect of strength training. The connective tissues around the muscles are not flexible enough and the muscle becomes short and stiff. Sometimes this process can also lead to imbalance between different muscle groups, due to inactivity after injury or over use. Poor flexibility in certain joints can mean that other joints must work harder to compensate for the inflexible area, and this can lead to unnecessary wear and tear.

Different stretching methods

There are several different ways to stretch which can be included in flexibility training. Whatever way you are stretching, it should involve a slow and controlled movement without jerking.

- Gentle stretching involves taking a muscle or joint slowly through its full range of

movement then back to the resting position. This type of stretching is ideal to include in a warm up routine and to maintain general flexibility.

- Stretch-hold stretching involves taking a muscle to the end of its range of movement then holding it in that position for 20-30 seconds. After a few seconds rest the stretch can be repeated 2-3 times.
- Tense-relax-stretch stretching is also a common stretching method. It involves tensing the muscle to be stretched before carrying out the actual stretch:

1. Get into the stretch position 2. Tense the muscle to be stretched for 5 seconds 3. Release the tension and relax for 3 seconds 4. Stretch the muscle so that it is at the end of its range of movement 5. Hold the stretch for 15 seconds

Does everyone need flexibility training?

If we mean flexibility training in the form of stretching before or after competition or training, then everyone needs to stretch. This type of maintenance stretching helps prevent injuries and improves performance. However we only need to dramatically increase flexibility when we have shortened inflexible muscles.

Race Analysis Form for Orienteering

Name: Jamie Stevenson Date: 18/07/2000 Race: WC Prologue Finland Physical Shape: 9 Health: 10 Motivation: 10 Technically prepared: 8 Harmony: 7 Concentration: 9 Mentally prepared: 9 Confidence: 8 Want to perform 100%: 10

Use the 0 – 10 RPS scale that you use to rate your "Perceived Shape" 10 Excellent performance, excellent shape, everything went brilliantly 5 Acceptable performance, acceptable shape, everything was more or less OK 0 Extremely bad performance, totally powerless, everything went extremely badly

The control The Leg Overall

Control

Trail

Number

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Route

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Total 0.45 9 0.10 9 9 25.08 23.15 10th, 3.9km, 6.4 mins/km

Acclimation, Hydration & Recovery A guide to competing in hot environments

Dr. David Bailey Physiologist

English Institute of Sport Loughborough University

Preparation & Recovery

Preparation

- environment (knowledge, acclimation)
- training (volume, duration, time of day)
- travel (jet lag)
- competition (time of day, duration)

Recovery

- Hydration
- Nutrition
- Recovery strategies

Overview

- Acclimation to hot/humid conditions

– Environmental conditions – Exercise in a hot/humid environment – Maximising adaptations

- Importance of water – Affect of dehydration on performance

& recovery – Monitoring hydration status

- Recovery strategies in hot environments – Nutritional strategies – Non-nutritional strategies

Environmental Conditions

Nagoya London

Temperature 32°C (16-38°C)

Humidity 92% am / 66% pm

Temperature 21°C (6-33°C)

Humidity 76% am / 62% pm

Maintaining heat balance

Heat loss in hot environments

30°C, 70%rh

E

R + C

32°C, 20%rh

E

R + C

REST

E R + C

E

R + C

Heat loss in hot environments

30°C, 70%rh 32°C, 20%rh

EXERCISE

Heat & Performance

- Type of exercise

– decreased endurance – limited affect on maximal strength

- Hyperthermia – Shift towards non-aerobic metabolism – Faster rate of muscle/liver glycogen usage (carbohydrate) – Increased cardiovascular strain (venous pooling)

Heart Rate x Stroke Volume = Cardiac Output

- Dehydration

– Decrease in body weight (2 – 8%) – Increase sweating (0.8 – 1.4 L/h, maximum recorded 3.7L/h) – Rate of water absorption (0.8 – 1.2L/h)

- Hypohydration – dehydration >4h without rehydration

Heat Acclimatisation

- Preparation for competition in a hot environment
- What are the beneficial adaptations?
- Regular exposure to heat/humidity over 10-14days;

↓ Cardiovascular strain (3-6 days)
 ↓ Perceived exertion (3-6 days)
 ↑ Blood volume (3-6 days)
 ↓ Body temperature (4-8 days)
 ↑ Sweat Rate (8-14 days)
 ↓ Electrolyte loss (5-10 days)

- Improved heat transfer from core to the skin
- Improved cardiovascular function (dehydration & blood

pooling)

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39.0

39.5

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140

150

160

170

180

Sweat

loss

s (l/

min

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1.1

1.2

1.3

1.4

1.5

Physiological adaptations to heat

Maximising Adaptations

- Exercising in similar conditions ($\pm 2^{\circ}\text{C}$)

– $>50\%$ $\text{VO}_{2\text{max}}$ (60-90% HR_{max}) – Between 90-100min – Gradual exposure over 10-14days – Exercise in groups or pairs – Exercise in environmental chamber – Wear more insulated clothing – Monitor physiological responses

- Exercise at competition venue

– Adapt training programme (high intensity work at cooler times) – Train at similar time to competition – Monitor physiological responses

Monitoring Acclimation

- Assess changes in body weight

– Weigh pre & post-exercise (calculate difference) – Add weight of clothing if wet – Add weight of fluid consumed during exercise (1litre = 1kg) – Subtract urine loss during exercise – Sweat rate can be calculated

- Example

Pre-exercise = 70kg Post-exercise (60mins) = 69kg Clothing weight 1.0kg 1.5kg Drank 500ml = 0.5kg Change in weight = 2kg Fluid replacement = 2 litres

Importance of Hydration

- Virtually impossible to maintain

euhydration during exercise in hot/humid conditions

- Not uncommon to experience 2-8% loss in body weight
- Small body weight losses (2-4%) can result in reduced exercise capacity/tolerance
- Dehydration leads to; $\downarrow$ blood volume/blood flow $\downarrow$ waste removal $\uparrow$ cardiovascular strain $>$ risk of heat illness
- Maintaining hydration minimises these detrimental responses

Dehydration & Performance

Monitoring Hydration Status

Consider fluid loss during exercise + whole day

– Mild climate – 2.5L/day – Hot climate – 10L/day

Monitor physiological changes; – Body weight – Body temperature (difficult) – Urine osmolality (very difficult) – Urine colour (colours 1,2 & 3)

Continual monitoring is important – Rehydration isn't instantaneous – Monitor urine colour >2 times post hydration – Consider food intake

Hydration Strategies

ACSM stand point (1996)

- $>2\text{h}$ pre-exercise consume 500ml water
- During exercise drink early
- Drink at rates equal to your sweat rate
- Practice drinking at these rates
- Try to consume cooled fluids

General advice

- Be realistic
- Avoid gastrointestinal distress
- Consider event duration
- >1h hydration during exercise is essential
- Greater emphasis on rehydration
- Maximise recovery

Fluid Replacement Beverages?

Water vs. Sports Drinks;

- Much debate (commercial interests)
- Depends on;

– Duration – Intensity – Urgency – Recovery time

- Plain water haemodilution
- Sports drinks contain;

– Electrolytes (Na⁺, K⁻, Cl⁻) – Carbohydrate

- Maximise recovery

Recovery Strategies

Nutritional

- Hydration is essential
- Repletion is as important
- Source of electrolytes
- Consider recovery period
- Amount of carbohydrate
- Type of carbohydrate
- Carbohydrate & protein mixtures

Muscle

glycogen

1 2 3 4 5 6 7 8h

Alternative Recovery Strategies

Non-nutritional Cryotherapy; Cold water immersion

↑ Cooling ↓ Muscle soreness ↑ Muscle function ↓ Muscle damage

Contrast therapy ↑ Blood flow ↑ Waste product removal

Cooling (pre vs. post)

Warning Signs of Heat Disorders

Key Points to Remember

Preparation is key

Allow sufficient time for acclimation

- Preparation is key

- Allow sufficient time for acclimation

- Maintain good hydration both during exercise & at rest
- Monitor hydration status regularly
- Continually assess tolerance to hot/humid condition
- Replace lost fuel as well as fluid
- Support fellow competitors/training partners
- Be aware of signs of hyperthermia

Thanks & good luck in Japan!

Any questions?

(David.Bailey@eis2win.co.uk)

HOT COMPETITION

Performance Impairment

Loss of fluid, as well as

wider distribution

throughout the body of

the already depleted

fluid volume means the

heart has to work harder

in order to continue to

supply the muscles and

organs with blood.

Sweating Dehydration

Cardiovascular Insufficiency

Vasodilatation

Overheating Impaired Performance

Dehydration

1-5% body mass loss due to dehydration:

5-50% reduction in endurance

performance; 10-22% reduction in

VO₂max

Increases temperature due to decreased

sweating and cutaneous blood flow :

3.7% decrement in 1500m pace (6s), 6-

7% decrement in longer events

Benefits of heat acclimatisation are negated

The risk of heat illness is increased

(Armstrong et al 1985; Craig & Cummings 1966; Pinchan et al, 1988)

From: Rowell L B. Human circulation. Regulation during physical

Hyperthermia & Central Fatigue

There is a diminished central drive to exercise in hyperthermic individuals

Cerebral blood flow falls due

Hyperventilation

Fall in CO and BP

Brain temperature increases

Brain sensitive to heat

Affects cerebral cortex

Nielsen & Nybo, 2003)

Heat & Performance

Marathon performance declines 1 minute for each 1 °C increase in air temperature above 15 °C (Maughan, ABC of Sports Med)

Compromised muscle and hepatic blood: earlier onset of anaerobic metabolism and blood lactate accumulation

Muscle glycogen utilisation is increased: fatigue occurs earlier during prolonged moderate exercise in the heat

Variables that affect response to heat

Environmental conditions

Body size (mass, skinfold thickness)

State of training / sudden increase in tempo

Degree of acclimatisation

Hydration status

Clothing worn

State of Health : fever, viral illness, cold, GI disturbances

Genetic profile: responsiveness to heat

Genetic disorders: malignant hyperthermia

Skin disorders - sunburn over 5% of body surface impairs thermoregulation for 21 days

Use of medication - diuretics, antihistamines

Sweat gland dysfunction (e.g. prickly heat)

Salt depletion

Age

Heat, Fitness & Performance:

Aerobically fit individuals are able to perform for longer in hot environments, and tolerate higher levels of hyperthermia than less fit individuals but

Abnormally high core temperatures impair exercise performance in all individuals in the heat, irrespective of fitness

Fatigue generally occurs with core temperatures between 38 - 40 °C (Hales et al, 1996 & Nielson et al 1997)

How to acclimatise

Raise your body temperature to stimulate sweating

This must be done in representative temperatures (30-35°C, 100min per day optimal)

It is important to exercise during these exposures (but the mode is unimportant and intermittent exercise is OK)

Acclimatisation is specific to the climate and activity level i.e. The best would be to train in the same conditions as you will experience on race day.

You should allow 10-14 days to acclimatise with no more than 3 days elapsing between successive exposures. 66-75% of changes occur in 4 to 6 days

Fitter individuals can acclimatise more quickly (7-10 days)

Even fit individuals need to exercise in a hot environment

On return to a temperate climate, the major benefits are retained for a week, 75% are then lost within 3 weeks

Acclimatisation

Aldosterone secretion increases and increases sodium & chloride reabsorption in sweat ducts and renal tubules. This results in lowered salt content in sweat (e.g. sweat sodium reduced from 50 to 25mmol.L⁻¹) and increased osmotic retention of water, producing increased plasma volume

Increased sensitivity to aldosterone

Less cardiovascular strain, perfusion pressure better maintained

More effective distribution of cardiac output

Improved cutaneous blood flow

Earlier onset and greater rate of sweating & gland recruitment

Lower resting deep body temperature

Lower skin and deep body temperatures for a given level of exercise

Improved physical work capacity

Increased comfort

Decreased reliance on carbohydrate metabolism

Reducing the impact of the Environment

Acclimatisation

Hydration/Rehydration

Cooling (Clothing, ice baths)

Cooling stations: fans, shade, source of cold water & drinks

Maintaining comfort (e.g. In accommodation, at prestart, after the race)

Pacing strategy

Clothing

Should be

Lightweight

Light in colour

Breathable/open weave

Loose

Absorbent

The main function of clothing should be to assist the evaporation of sweat

WOC 2008 – The Czech Republic

The warmest month of the year in the Czech Republic is July. On average, the summer temperatures are about 20°C higher than during winter. Especially in the last decade, temperatures above 30°C are not unusual. Summer is also characterized by rains and storms.

What is the heat stress in Czech?

July Data

Average High Temperature 22.2 °C

Average Low Temperature 12.2 °C

Average Mean Temp 17.2 °C

Average Morning Relative Humidity 81%

Average Afternoon Relative Humidity 54%

Record High 36.1 °C

Record Low 5 °C

Heat Stress Index – Temperatures in Celsius

Heat Cramps

Heat Exhaustion

Heat Stroke

RH

(%)

21.1 23.9 26.7 29.4 32.2 35 37.8 40.6 43.3 46.1 48.9

0 17.7 20.6 22.8 25.6 28.3 30.6 32.8 35 37.2 39.4 41.7

10 18.5 21 23.5 26.7 29.4 32.2 35 37.8 40.6 43.9 46.7

20 19 21.6 24.5 27.8 30.6 33.9 37.2 40.6 44.4 48.9 54.4

30 19.4 22.8 25.6 28.9 32.2 35.6 40 45 50.6 57.2 64.4

40 20 23.3 26.1 30 33.9 38.3 43.3 50.6 58.3 66.1

50 20.6 23.9 27.2 31.1 35.6 41.7 48.9 57.2 65.6

60 21.1 24.4 27.8 32.2 37.8 45.6 55.6 65

70 21.1 25 29.4 33.9 41.1 51.1 62.2

80 21.7 25.6 30 36.1 45 57.8

90 21.7 26.1 31.1 38.9 50

Training Concepts by Steve Vernon

Aerobic runs – Ranging from slow and easy ‘recovery’ runs to runs approaching higher intensities but below anaerobic threshold training.

- Great for recovery runs and early season aerobic build up. - Different intensities should be used. - Medium intensity aerobic runs should form the basis of your training for much of the year.

Sub-Threshold runs – Also known as marathon paced runs. These are slightly slower than threshold but a lot more comfortable and are at the top end of the aerobic run spectrum.

- They can give all the benefits of the threshold run without going over the top and pushing too hard. - These runs can be over 30mins and up to around 60mins. - They can be run 1 to 2 times per week in the endurance build up phase.

Threshold runs – Also known as ‘Tempo’ runs. Often described as comfortable hard and is very close to the maximal pace a well conditioned runner can run for 60mins.

- These runs should be 20 – 30mins max. - Threshold is approximately 85% of your maximum heart rate.

VO₂ Max Intervals - VO₂ Max intervals are very effective when performed for 2 – 5 minutes, with equal or shorter time recoveries. For interval sessions at VO₂ Max pace each repetition has a recovery generally equal to or shorter than the duration of the repetition. VO₂ Max speed can be held for 8 – 10mins, therefore 3,000m is 100% VO₂ Max.

- 5,000m pace (95% VO₂ Max pace) is very safe and effective, yielding all the benefits without the dangers of overdoing it at 100% VO₂ Max work (3,000m) pace. Any faster we are training at an even higher anaerobic (Lactic) rate with increased acidosis.

- VO₂ Max intervals should be run after the bulk base phase but using a progression into them via upper limit threshold intervals.

- Example VO₂ Max interval session - 6 x 1,000m (2min jog recovery).

Lactic Repetitions – These are intervals that are considerably faster and shorter than VO₂ Max intervals with a much needed recovery that is far longer than the interval duration.

- They are run at glycolytic anaerobic pace (800m – 1500m pace). - They are very anaerobic in their effect and should never be run in an aerobic base period. E.g. 8 x 400m (3min recovery).

Leg-Speed Drills – These are very short intervals (generally less than 10 sec) that use the alactic anaerobic system where little acidosis is possible.

- They are very effective and safe to use year round as long as you use a good recovery walk or jog of over 30 seconds.

- They increase power, muscular efficiency and speed. - Ensure you are not too tired when performing these. E.g. 8 x 10sec hill sprints.

Optimising Performance

Nutrition

Aim of session

- Recovery nutrition – What & When

– Are you eating enough ?

- Body composition – Do you need to measure it ?
- Supplements – Do you need them

– Suppliers

Fuelling your training

10-124-6 Hours/Day

7-122-4 Hours/Day

5-71-2 Hours/Day

4-5Up to 1 Hour/Day

Carbohydrate (g/kg/d) Training Duration

(Burke et al. 2003)

Fuelling your training

1.4-1.7 ?> 4 hour/day

1.2-1.42-4 hour/day

1.0-1.21-2 hour/day

0.75Up to 1 hour/day

Protein (g/kg/d) Training Duration

(Tipton & Wolfe 2003)

Post exercise window

- Start refuelling as soon as practical
- Carbohydrate & Protein recovery snacks

– Protein acts as a satiety factor

– 5:1 Ratio of CHO:PRO

– CHO: 1.0-1.2g kg of body weight

(Burke et al. 2003)

Are you eating enough to improve performance ?

- Signs that you might not be
 - Training hard / high volumes but not getting 'fitter'
 - Recurrent injuries / illnesses
 - Disrupted menstrual cycle
 - General fatigue

But if I am not eating enough surely I must be losing weight ?

Consistently < 500 kcals per day – disruption to hormonal & metabolic balance (Loucks 2005)

Body composition

- Height / Weight
- Skinfolds
 - Sum of four
 - Bicep, tricep, subscapular, supraspinale
 - Bicep, tricep, subscapular, supraspinale, iliac crest, abdominal, thigh, calf
 - Girths
 - Upper arm, waist, hips, thigh, calf
 - Who measures
 - Same person for repeated measures
 - ISAK qualification

Supplements

Supplements

- In May 2003, the International Olympic Committee published the results of a study which found that 15% of the 634 supplements it tested contained banned substances, including nandrolone.
- Listed on a country-by-country basis, the Netherlands saw the highest levels of positive products with just over one in four of the tested substances containing a banned substance. The UK had the third highest rate, with 18.9% providing a positive result.

Do you need them ?

- Should be able to get vitamins & minerals from food
- CHO drinks / gels / bars
- Protein replacement products

Running stability exercises – Level 1 1/ Pelvic tilt Lie on the bed, knees bent and pointing to the ceiling Flatten back, tilt pelvis back Lift pelvis off the floor Avoid pushing on your feet and cramping hams Lower back and pelvis down slow

2/ Deep abdominal work Use as a base for all exercise Lie on your back , lumbar spine in neutral, feet pointing forward, knees pointing up Breathe in (expanding your chest) Breathe

out and relax all abs Gently contract lower abs (as stopping going to the loo) till you feel contraction under fingers Leg lift (no pressure on the other leg and no arching back)

Sequence:

- B in - out - contract - B in - out leg lift - B in - out leg down - 23-30 on each leg and start with R leg lift - daily

3/ Dynamic standing Breathe in - grow tall Breathe out - maintain height and relax your rib cage down Bring your weight over your heels Take your whole body over the middle of your feet Keep looking ahead Move your arms as in running Progress: Find the position as above Tip toes Add running arms

WRONG

CORRECT

4/ Clams Lie on your side, straight back, knees bent Find spine neutral (arch your back in/out and find middle in between these points) Contract Lower Abs Lift your knee up, while maintaining ankles together Progress: Lift the uppermost leg straight up

5/ Lunges Find dynamic standing posture Lunge forwards with one leg while maintaining your back straight and pelvis in neutral. Return back to starting position Progress: Within the last set maintain the lunge position and pulse up and down 10-15 times before returning back to starting position

WRONG

CORRECT

Week 4

Abdominal Control

Exercise Technique Double leg and arm stretch with gym ball Perform 2 sets × 15 on each leg.

Find neutral spine and contract LAbs. Assume a position as in the picture A Breathe in-out - stretch both legs out and one arm back while keeping ribs down (B) Breathe in - out and return to starting position

Exercise Technique Knee fall out Perform 2 sets × 15 on each leg.

Find neutral spine and contract LAbs. Breathe in-out and let one leg to fall out while keeping back and pelvis still Breathe in - out and return to starting position (think of an elastic band attached from your knee to tummy button)

Gluteal Muscles (week 4)

A B

Exercise Technique Shoulder Bridge with gym ball +/- leg lift Perform 2 sets of 15

Place legs on the gym ball Find neutral spine and contract lower abs. and buttock muscles Breathe in-out, lift your pelvis off the floor (A) Breathe in-out, return to the floor alternatively Breathe in-out and lift one leg off the ball while maintaining the bridge (see picture B) Breathe in - out and return leg back on the ball and pelvis down

Exercise Technique Swimming legs on gym ball Perform 2 sets of 30 kicks (alternating the legs) Don not move your trunk from side to side

Lie over the top of the gym ball (see picture) Find a neutral spine contract LAbs and buttock muscles Pulse legs in a swimming fashion while maintaining your back still

Leg and Trunk Strengthening (week 4)

Exercise Technique Hip lifts on gym ball

Assume the position as in picture A

A B

Perform 2 sets of 15 repetitions

Breathe in-out and bring your pelvis up, using your abdominal and front hip muscles Keep your back straight, motion should only occur at your hips Breathe in-out and return to the starting position

Exercise Technique Gluteal muscles level 4 Perform 3 sets of 15 repetitions on each leg

- Do not twist or side bend your spine while

lifting your leg.

Keep spine straight and bend knees Find spine neutral contract L.Abs Lift the uppermost leg straight up to neutral position and bring it slightly back Pulse leg up/down

Standing drills (week 4 - running track)

Exercise Technique

A B

Lunge jump Perform 3 sets of 15 repetitions on each leg Do not lean forwards or back while performing the lunge Your weight should be distributed evenly on both feet

Find dynamic standing posture Jump up and land in a lunge position, hold for 5 seconds and return to standing. Hands by your side Progress: close eyes

Exercise Technique High knee with heel raise

Cover 20-40m distance 2 times Do not bend you backwards or slouch while lifting the leg

Find the dynamic standing position as in week one Raise your knee to hip level (parallel with the ground and foot like a fish hook) and heel raise on supporting leg Bring leg back down stepping forwards

Week 5 - Abdominal Control

Exercise Technique Supine Running Perform 2 sets × 15 on each leg. Progress: add alternating arms

Find neutral spine, place both leg on the gym ball and contract L.Abs. (as in picture A but with arms down) Breathe in-out - stretch one leg out while keeping ribs down (B) Breathe in - out - return to starting position

Repeat with opposite leg

Exercise Technique Knee fall out (level 2) Perform 2 sets × 15 on each leg.

assume position as in picture A and find neutral spine and contract lower abs Breathe in-out and let both legs to fall to one side control movement through the centre Breathe in - out and return to starting position (think of an elastic band)

Gluteal Muscles (week 5)

Exercise Technique Reverse bridge and leg lift Perform 2 sets of 15 on each leg

Assume the position as in picture A Find neutral spine and contract lower abs. and buttock muscles Breathe in- out - lift leg off the floor Breathe in-out - return to the floor Repeat with other leg

Exercise Technique Swimming legs on gym ball (arm parallel hold) Perform 2 sets of 30 kicks (alternating the legs) Don not move your trunk from side to side

Lie over the top of the gym ball (see picture) Find a neutral spine contract lower abs and buttock muscles Pulse legs in a swimming fashion while maintaing your back still

Leg and Trunk Strengthening (week 5)

Exercise Technique Front plank hold +/- ball Side plank hold +/-ball Hold up to one minute

Assume the position as in pictures Progress: lift one leg and hold

Exercise Technique Gluteal muscles level 4 Perform 3 sets of 20 repetitions on each leg Do not twist or side bend your spine while lifting your leg.

Keep spine straight and bend knees Find spine neutral contract L.Abs Lift the uppermost leg straight up to neutral position and bring it slightly back Pulse leg up/down

Standing drills (week 5 - running track)

Exercise Technique

Lunge jump (alternating and landing on a parallel line) Perform 2 sets of 30 hops (includes both legs) Do not lean forwards or back while performing the lunge Your weight should be distributed evenly on both feet

Find dynamic standing posture Jump up and land in a lunge position, hold for 5 seconds, jump up and land into alternating lunge and hold 5 seconds Hands by your side to balance

Progress: close eyes

Exercise Technique High knee with jump

Cover 20-40m distance 2 times Do not bend you backwards or slouch while lifting the leg Find the dynamic standing position as in week one Jump up and raise your knee to hip level (parallel with the ground) Hop forwards Repeat on other leg

Week 6

Abdominal Control

Exercise Technique Supine Running Perform 2 sets × 15 on each leg.

Find neutral spine and contract lower abs. Lift knees above your hips Breathe in-out - stretch opposite arm and opposite leg out while keeping ribs down Breathe in - out - return to starting position Repeat with opposite leg

Gluteal Muscles (week 6)

Exercise Technique Reverse bridge and leg lift and knee bend Perform 2 sets of 5-10 repetitions on each leg

Lie over the back of the ball Find neutral spine and contract lower abs. and buttock muscles Breathe in- out - lift leg off the floor and bend knee Mimic running action with that leg Repeat with other leg

Exercise Technique Swimming legs on gym ball (arm parallel hold) Perform 2 sets of 30 kicks (alternating the legs) Don not move your trunk from side to side

Lie over the top of the gym ball (see picture) Find a neutral spine contract lower abs and buttock muscles Pulse legs in a swimming fashion while maintaing your back still, concentrate on large hip motion

Leg and Trunk Strengthening (week 6)

Exercise Technique

Front plank hold with ball Side plank hold Hold up to one minute once

Assume the position as in pictures lift one leg and hold

Standing drills (week 6 - running track)

Exercise Technique

walk - run drill

Cover 20-40m distance 3 times Maintain long straight spine, maintain arm swing,

Find the dynamic standing position as in week one Begin to move arms, add legs to walk, maintain height and begin to run, continually speeding up to comfortable pace

Homework Repeat this session' s exercises two more times before next weeks session

Exercise Technique Prone running with a ball Perform 2 sets of 10 reps on each leg

Assume position as in picture A Bring one leg up towards your chest Mimic running action with both legs (picture B)

A B

1. Warm up - jog 2. Go through stretching routine before starting stability and strength 3.

Lower abdominal routine as per last session 4. Gluteal routine as per last session 5. Leg and

trunk routine as per last session 6. Standing drills Concentrate on accurate execution and alignment of the drills, plus correct muscles working in key exercises!

INTERVENTION

Week 1 - Abdominal Control

Exercise Technique Lower Abdominals contraction Perform 2 sets × 10. Hold each repetition for 10 seconds

Breathe in - to prepare Breathe out - while initiating lower abdominal (LAbd.) contraction

Exercise Technique 100' s (leg lift) Perform 2 sets × 10 on each leg.

Find neutral spine Breathe in - to prepare Breathe out - while initiating LAbd contraction and lift knee over the hip Breathe in - and maintain the position Breathe out - AND return back to starting position

Week 1 - Gluteal Muscles

Exercise Technique Bridge Perform 2 sets × 10

Find neutral spine Activate LAbd Tighten your buttock muscles Lift your pelvis off the ground

Exercise Technique Arrow Perform 2 sets × 10 hold each repetition 10 seconds

Find neutral spine Activate LAbd Tighten your buttock muscles Lift shoulders and arms off the ground Lift head off the ground

Week 1 - Standing drills

Exercise Technique Dynamic Standing - Running Arms

Perform 2 sets × 1 min

Breathe in - grow tall Breathe out - maintain height and relax your rib cage down Bring your weight over your heels Take your whole body over the middle of your feet Keep looking ahead Move your arms as in running

Relaxed stance Dynamic standing posture

Running arms while maintaining correct posture

Week 1 - Leg and Trunk Strengthening

Exercise Technique Scissors (Hip Flexors and Abdominals) Perform 3 sets of 10-15 repetitions on each leg Have 2 minutes rest in between the sets Avoid moving the legs too far out if you are unable to maintain your back still.

Find neutral spine Activate LAbd Lift one leg Flatten your spine against the floor and contract all your abdominals Lift the other leg off the ground Breathe in - to prepare Breathe out - straighten one of your legs Breathe in - return to starting position Breathe out - straighten other leg ie. move legs in a scissor action while your your back stays still.

Exercise Technique Clams (Gluteal muscles)

Perform 3 sets of 10-15 repetitions on each leg Do not twist or side bend your spine while lifting your leg.

Find spine neutral Contract LAbd Lift your knee up, while maintaining ankles together

Exercise Technique

Lunge Perform 3 sets of 10-15 repetitions on each leg Do not lean forwards or back while performing the lunge Your weight should be distributed evenly on both feet

Find dynamic standing posture Lunge forwards with one leg while maintaining your back straight Return back to starting position (arms can be by the side of the body or as in picture)

Week 2 - Abdominal Control

Exercise Technique One leg stretch Perform 2 sets × 15 on each leg.

Find neutral spine Breathe in - to prepare Breathe out - initiate LAbd contraction and lift knee over the hip Breathe in - and maintain the position Breathe out - and straighten leg out while maintaining back in neutral Breathe in - bring leg back over hip again Progress: Extend leg further in second set

Week 2 - Gluteal Muscles

Exercise Technique Shoulder Bridge +/- leg lift Perform 2 sets × 10/ each leg 10 sec holds
Use small ball

Place a ball between your knees Find neutral spine Activate LABd Tighten your buttock muscles Breathe in - to prepare Breathe out - Lift your pelvis and back off the ground

Progress: Breathe in - to prepare Breathe out - lift one leg and hold while breathing

Exercise Technique Arrow Perform 2 sets × hold position for 10-20 seconds while pulsing your legs Do not move your trunk from side to side

Find a neutral spine Activate LABd Tighten your buttock muscles Lift shoulders off the ground Lift head off the ground Pulse your legs (swimming style) slowly

Week 2 - Leg and Trunk Strengthening

Exercise Technique Scissors and crunch Perform 2 sets of 10-15 repetitions on each leg
Have 2 minutes rest in between the sets Avoid moving the legs too far out if you are unable to maintain your back still.

Breathe in - to prepare, Breathe out - contract LABs Lift your head, supported by your hands
Lift one leg and then the other leg (as in picture A but hands behind head) Breathe in-out, straighten right leg out while rotating trunk to the left knee Breathe in - bend your right leg and bring right shoulder down Breathe out - repeat to the same side

Exercise Technique Gluteal muscles level 2 Perform 3 sets of 10-15 repetitions on each leg
Do not twist or side bend your spine while lifting your leg.

Find spine neutral Keep spine and head straight Contract LABd Breathe out - Lift the uppermost leg up Breathe in return to neutral position

A B

Week 2 - Standing drills

Exercise Technique Dynamic Standing - Running Arms with added weights

Perform 2 sets × 1 min

Find the position as in week one Add weight in your hands Do not grip the floor with your feet, use abdominals.

Exercise Technique

Lunge and pulse Perform 3 sets of 10-15 repetitions on each leg Do not lean forwards or back while performing the lunge Your weight should be distributed evenly on both feet
Find dynamic standing posture Lunge forwards with one leg while maintaining your back straight Maintain the lunge position and pulse up and down 10-15 times before returning back to starting position Arms by the side of the body or as in picture

Week 3

Abdominal Control

Exercise Technique One leg stretch Perform 2 sets × 15 on each leg.

Find neutral spine and contract lower abs. Lift one and other leg off the floor Lift your head and arms off the floor as in picture Breathe out - stretch one leg out Breathe in - return to starting position Repeat with alternative leg stretch

Gluteal Muscles (week 3)

Exercise Technique Shoulder Bridge and one leg lift Perform 2 sets × 15 on each leg Use small ball

Place a ball between your knees Find neutral spine and contract L abs. Hands across your chest (or by your side) Lift your pelvis and back off the ground Breathe in - out and lift one leg off the floor Breathe in - lower pelvis down Breathe out - lift pelvis up

Exercise Technique Arrow and swim Perform 10 × 20 second holds while pulsing your arms and legs Don not move your trunk from side to side Use pillow under stomach if your back is uncomfortable

Find a neutral spine contract lower abs and buttock muscles Lift shoulders off the ground, bring arms above head and lift head off the ground (or as in picture) Pulse your legs and arms (swimming style) slowly

Leg and Trunk Strengthening (week 3)

Exercise Technique Scissors and crunch Perform 3 sets of 10-15 repetitions on each leg Have 2 minutes rest in between the sets Avoid moving the legs too far out if you are unable to maintain your back still.

Breathe in - to prepare, Breathe out - contract LAbs Lift your head, supported by your hands Lift one leg and then the other leg (as in picture A but hands behind head) Breathe in-out, straighten right leg out while rotating trunk to the left knee Breathe in - bend your right leg and bring right shoulder down Repeat again in opposite direction

Exercise Technique Gluteal muscles level 3 Perform 3 sets of 10-15 repetitions on each leg Do not twist or side bend your spine while lifting your leg.

Keep spine straight Find spine neutral contract LAbs Lift the uppermost leg straight up to neutral position Lift the leg up further while pointing toes up and bring leg back to neutral while pointing pointing toes down